

Synthetic Data Generation for Performance Increase of Waste Classification

Master's Thesis

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1 Abstract

With climate change becoming an increasingly critical problem, sustainable consumption is more necessary than ever. Proper waste sorting is therefore essential to make the best use of the resources we have available. ARIS Robotics specialises in providing machines that assist the public in sorting waste correctly, which requires well-trained waste classification models. This project explores image generation algorithms through diffusion models and whether they can be used to improve downstream performance on the ARIS waste classification dataset.

As Diffusion models are a complex field, the necessary information required to understand the building blocks of diffusion models is covered, from the basic principles of diffusion-based image generation to the different ways of conditioning and controlling the generation process. This project implements state-of-the-art diffusion models for dataset augmentation. Base diffusion models, which learn the image dataset distribution from scratch, and foundation models, which utilise pretrained Stable Diffusion models to have foundational understanding of image structure and semantic context. All implemented diffusion models are trained and evaluated in a side-by-side comparison of downstream classification accuracy across varying amounts of training data and levels of augmentation. This project finds that both the base conditional diffusion model and the foundation models Diff-Aug and Diff-Mix generate images that are comparable to the original dataset's distribution. These models improve downstream performance on the ARIS dataset; however, they do not improve it beyond their current Copy-Paste data augmentation approach. An analysis of the ARIS dataset reveals a hidden bias that affects the downstream performance metric. Although this is held constant across all DiffDA methods, it partially redirects the learned information from the waste objects in question to the machines from which the waste images were captured. This project concludes that DiffDA is a valid way of augmenting ARIS' dataset and that DiffDA methods, specifically Diff-Aug and Diff-Mix, can simplify aspects of their data augmentation process by reducing the dependency on image segmentation masks. Future work includes exploring methods to combine DiffDA with ARIS' Copy-Paste method and to make DiffDA invariant across different machines and lighting conditions.

2 Resumé

Klima-krisen bliver et større og større problem, bæredygtigt forbrug er derfor mere vigtigt end nogensinde før. Effektiv affaldssortering er derfor kritisk for at udnytte de ressourcer, vi har til rådighed, bedst muligt. ARIS Robotics specialiserer sig i at forsyne genbrugsstationer med maskiner, der hjælper borgere med at sortere deres affald korrekt. Det kræver veltrænede klassificeringsmodeller, der kan skelne mellem forskellige affaldstyper. Dette speciale vil udforske generative billedalgoritmer gennem diffusionsmodeller og undersøger, om de kan bruges til at forbedre præstationen af downstream-opgaver på ARIS' affaldsklassificerings datasæt.

Diffusionsmodeller er et komplekst felt. Det er derfor nødvendigt at forstå byggestenene bag diffusionsmodeller fra de basale principper bag diffusion-baseret billedgenerering til de forskellige måder, hvorpå billedgenerering kan styres. Dette projekt implementerer førende diffusionsmodeller inden for dataaugmentering. Basisdiffusionmodeller, der lærer en billedfordeling fra bunden, og foundationmodeller, der udnytter prætrænede Stable Diffusion-modellers fundamentale forståelse af billedestruktur og semantisk kontekst. Alle implementerede diffusionsmodeller er trænet og evalueret med sidestående sammenligninger af klassificerings nøjagtighed på tværs af variende mængder af træningsdata og augmenteringsniveau.

Dette projekt påviser, at både basis diffusionsmodeller og foundationmodels Diff-Aug og Diff-Mix genererer billeder, der er sammenlignelige med det originale datasæts billede fordeling. Disse modeller forbedrer downstream-opgaver på ARIS' datasæt, men de forbedrer ikke ARIS' nuværende data augmenterings metode, Copy-Paste. En analyse af ARIS' datasæt afslører skjult bias der berører downstream-opgave metrikken. Selvom dette fænomen er holdt konstant gennem alle DiffDA metoderne, får det lærte information fra affaldsgenstandene indflydelse fra maskinen som de var indsamlet fra.

Dette projekt konkluderer at DiffDA er en gyldig måde at augmentere ARIS' datasæt og at DiffDA metoder, specifikt Diff-Aug og Diff-Mix, kan simplificere aspekter af data augmenterings processen ved at reducere ARIS' afhængighed af segmenteringsmasker. Fremtidigt arbejde inkluderer udforskning af metoder til at kombinere DiffDA med ARIS' Copy-Paste metode og måder, hvorpå DiffDA kan gøres invariant på tværs af maskiner og lysforhold.

Acknowledgements

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As the theme of this project is heavily connected to modern AI, transparent acknowledgment of AI use is in order. GitHub Copilot has been used as a coding tool to accelerate programming and has aided in formatting LaTeX tables and figures. Google's AI-assisted search feature has been used as a powerful search engine, and Grammarly has been used for grammar checking in the report-writing process.

The project concludes a 40 ECTS Master of Science in Robot Systems with a specialization in Advanced Robotics Technology at the University of Southern Denmark.

List of Abbreviations

ARIS	ARIS Robotics.	LAION	Large-scale Artificial Intelligence Open Network.
CIFAR	Canadian Institute for Advanced Research.	LDM	Latent Diffusion Model.
CLIP	Contrastive Language–Image Pre-training.	LoRA	Low-Rank Adaptation.
DA	Data Augmentation.	ODE	Ordinary Differential Equation.
DDIM	Denoising Diffusion Implicit Model.	SD	Stable Diffusion.
DDPM	Denoising Diffusion Probabilistic Model.	SDEdit	Stochastic Differential Editing.
DiffDA	Diffusion-based Data Augmentation.	SOTA	State-of-the-Art.
DM	Diffusion Model.	TI	Textual Inversion.
FID	Fréchet Inception Distance.	UN	United Nations.
GAN	Generative Adversarial Network.	VAE	Variational Autoencoder.
GPU	Graphics Processing Unit.	ViT	Vision Transformer.
KL	Kullback–Leibler.	VQ	Vector Quantization.
		VRAM	Video Random Access Memory.
		WADE	Waste Detective.

3 Introduction

In accordance with the 12th United Nations (UN) Sustainable Development Goal, which promotes sustainable consumption and production patterns [1], proper recycling and waste sorting are critical to the future of our world. ARIS Robotics helps in this endeavour by providing waste classification machines for recycling centres to help citizens sort recyclable materials correctly. A computer vision system combined with a classification neural network takes an input image of a piece of waste, and based on previous training on a dataset containing images of available waste classes, classifies the piece of waste as the most probable matching class. This method works reliably on classes with sufficient data for training, such as differentiating between glass wool and rock wool [2]. However, for uncommon or newly arising waste classes due to legislative changes, thousands of example images do not exist for each new class. The manual collection of enough data points to train these new classes is considered a high investment requiring expensive manual labour, making manual data collection an unattractive solution.

Currently, classical data augmentations, which rely on simple image processing techniques to modify the original data, have been successful in training models that generalize to hue shifts, brightness changes, geometric transforms, and similar perturbations [3]. Classical approaches therefore retain semantic coordination well but struggle to diversify beyond the aforementioned transformations. With the advent of modern image-generation networks, the idea of synthetically generating training datasets for image classification has emerged as *Diffusion based Data Augmentation* (DiffDA) [4, 5]. While classical dataset augmentations reuse significant portions of the original images, DiffDA methods aim to understand the semantic content of the original dataset and use the semantic understanding to generate new permutations of geometries and textures unseen in the original dataset.

DiffDA methods are therefore fundamentally different from classical approaches; several considerations should be taken into account when choosing a new direction for data augmentation. Which kind of diffusion model should be chosen? Should it be trained from scratch or pretrained? Should it synthesize completely new images, or should it be guided by an input image? And when a method is finally chosen, how much synthetic data should be chosen? And does it depend on the dataset size? This thesis attempts to answer these questions, among others, for the industrial waste sorting use case of ARIS Robotics.

Related works

The field of neural network-based data augmentation remains an active area of research. The topic of waste sorting has also been addressed by Bacchin et al., who propose WasteGAN [6] as a complete waste-sorting pipeline. WasteGAN utilises a Generative Adversarial Network to synthetically augment the training dataset, which they then use to train a segmentation model. With the segmentation model, grasp point predictions can be made from the segmentation masks, enabling a robotic manipulator to pick and move. They found that GAN-based data augmentation did lead to downstream performance improvements; however, they note an upgrade from GANs to DiffDA as a direction for future work, which this work aims to explore.

Previous DiffDA attempts focus on large-scale publicly available datasets such as CIFAR100, ImageNet, Caltech101, and more. A large number of DiffDA methods utilise pretrained diffusion pipelines such as Stable Diffusion models [7]. He et al. conducted one of the earliest explorations of DiffDA using Real Guidance and concluded that synthetic data is beneficial for zero-shot and few-shot classification, achieving the best results on CIFAR100 [8]. Trabucco et al. build upon Real Guidance and propose DA-Fusion. They use the same image guidance but add a fine-tuning step, making their method more domain-agnostic. Wang et al. introduced Diff-Aug, Diff-Mix, and Diff-II as DiffDA methods using fine-tuned SD models [9]. They achieved improved results with Diff-Mix across six different datasets, performing an average of 0.5% above the baseline classification. Notably, the common practice in the scientific community for evaluating DiffDA methods does not account for uncertainties in model performance. This makes the methodology questionable when they state they achieve a 0.1 – 0.5% performance increase over the baseline without arguing for the statistical significance. This work will, however, account for uncertainties by accounting for diffusion and downstream model variance

Diffusion-based data augmentations have a wide range of applications in computer vision due to their high level of controllability. Several studies explore how bounding boxes from object detection can further guide the diffusion process for downstream tasks [10, 11, 12, 13]. Similarly, other work explores how segmentation masks can be used as guidance [14, 15], which shows notable improvements. However, as this work is limited to image classification, these methods will be used only as inspiration.

Problem Statement

The problem which will be investigated in this Master thesis is stated as follows:

Can image based datasets be artificially inflated via modern image generative algorithms to improve downstream performance?

As this statement is broad, this project will tackle answering the question with the following research tasks:

- Investigate methods for augmenting datasets via diffusion model image generation algorithms and compare them to classical data augmentation approaches.
- Analyse whether image generation models trained purely on the original dataset can lead to downstream performance increase.
- Investigate whether image generation algorithms generally lead to downstream performance increases or under which criterion they lead to performance increases.

Project Constraints

The development of this project will have access to two 3090 GPU computers with 24GB of VRAM. This is the limiting factor in model size, training time, and inference time. To limit the scope of this thesis, only data augmentation methods for image classification will be considered.

Readers Guide

The following section describes the reading process of this report. Section 4 describes the relevant theory and literature required for understanding and implementing diffusion models. Section 5 covers the implementation of the selected DiffDA methods, the ARIS waste classification dataset, and the nuances behind it, followed by the experimental setup used for fine-tuning and evaluating the DiffDA methods. Section 6 covers the qualitative and quantitative results of the experimental phase of the project. Section 7 presents a retrospective discussion of DiffDA methods applied to industrial use cases, based on the results found. Finally, the project is concluded in Section 8.

4 Diffusion models

This section covers the fundamentals of diffusion-based image generation and the previously attempted applications of DiffDA. As diffusion-based image generation is a very young field of research, this section is structured according to the relative publication timings of the respective milestones within the DiffDA field, as shown in Figure 4.0.1

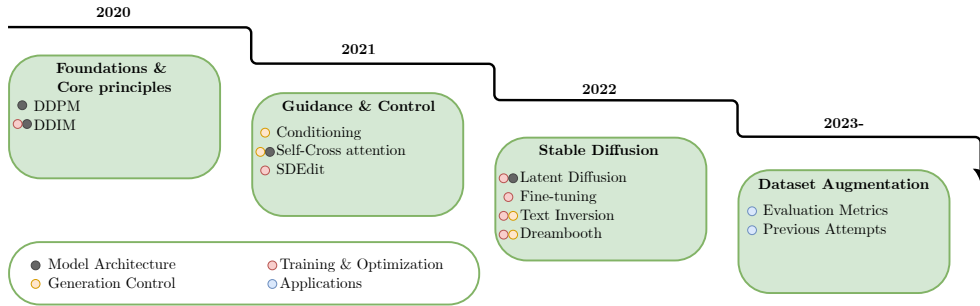


Figure 4.0.1: Diffusion image generation historical timeline of subjects covered in this section.

Initially, the fundamentals of diffusion models are covered in Section 4.1. As the base diffusion model is not intrinsically controllable, Section 4.2 covers the key methods used for guiding diffusion processes. Section 4.3 covers Stable Diffusion and its relevance in the field of dataset augmentation. With an understanding of the theory behind diffusion-based image generation, Section 4.4 covers previous uses of DiffDA, classic data augmentation approaches, as well as metrics used for evaluating diffusion models.

4.1 Fundamentals & Core Principles

DDPM

Denoising diffusion probabilistic models (DDPM) were popularized in 2020 by Ho et al. [16]. At the time, they achieved state-of-the-art image synthesis results on the CelebA-HQ and CIFAR-10 datasets and revolutionized image-generation models. The core idea of diffusion models is to synthesize an image

$\tilde{x}_0$ from the training distribution $q(x_0)$ from a fully corrupted image x_T sampled from a Gaussian distribution $x_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. Diffusion models are based on two core procedures, forward diffusion and reverse diffusion. Forward diffusion serves to gradually corrupt the image, while reverse diffusion reconstructs an image from pure Gaussian noise. A visualization of both the forward and reverse diffusion processes is shown in Figure 4.1.1.

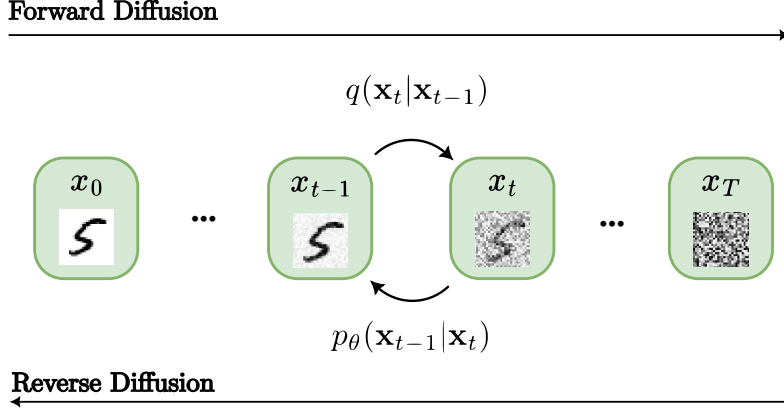


Figure 4.1.1: Forward and reverse diffusion illustration. $x_0, x_t, \dots, x_T$ represent the original image as it gradually gets destroyed by Gaussian noise. $q(x_t|x_{t-1})$ is the probability of the sampling x_t given x_{t-1} . $p_\theta(x_{t-1}|x_t)$ represents the probability of sampling x_{t-1} given x_t and parameters θ .

In the forward diffusion process, a small amount of noise is added iteratively through the β schedule $\beta_1, \beta_2, \dots, \beta_T$ to x_0 until the original image is fully corrupted. The subscript defines the noise timestep with $0 < \beta_t < 1$ and $t \in \{0, 1, 2, \dots, T\}$. At the 0th timestep, only the pure, noise-free image is present, whereas at timestep T the image is fully corrupted. The forward diffusion process is defined by Ho et al. as:

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t \mathbf{I}) \quad (4.1)$$

Where $q(x_t | x_{t-1})$ is the probability distribution of x_t given x_{t-1} . When writing eq. 4.1 in its sampling form (eq. 4.2) it can be clearly seen that each step in the forward diffusion process only depends of the previous step.

$$x_t = \sqrt{1 - \beta_t} x_{t-1} + \sqrt{\beta_t} \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \mathbf{I}) \quad (4.2)$$

Notice that at small values of β , x_t will be closer to x_0 , while β values closer to 1 will cause x_t to be predominantly Gaussian noise. This causes x_t to gradually converge towards a Gaussian distribution for each new timestep t . A core property of DDPM is that each noise step only depends on the previous

step and is therefore a Markov chain. Eq. 4.3 shows how the fully corrupted image is generated from T timesteps of small noise steps.

$$q(x_{1:T} | x_0) = \prod_{t=1}^T q(x_t | x_{t-1}) \quad (4.3)$$

Where $q(x_{1:T} | x_0)$ is the probability distribution of the noise trajectory from x_1 to x_T given the fresh image sample. Ho et al. defines α_t and $\bar{\alpha}_t$ as:

$$\alpha_t = 1 - \beta_t \quad (4.4)$$

$$\bar{\alpha}_t = \prod_{s=1}^t \alpha_s \quad (4.5)$$

Where α_t is the complementary to β , it can be thought of as the amount of signal retained from the previous sample, $\bar{\alpha}_t$ is the cumulative product of all α_s up until timestep t , it effectively describes the total amount of signal retained in x_t . By using the α_t notation then eq. 4.2 becomes:

$$x_t = \sqrt{\alpha_t} x_{t-1} + \sqrt{1 - \alpha_t} \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \mathbf{I}) \quad (4.6)$$

Ho et al. show that eq. 4.6 can be reformulated into eq. 4.7. They do this by recursively substituting x_{t-1} with x_t from eq.4.6 until $t = 0$ which finally reveals 4.7.

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I}) \quad (4.7)$$

Eq. 4.7 is significant because it shows that any noisy sample can be sampled as a mixture of the original image x_0 and random Gaussian noise ϵ given timestep t . It can be equivalently written as a probability distribution:

$$q(x_t | x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t) \mathbf{I}). \quad (4.8)$$

This shows that the corrupted image x_t is sampled from a distribution whose mean and variance are a scaled version of the original image distribution. Where the mean approaches 0 while the variance approaches 1 as the number of timesteps increases.

Using eq. 4.7 and 4.8, forward diffusion can thereby provide a controlled method for gradually destroying an image until it is entirely Gaussian noise. With fine-grained control of the image corruption and a large dataset of an image distribution, a neural network could be trained to approximate the reverse diffusion process $p_\theta(x_{t-1}|x_t)$. A U-Net is used to estimate the noise at a

given timestep. Ho et al. show that the training objective for estimating noise can be simplified to an L2-loss function between the sampled and predicted noise (eq. 4.9) [16].

$$\mathcal{L}_{DM} := \mathbb{E}_{x, \epsilon \sim \mathcal{N}(0, \mathbf{I}), t} [\|\epsilon - \epsilon_\theta(x_t, t)\|_2^2] \quad (4.9)$$

Where $\mathbb{E}_{x, \epsilon \sim \mathcal{N}(0, \mathbf{I}), t}$ is the expected loss value given the image, Gaussian noise and the timestep t , ϵ is the sampled noise and $\epsilon_\theta(x_t, t)$ is the predicted noise. With a repeatable, scalable training loop, it is possible to train diffusion models to synthesize images from arbitrary image distributions given a representative dataset. Algorithm 1 shows the core training steps in the training loop.

Algorithm 1 Diffusion model training loop [16]

```

1: repeat
2:    $x_0 \sim q(x_0)$ 
3:    $t \sim \text{Uniform}(\{1, 2, \dots, T\})$ 
4:    $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
5:    $\bar{\alpha}_t = \text{Scheduler}(t)$ 
6:    $x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$ 
7:    $\mathcal{L}_{DM} = \mathbb{E}_{x, \epsilon \sim \mathcal{N}(0, \mathbf{I}), t} [\|\epsilon - \epsilon_\theta(x_t, t)\|_2^2]$ 
8:    $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}_{DM}$ 
9: until convergence

```

First, an image, timestep, and random Gaussian noise are sampled. Then, forward diffusion is performed, and the loss is calculated as a measure of the model’s ability to predict the actual noise added. Finally, the model parameters are updated by taking a learning rate scaled, η , gradient step towards lower loss. This process is repeated until convergence or training termination. By randomly sampling timesteps uniformly, the model learns to make qualitative guesses about the final image at any timestep; however, a model prediction from the 1000th timestep rarely resembles an image from the training distribution. Ho et al. show that a reverse denoising step can be formulated to estimate an image from the training distribution through an inference loop. The reverse denoising step is formulated as:

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(x_t, t) \right) + \sigma_t \mathbf{z} \quad (4.10)$$

Where $\mathbf{z}$ is a fresh sample of Gaussian noise $z \sim \mathcal{N}(0, \mathbf{I})$ and σ_t serves to scale $\mathbf{z}$ to a standard deviation of σ_t which is determined by the schedulers [16]. In the inference loop, an image prediction is made at each iteration. As the model learns to predict noise and not the final image, the noise prediction needs to be

subtracted from the original Gaussian noise x_T . Adding fresh noise according to the scheduler at every inference step causes the diffusion model to reliably synthesize images from the training distribution. Algorithm 2 shows the main steps of inference.

Algorithm 2 Diffusion model inference loop [16]

```

1:  $x_T \sim \mathcal{N}(0, \mathbf{I})$ 
2: for  $t = \{T, T-1, \dots, 1\}$  do
3:    $\bar{\alpha}_t, \alpha_t, \sigma_t = \text{Scheduler}(t)$ 
4:   if  $t > 1$  then
5:      $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I})$ 
6:   else
7:      $\mathbf{z} = 0$ 
8:   end if
9:    $x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left( x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(x_t, t) \right) + \sigma_t \mathbf{z}$ 
10: end for
11: return  $x_0$ 

```

When doing inference, the first step is to sample pure Gaussian noise, $\epsilon \sim \mathcal{N}(0, \mathbf{I})$, for timestep T . Then the model attempts to predict the added noise, after which the noise prediction is scaled according to the scheduler. Finally, the new noise is sampled as $\mathbf{z}$ and scaled according to the scheduler, after which a new iteration can begin. An illustrative visualisation of the inference loop can be seen in Figure 4.1.2.

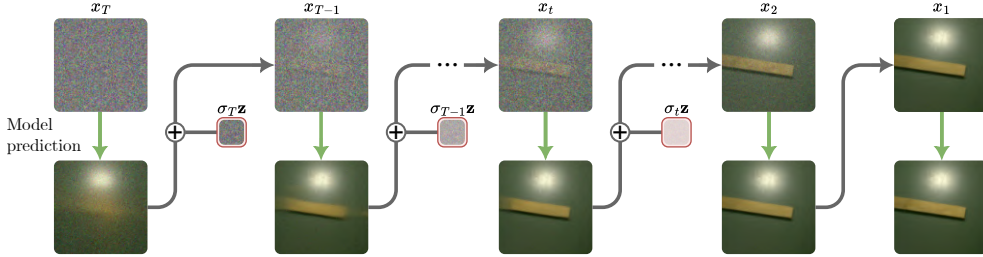


Figure 4.1.2: Visual illustration of diffusion image inference.

Together the forward diffusion and reverse diffusion process lays the foundation of training diffusion models. The forward diffusion process offers a controlled method for gradually corrupting an image into pure Gaussian noise. By randomly corrupting images with different levels of noise it is possible for a neural network to approximate the reverse diffusion process to synthesize images from the learnt image distribution by denoising samples of pure Gaussian noise. The next section will go into more detail on the schedulers and advancements that have significantly streamlined the training and inference processes.

Schedulers

A key part in diffusion models is the noise scheduler, which determines the level of noise applied to x at timestep t . Based on standards established in the original DDPM paper, the maximum timesteps T where the image is 100% noise has a value of $T = 1000$ [16]. For practical purposes, this means that every generated image requires 1000 denoising iterations using the U-Net denoising model through the Markov chain, resulting in long inference times. Song et al. introduced the Denoising Diffusion Implicit Model (DDIM) to accelerate the inference time of DDPMs significantly [17]. They generalize DDPMs to a non-Markovian diffusion process while retaining the original training objective by reformulating the problem as a first-order ordinary differential equation (ODE). With the diffusion process being formulated as an ODE, larger discrete timesteps can be taken instead of iterating across smaller timesteps. This allows the reverse diffusion process to be performed over 50 timesteps instead of the original 1000, leading to a ~ 20 -fold decrease in inference time. This optimization of inference time is significant and one of the key factors driving the development of diffusion models at a faster pace. Lu et al. further optimized the sampling process of diffusion models by introducing the DPM-Solver++ [18]. By formulating the diffusion process as a second-order ODE, faster image generation is achieved; however, the process experiences instability. This is combated by incorporating thresholding. DPM-Solver++ enables high-quality image generation with only 20-30 timesteps, further reducing inference times compared to DDIM. As DPM-Solver++ does not influence the training process, it can be applied at inference time. DPM-Solver++ is therefore the sampling scheduler used in this project.

4.2 Guidance & Control

In image generation algorithms there are two terms used for controlling the process of generating new images. *Guidance* refers to the process of guiding the image generation process exclusively at inference time and doing so without modifying the architecture of the denoising model [19]. *Conditioning* refers to a diffusion model trained on additional information besides the core training images [20]. The difference between conditional/unconditional diffusion is analogous to supervised/unsupervised learning in machine learning.

Conditioning

Diffusion image generation, as introduced by DDPM, focuses on the diffusion algorithm without highlighting methods for controlling and steering it to generate images guided by the user [16]. DDPM, as introduced, is considered an *unconditional* image generation algorithm, resulting in a model that is trained on a single dataset and, at inference time, generates images that represent

the training distribution it was trained on. *Conditional* image generation, originally introduced in Generative Adversarial Networks (GAN), refers to the control of the image generation process to influence the content of the generated image [20]. Conditional image generation allows a high level of control with many different modalities for guidance. The following list summarizes key conditioning types [7].

- **Class Label Conditioning:** Models using class label embeddings can allow a diffusion model to generate images of specific classes. This is a rudimentary method that can be used for classification datasets.
- **Masked Conditioning:** As class label conditioning only allows control of the distribution the data is drawn from, incorporating another modality allows control of the visual structure of the image. Masked conditioning can restrict the model to only generate in a certain region of an image.
- **Image Conditioning:** Popularized by ControlNet, image conditioning allows higher-level visual control of generated images [21]. Canny edge maps, depth maps, segmentations, human poses, and more are used to have fine-grained control of the image generation process.
- **Text Conditioning:** Pipelines such as Stable Diffusion use text prompts such as CLIP embeddings in order to guide the image generation process [22]. Image generation is performed via cross-attention between image and text embeddings.
- **Multi-Modal Conditioning:** Multiple conditioning methods can be combined to allow control of both visual structure and semantic content, such as combining text conditioning with segmentation maps or combining a reference image with a class label. Combining a reference images with a class label is used to generate an image that has the structure of the reference image, whilst semantically adhering to the class label, as seen in Diff-Mix [9].

With the general modes of conditioning covered, the following sections will cover the theory of the relevant conditioning methods for dataset augmentation.

Class Label Conditioning

For diffusion-based data augmentation of a dataset using class labels, class label conditioning is the logical approach to guided image generation. Class labels can be incorporated into a U-Net via utilizing a class embedding lookup table [23]. For $x \in \mathbb{N}, 0 \leq x < C$, the weights $W \in \mathbb{R}^{C \times D}$ are initialised, where C is the number of classes and D is the embedding dimension. The embeddings

are given by $e(x) = W_x$, where W_x is the x th row of the embedding matrix. For diffusion and U-Nets, the embeddings are incorporated via linear addition on every ResNet block found in the U-Net. By incorporating learnable class embeddings into the U-Net, one diffusion model can be trained to generate images of any given class. The loss function in eq. 4.9 can be updated to include the class label, as seen in eq. 4.11

$$\mathcal{L}_{DM} := \mathbb{E}_{x, \epsilon \sim \mathcal{N}(0,1), t, c} \left[\|\epsilon - \epsilon_\theta(x_t, t, c)\|_2^2 \right] \quad (4.11)$$

Masked Conditioning

Since some image datasets in industrial settings are highly structured, this aspect can be utilised to guide the diffusion process. The ARIS dataset images contain a green background with a single object on it. With a segmentation mask that separates the object from the background, the U-Net learns to denoise the object differently from the background [21]. For the implementation, masked conditioning can be incorporated either as a hidden layer or directly appended to the input images of the U-Net. The noisy image $x_t \in \mathbb{R}^{B \times 3 \times H \times W}$ is concatenated with the mask image $M \in \mathbb{R}^{B \times 1 \times H \times W}$ leading to the noise space representation $z_t = (x_t, M)$ where $z_t \in \mathbb{R}^{B \times 4 \times H \times W}$. With the mask concatenated with the noisy input image, the U-Net is guided to generate the masked areas differently compared to the rest of the image.

Self- & Cross-attention

Since a denoising U-Net does not guarantee visual cohesion in the generated images, attention mechanisms are introduced to enhance cohesion [24]. Attention mechanisms use query, key, and value matrices; Q , K , and V , which are extracted features from the image and/or embedding. Figure 4.2.1 shows the self- and cross-attention processes.

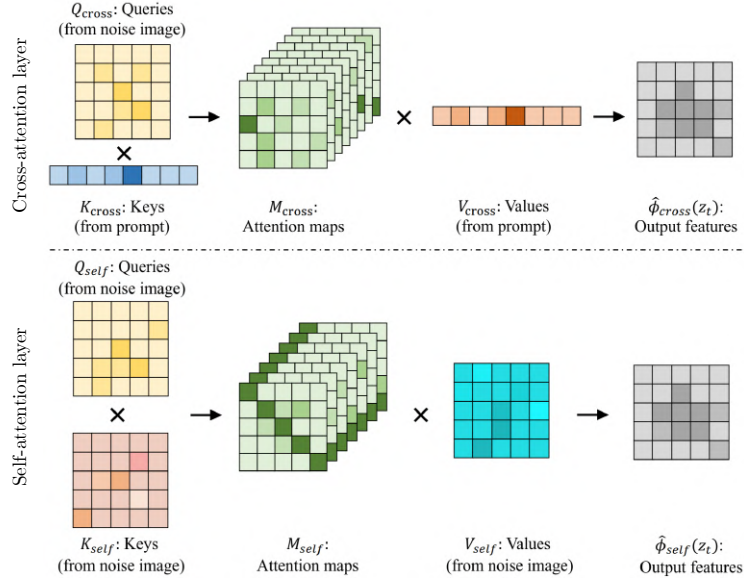


Figure 4.2.1: Self- and Cross-attention visualisation as seen in [24]

Self-attention uses a cross product between two instances of spatial features extracted from the image [24]. This mathematically correlates all pixels with one another, thereby informing the global geometry of an image feature. Self-attention therefore is used to understand and preserve spatial structure in images. The equations for self-attention are displayed as follows.

$$Q_{\text{self}} = W_q(\phi_{\text{self}}(z_t)) \quad (4.12)$$

$$K_{\text{self}} = W_k(\phi_{\text{self}}(z_t)) \quad (4.13)$$

$$V_{\text{self}} = W_v(\phi_{\text{self}}(z_t)) \quad (4.14)$$

$$M_{\text{self}} = \text{Softmax}\left(\frac{Q_{\text{self}}K_{\text{self}}^T}{\sqrt{d_{\text{self}}}}\right) \quad (4.15)$$

$$\hat{\phi}_{\text{self}}(z_t) = W_o(M_{\text{self}}V_{\text{self}}) \quad (4.16)$$

Where W_q, W_k, W_v and W_o are trainable linear projection matrices. $\phi_{\text{self}}(z_t)$ represents the spatial features of the noisy image z_t . d_k denotes the dimension of the keys and queries. $\hat{\phi}_{\text{self}}(z_t)$ is the output feature map of self-attention. Cross-attention modifies the key and value matrices to be derived from an embedding [24]. The embedding for this project will be implemented as text embeddings, although any other embedding can be used. Equations for cross-

attention are displayed as follows:

$$Q_{cross} = W_q(\phi_{cross}(z_t)) \quad (4.17)$$

$$K_{cross} = W_k(\tau_\phi(y)) \quad (4.18)$$

$$V_{cross} = W_v(\tau_\phi(y)) \quad (4.19)$$

$$M_{cross} = \text{Softmax}\left(\frac{Q_{cross}K_{cross}^T}{\sqrt{d_{cross}}}\right) \quad (4.20)$$

$$\hat{\phi}_{cross}(z_t) = W_o(M_{cross}V_{cross}) \quad (4.21)$$

Where the listed values follow similar structure as in self-attention. When cross-attention is used with text embeddings, the output feature map $\hat{\phi}_{cross}(z_t)$ correlates every pixel with an intensity tied to every word in the text embedding.

Image Guidance

The previously mentioned conditioning methods learn a representation of the underlying dataset distribution and can generate a random sample from that distribution given the conditioning input. For many use cases, having one specific input image and wanting near-identical clones of that image with small permutations is desired; such an image is often referred to as a guide image. Image-2-Image models address this use case with Stochastic Differential Editing (SDEdit), which utilises guidance through images to generate images that closely adhere to the guide image’s content [25]. Because this method uses guidance rather than conditioning, it does not require task-specific model training or loss functions. SDEdit proposes the use of eq. 4.7 to determine an initial starting guess for generating images. s defines the strength of augmentation, where $0 \leq s \leq 1$. Eq. 4.22 shows the process of applying noise to the guide image x_0 at initial timesteps t_0 through SDEdit.

$$x_{t_0} = \sqrt{1-s}x_0 + \sqrt{s}\epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I}) \quad (4.22)$$

When $s = 0$, no noising/denoising occurs, leaving the original guide image. When $s = 1$, the guide image has 100% noise applied to it, leaving none of the original image. This results in unguided diffusion. Therefore, s should be chosen as $0 < s < 1$ to guide the image generation process. Figure 4.2.2 shows the generation process of diffusion models guided by SDEdit as well as the influence of s in SDEdit.

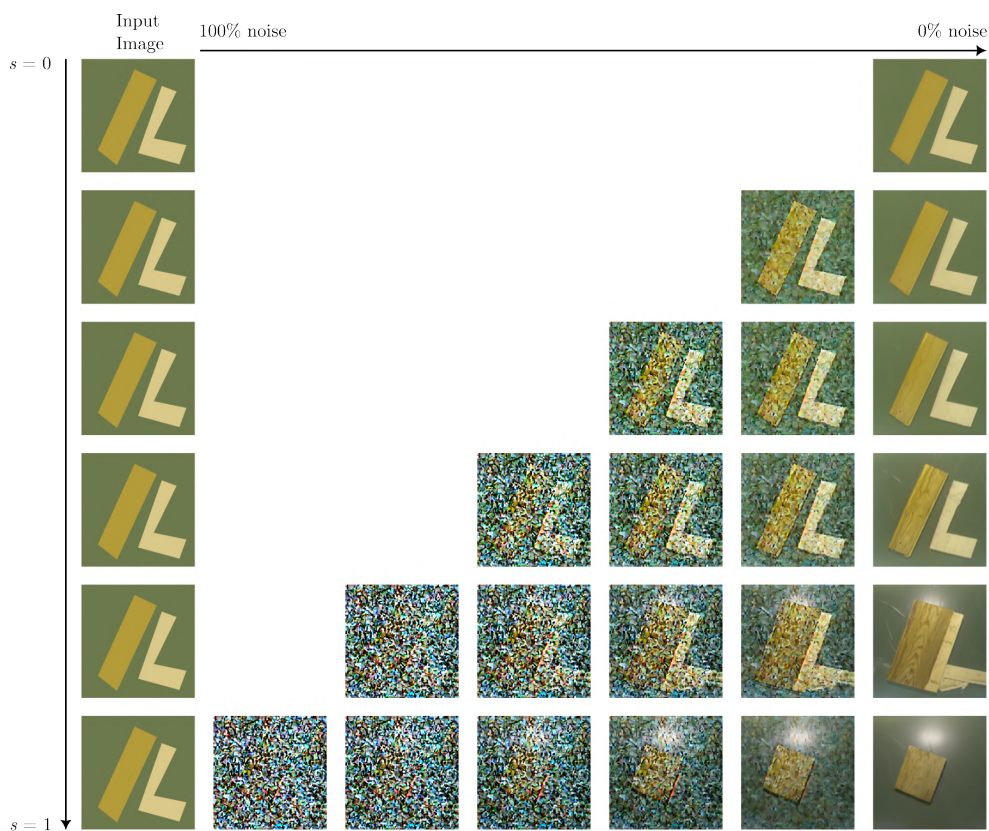


Figure 4.2.2: SDEdit used at varying s values on the same guide image. $s \approx 0$ results in images close to the guide image, while $s = 1$ is completely unguided by the original image.

More methods for conditioning and guidance of diffusion models exist; however, these are the relevant methods for controlling diffusion towards data augmentation of classification datasets. The next Section will delve into Stable Diffusion, how it works, its architecture, and different means to fine-tune it.

4.3 Stable Diffusion

Previous sections cover the foundations of diffusion models and how to control the image generation process. To fully understand the domain of diffusion-based image generation, Stable Diffusion - the core pillar of open-source pre-trained diffusion models, will be covered [22]. Stable Diffusion uses text prompts to guide image generation, a process referred to as text-to-image. Stable Diffusion can also be combined with SDEdit, allowing images to be used as input for the guidance process. The key advantage of using pretrained Stable Diffusion models is the additional data they have been trained on, which enables them to generate high-quality, visually coherent images. The Stable

Diffusion 1.5 model is trained on the 5 billion image-text pairs from LAION-5B [22]. For classification-based downstream tasks where images are categorized into deterministic class labels, text-to-image models are overly flexible and complex for the input modality relative to the requirements. Converting an integer class label to an embedding for conditioning is simple, as covered previously, but converting text to embeddings is algorithmically more challenging. Removing the text embedding from Stable Diffusion and replacing it with a direct integer class embedding would be most fitting for the use case. However, removing the pre-trained Stable Diffusion text embedding would remove the underlying embedding and natural language understanding of the CLIP embeddings. Instead, class labels are embedded through a text prompt. The following section highlights the key algorithms used for applying Stable Diffusion to image classification datasets.

Latent Diffusion

The computational requirements for diffusion models increase exponentially with image resolution. Early state-of-the-art diffusion models, which outperform GAN models, require 150-1000 V100 GPU days to train [26]. Historically, the core Stable Diffusion 1.5 model used for pre-training used 150000 A100 GPU-hours of training with an estimated CO_2 footprint of $11TCO_2$ [22]. In order to enable diffusion model training on GPUs with limited computational resources latent diffusion is used [22]. Latent diffusion models (LDM) promise a perceptually equivalent, but computationally efficient pipeline compared to baseline diffusion models. Instead of training and generating images in pixel space, latent diffusion encodes images via a pretrained variational autoencoder (VAE) into a latent space with drastically reduced dimensionality. With latent diffusion, training is divided into two stages. Firstly, in the *perceptual compression* stage, images used for training are passed through an autoencoder and mapped to a latent space. In the second stage, *semantic compression*, the diffusion model is trained. Given an image $x \in \mathbb{R}^{H \times W \times 3}$, the encoder $\mathcal{E}$ encodes x into a latent space representation, $z = \mathcal{E}(x)$, where $z \in \mathbb{R}^{\frac{H}{f} \times \frac{W}{f} \times c}$, c is the latent dimensionality and f is a scaling factor. For an autoencoder with $(f, c) = (8, 4)$, the encoding of $z = \mathcal{E}(x)$ is a $48\times$ reduction in size. The Decoder $\mathcal{D}$ reconstructs the image $\tilde{x}$ from the latent as $\tilde{x} = \mathcal{D} = \mathcal{D}(\mathcal{E}(x))$. Instead of the diffusion-generated images having the properties of eq. 4.9, they are drawn from the latent distribution in eq. 4.23

$$\mathcal{L}_{LDM} := \mathbb{E}_{z \sim \mathcal{E}(x), \epsilon \sim \mathcal{N}(0,1), t} \left[\|\epsilon - \epsilon_{\theta}(z_t, t)\|_2^2 \right] \quad (4.23)$$

Stable Diffusion uses the AutoencoderKL VAE because it assumes Gaussian properties for the latent space [22]. AutoencoderKL maps images into

latent space via a diagonal Gaussian distribution, $z \in \mathcal{N}(0, \mathbf{I})$. This is beneficial for diffusion processes since new images are generated from a Gaussian distribution. Due to the latent representation retaining a spatial connection to pixel-space, image conditioning such as segmentation masks can still be applied.

Stable Diffusion Model Architecture

Stable Diffusion builds upon the same fundamentals already covered, but adds text embedding to open up directions of customisation such as style and semantic content. Different Stable Diffusion models have been released iteratively, becoming higher resolution and more visually consistent. Currently, SD3.5 is the latest release, capable of generating images up to 1024×1024 in resolution [4]. Figure 4.3.1 shows the visual difference of SD1.5, SD2.1, and SD3.5 using Real Guidance to generate permutations of reference "bear" images from the CIFAR-100 dataset. Notably, the geometric structure of the bears is more realistic in the latter models.



Figure 4.3.1: "bear" images generated through Real Guidance with different Stable Diffusion backbones [4].

Although images generated with newer backbones are visually sharper and higher-fidelity, Li et al. concluded that newer, larger Stable Diffusion backbones do not yield gains in classification performance [4]. Their conclusion is drawn based on the data shown in Table 4.3.1. Based on this conclusion, SD1.5 serves as the backbone in this project.

Table 4.3.1: Top-1 accuracy (%) of Diff-Aug and Diff-Mix with different generative backbones [4].

Method	Caltech-101	CIFAR-100	Birds (5-shot)	Birds (10-shot)
Diff-Aug (SD1.5)	71.67	67.73	60.36	71.71
Diff-Aug (SD2.1)	71.27	68.19	48.73	64.01
Diff-Aug (SD3.5)	71.52	66.97	44.65	58.13
Diff-Mix (SD1.5)	75.62	71.45	62.96	73.26
Diff-Mix (SD2.1)	75.35	71.57	51.14	67.75
Diff-Mix (SD3.5)	75.21	70.64	46.10	60.27

The U-Net found in SD1.5 uses a 4-layer U-Net with cross-attention layers on the outer 3 U-Net layers, as shown in Figure 4.3.2. The key modifications used in the Stable Diffusion architecture compared to previous methods are the inclusion of text tokenizers and embeddings. The initial z_T is the latent normal distribution of noise, where the U-Net iteratively denoises the latent space variable across T .

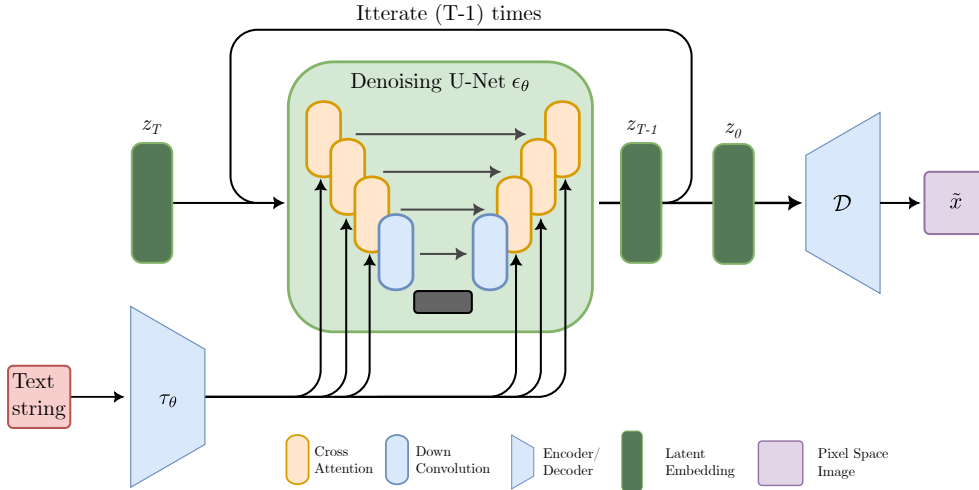


Figure 4.3.2: Stable Diffusion generation pipeline

CLIP embeddings

Stable Diffusion, Dall-E, and other text-to-image models utilize *Contrastive Language-Image Pre-training* (CLIP) embeddings to guide image generation. CLIP embeddings are combined with cross-attention to guide the image generation process based on a text prompt [27]. As emphasized in the abbreviation of CLIP, it utilises contrastive learning to learn a semantic connection between images and text. In order to translate a text string into an embedding understood by neural networks, tokenizers and embedding lookups are used. A Tokenizer takes a text string, y , as input and converts every word or subword into an integer based on a predetermined lookup table. After tokenization, ev-

ery token is passed through a trainable lookup table converting every word to a latent representation, v_{θ_n} , where θ_n represents the ID of the n'th word and $v_{\theta_n} \in \mathbb{R}^{768}$. A self-attention-driven text transformer encodes the latent representations of words into one combined embedding of the whole text string $V = \tau_{\theta}(y)$, where τ_{θ} is the trainable text transformer. V is limited to 77 tokens and is zero-padded to always have the dimension $V \in \mathbb{R}^{77 \times 768}$. This text embedding process is shown in Figure 4.3.3. For training purposes only, an additional layer is added to the text-embedding model. V is convolved across all tokens to create a latent representation of the whole text prompt with the dimensionality of $\mathbb{R}^{768}$

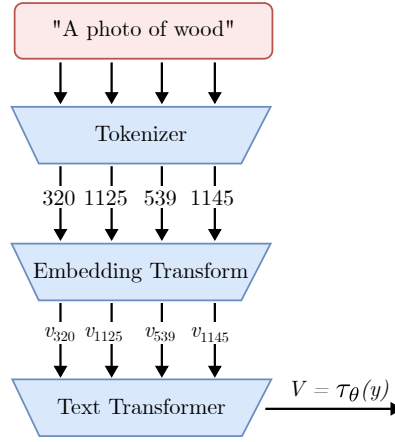


Figure 4.3.3: Example text embedding of y : "A photo of wood".

A *Visual Transformer* (ViT) is used to map a given image x to a latent representation X where $X \in \mathbb{R}^{768}$ [28]. Since the compressed text-embedding model and the ViT share the same latent dimensionality, they can be trained via contrastive learning. Once trained, the trained text-embedding model, without the added 1D convolutional layer, can be used in cross-attention to condition the image generation process, as mentioned in Section 4.2. A given U-Net containing cross-attention layers, trained with a fixed text-embedding model, becomes bound by that specific text-embedding model. This means text-embedding models are not freely interchangeable once used in the conditional training of a denoising U-Net. The ViT trained through CLIP is discarded because its input dimensionality does not align with the varying latent dimensions in the U-Net.

Fine-tuning

As the pre-trained text-embedding model and denoising U-Net are trained on matching text-image pairs from LAION 5B, Stable Diffusion generates images visually consistent with those from LAION 5B [22]. However, for all image-based classification tasks, where the dataset is sorted by class, both the visual

consistency of the images and the semantic meaning of the class labels will not align with pre-trained diffusion models. Therefore, pretrained text-to-image models need to be fine-tuned to both match the visual consistency of general images in the dataset and provide the text-embedding model with a use-case-specific understanding of class labels. The following sections explain the methods used for fine-tuning the text-embedding model and the denoising U-Net.

Textual Inversion

For dataset augmentation using text prompts, the text embedding model needs an understanding of the class labels. When prompted with the text "*A photo of wood*", where "*wood*" is a class label, the text embedding model will associate "wood" with its prior understanding of that word. Originally introduced by Gal et al., textual inversion aims to adapt the text embedding model to modify its understanding of a word or add a new word, y , to its vocabulary [29]. Textual inversion forgoes any fine-tuning of the text transformer or the denoising U-Net by freezing all model weights during training. The only unfrozen parameters are found in the embedding lookup that maps the token ID of y , $\langle * \rangle$, to the latent space representation $v_* \in \mathbb{R}^{768}$. The LDM loss function for estimating noise is given by:

$$\mathcal{L}_{LDM} := \mathbb{E}_{z \sim \mathcal{E}(x), y, \epsilon \sim \mathcal{N}(0,1), t} \left[\|\epsilon - \epsilon_\theta(z_t, t, \tau_\theta(y))\|_2^2 \right] \quad (4.24)$$

where t is the timesteps, z_t is the latent image with noise applied to time t , $\tau_\theta(y)$ is the text embedding of y , ϵ is the unscaled noise sample, and ϵ_θ is the denoising network. v_* is found by optimising the v that minimises eq. 4.24 as shown in eq. 4.25.

$$v_* = \arg \min_v \mathbb{E}_{z \sim \mathcal{E}(x), y, \epsilon \sim \mathcal{N}(0,1), t} \left[\|\epsilon - \epsilon_\theta(z_t, t, \tau_\theta(y))\|_2^2 \right] \quad (4.25)$$

Figure 4.3.4 shows the estimation of $\epsilon_\theta(z_t, t, \tau_\theta(y))$, used to optimize v_* . Gal et al. suggest that as few as 3-5 images of a given class are required for an accurate estimation of v_* .

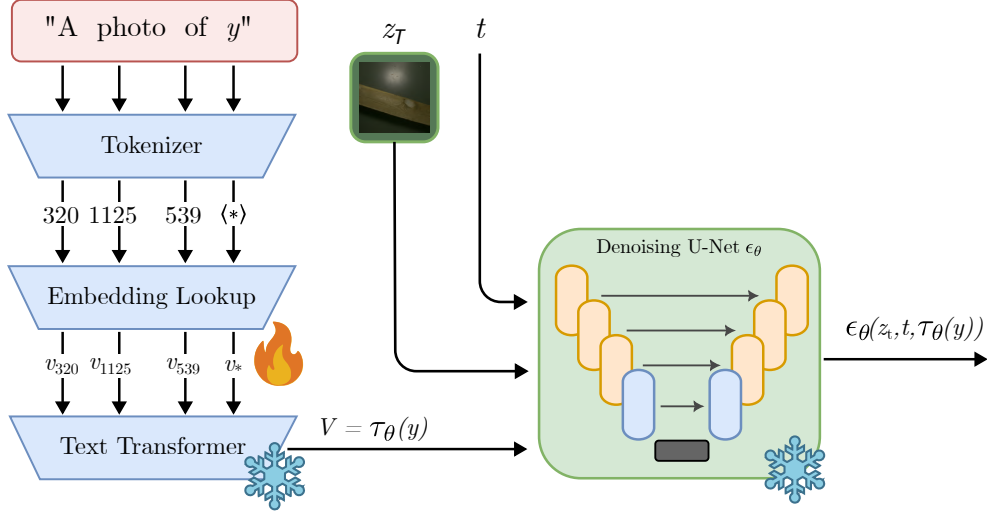


Figure 4.3.4: Noise estimation ϵ_θ of the latent image z_t at timestep t conditioned with text embedding $\tau_\theta(y)$. Both the text transformer and the denoising U-Net are frozen. Textual inversion optimizes v_* , the latent representation of y , to minimise the noise estimation loss function.

Although textual inversion trains the text embedding model to understand the semantics of any given class label, at no point has it fine-tuned the denoising U-Net. This leads to images that are semantically closer to the target class label; however, the visual appearance of the synthetic images can still leave a distributional gap if the Stable Diffusion model was not trained on the target distribution.

LoRA

Another option for diffusion fine-tuning is to fine-tune the models used to generate images via text-to-image. As both the text transformer and the U-Net were frozen in textual inversion, they cannot be fine-tuned to domain-specific image features such as the dataset presented in this industrial use case. Dreambooth, introduced by Ruiz et al.[30], solves this by defining a training process that updates the weights of both the text transformer and the denoising U-Net [30]. This, however, is computationally expensive and requires retraining of the 859M parameters in Stable Diffusion 1.5. *Low Rank Adaptation* (LoRA) [31] introduces a fine-tuning method that compromises the fine-grained visual quality of Dreambooth but outperforms it in many aspects [31]. LoRA offers efficient memory usage, much faster training times, and the ability to merge multiple LoRA training runs at inference time for greater adaptability. For a pre-trained weight matrix $W_0 \in \mathbb{R}^{d \times k}$, an updated version is defined as $W_0 + \Delta W = W_0 + BA$ where $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times k}$ and the rank $r \ll \min(d, k)$. During training, only A and B are trainable, and W_0 remains

frozen.

$$h = W_0x + \Delta Wx = W_0x + BAx = (W_0 + BA)x \quad (4.26)$$

As the low-rank weight matrices can be added to the frozen pre-trained weight, this operation can be done before inference time, adding no additional computational bloat. For Stable Diffusion, LoRA adapters are added onto the cross-attention layers W_q , W_k , W_v , and W_o of the denoising U-Net and onto the self-attention projections of the text transformer [31].

4.4 Dataset Augmentation

The field of general dataset augmentation has already seen significant work. While classical image augmentation methods have their benefits of being a simple yet effective dataset augmentation method, they have their limitations, as they modification are directly derived from training images. Diffusion-based augmentation methods attempt to fill this gap; while more cumbersome to implement, they offer ways to diversify the dataset while preserving semantic information [32]. The following section covers the standard evaluation metric for quantifying DiffDA methods, previous attempts at dataset inflation, key classical image augmentation methods, and the image synthesis method currently used at ARIS, which will serve as a point of reference for the DiffDA methods.

Evaluation Metrics

Alimisis et al. [7] suggest that metrics for quantifying diffusion models can be broken down into two main categories: downstream model performance and perceptual quality and diversity. Firstly, the downstream performance metrics will be introduced, followed by an introduction to the FID score, which aims to quantify the perceptual realism of a dataset.

Downstream Model Performance

The main metric for evaluating diffusion models for data augmentation is the downstream performance of the chosen application. For classification tasks, a model such as a Residual Network (ResNet)[33] is trained on diffusion-generated images and evaluated on real images. Metrics such as accuracy, precision, recall, and F1-scores are used. These metrics indicate whether a direct improvement in performance has been achieved and enable direct comparison between different data augmentation methods. If class balance is important, the macro-averaged accuracy, or F1-score, provides metrics by weighting each class equally. F1-macro has the benefit of including the precision-recall trade-off of the F1-score; however, as the F1-score not only depends on the individual class performance but also misclassification of other classes, the F1-score does

not represent the true positives of a minority class well [34]. For this reason, the macro-averaged accuracy is used for this thesis. Macro accuracy, also referred to as balanced accuracy, is computed as follows:

$$\text{Macro Accuracy} = \frac{1}{C} \sum_{c=1}^C \frac{TP_c}{TP_c + FN_c} \quad (4.27)$$

Although the macro accuracy is balanced, it can misrepresent the model's overall performance because it places no emphasis on each class's size [35].

Perceptual Quality

Fréchet Inception Distance (FID), proposed in 2017 by Heusel et al. [36], replaced the Inception Score (IS) as the standard for quantifying perceptual quality and remains the standard in 2026. FID measures the distance between the latent distributions of two datasets passed through a pretrained Inception-v3 network. The Inception-v3 model is pretrained on ImageNet. When comparing a synthetically generated dataset to a dataset containing real images, commonly the training dataset, a low FID score indicates that the generated dataset closely resembles the reference dataset in terms of quality and diversity.

$$d_F(\mathcal{N}(\mu, \Sigma), \mathcal{N}(\mu', \Sigma'))^2 = \|\mu - \mu'\|_2^2 + \text{tr}(\Sigma + \Sigma' - 2(\Sigma\Sigma')^{\frac{1}{2}}) \quad (4.28)$$

As seen in eq. 4.28, the Fréchet distance assumes a gaussian distribution. Although this is not guaranteed in practice, the metric still holds validity.

Previous attempts at dataset inflation

Many of these publicly available datasets are not fine-grained because they contain class-specific features spanning the entire image. The advantage of using DiffDA on these datasets is that it fully explores the distribution of images within a class, making the classifier more robust to outliers. Li et al. explored the application of DiffDA on the Semi-iNat benchmark with the goal of analysing the downstream performance of DiffDA methods on datasets with varying levels of difference between classes [4]. The Semi-iNat dataset, for this use case, consists of 810 (sub)classes of insects with 5-80 images per class. Figure 4.4.1 shows the hierarchical structure of the Semi-iNat dataset and how the images become significantly more similar and fine-detailed at the *species* level compared to *order* level.

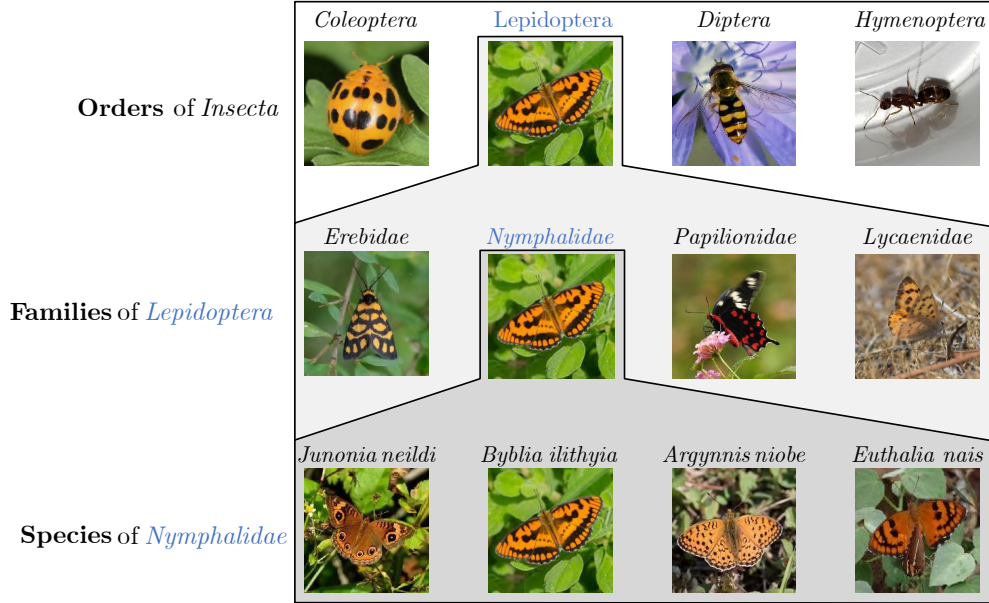


Figure 4.4.1: Hierarchical taxonomic annotations of the Semi-iNat dataset [37, 4].

Their findings, as shown in Table 4.4.1, show only very minor improvements at the *order* level of insect classification and a degradation of performance at the *species* and *family* level. They conclude that the fine-detailed subtle visual variations are too small for diffusion models to accurately reconstruct. Furthermore, they conclude that DiffDA can increase classification performance under certain conditions; however, the effectiveness strongly depends on the visual features to be learned and the uniqueness of semantic cues in the data. Their analysis provides an expectation of the performance of DiffDA methods on image classification tasks where details are small and nuanced [4]. For industrial use cases such as waste sorting of different types of wood based on the grain texture, the inter-class nuances might be small enough for DiffDA not to give significant general performance increases. The implications of this research will be further explored in the discussion.

Table 4.4.1: Top-1 and Top-5 accuracy (%) of different DiffDA methods on the Semi-iNat dataset under varying semantic granularities [4].

Method	Species		Family		Order	
	Top-1	Top-5	Top-1	Top-5	Top-1	Top-5
Baseline	41.88	64.15	47.85	70.79	54.96	79.43
Real Guidance	38.96	61.89	44.78	66.63	53.85	78.27
GIF	37.83	60.21	43.91	66.26	53.16	77.83
Diff-Aug	32.91	55.73	43.96	66.37	54.58	79.81
Diff-Mix	37.04	60.86	45.06	67.20	55.42	81.94

Classical data augmentation approaches

Data augmentation is a common preprocessing step in machine learning workflows. Generally, images contained in a training dataset do not fully represent the domain of available images [4]. Therefore, image augmentation methods are used to increase the diversity of images used for training, even with a limited dataset[38]. The goal of data augmentation is to create more images that differ in appearance but retain the same semantic content. Traditional approaches for image augmentation range from geometric transformations, such as image rotation, translation, flip, crop, scale, etc., to color space adjustment, such as hue shifting, blurring, sharpening, jittering, etc. [3]. The following section will cover the popular classic augmentation methods mixup and CutMix and highlight their benefits over traditional classic image augmentation methods. Secondly, the preferred data augmentation method by ARIS, Copy-Paste, will be explained.

Mixup and CutMix

The core idea of mixup and CutMix is to combine two training samples with a mixing ratio λ . When the labels are combined, they form soft targets rather than traditional one-hot encodings. Mixup introduced by Zhang et al. [39] in 2018 achieves this through a simple linear interpolation, making the approach data-agnostic:

$$\tilde{x} = \lambda x_i + (1 - \lambda)x_j, \quad (4.29)$$

$$\tilde{y} = \lambda y_i + (1 - \lambda)y_j \quad (4.30)$$

Where (x_i, y_i) and (x_j, y_j) are two randomly selected samples from the training data and $\lambda \in [0, 1]$ defines the mixing ratio. Despite its simplicity, it achieved state-of-the-art results at its time on the CIFAR-10, CIFAR-100, and the ImageNet-2012 image classification datasets.

The CutMix method, introduced by Yun et al. [40] in 2019, centers on swapping a randomly selected region from one training sample with that of another. The operation is defined in eq. 4.31 and 4.32. Notice that it is only the mixing method of CutMix (eq. 4.31) that differs from mixup; the label mixing (eq. 4.32) remains the same as mixup with a linear interpolation.

$$\tilde{x} = \mathbf{M} \odot x_i + (\mathbf{1} - \mathbf{M}) \odot x_j \quad (4.31)$$

$$\tilde{y} = \lambda y_i + (1 - \lambda)y_j \quad (4.32)$$

Where $\mathbf{M}$ defines a binary mask which specifies which part of the image belongs to sample x_i and x_j , and $\odot$ defines an elementwise multiplication. The area of the patch is defined by λ by scaling the width and height of the patch by $\sqrt{1 - \lambda}$.

CutMix achieved state-of-the-art results at the time of publication on CIFAR

classification and remains a popular classic data augmentation method for neural network-based computer vision tasks.

An illustration of the mixup and CutMix methods can be seen in Figure 4.4.2; an expanded dataset preview of the mixup and CutMix data augmentation methods can be found in Appendix B.3 and B.4.

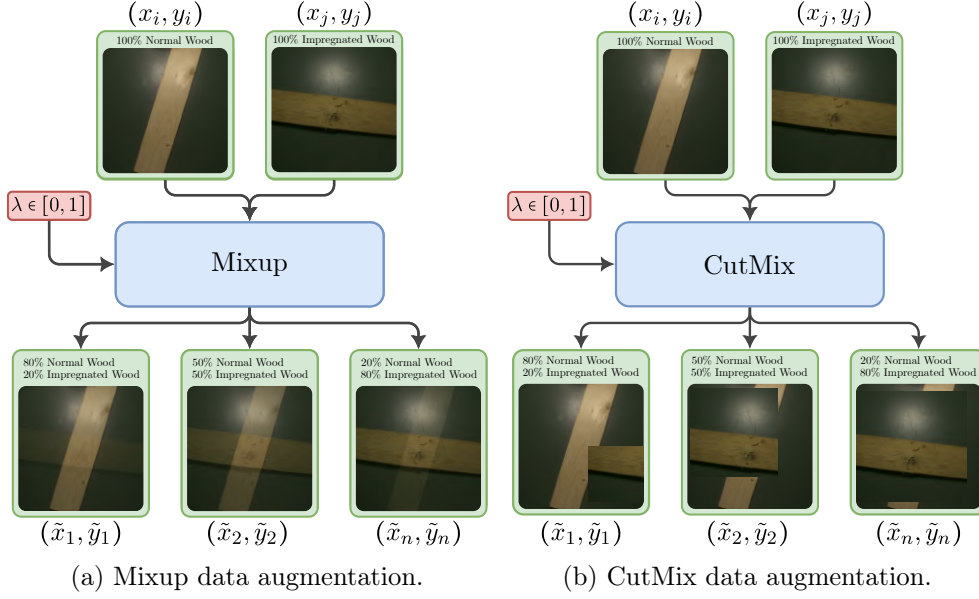


Figure 4.4.2: Mixup and CutMix data augmentation method illustration.

Similar to mixup and CutMix, ARIS proposes a Copy-Paste method for mixing multiple images into augmented training samples, which is introduced next.

Copy-Paste

The Copy-Paste method was first popularised in 2020 by Ghiasi et al. [41], as a data augmentation method for instance segmentation. The idea behind the method is simple: take segmentations from one sample and randomly paste them onto another sample. As ARIS Robotics is currently more focused on classification tasks, they have augmented the approach to fit their use case. Since all their machines share the same background for the images, they can use the shared background as their canvases and then paste segmentations of waste objects onto the fresh backgrounds. They additionally add opaque greyscaled stock images as an overlay on top of the new synthetic images as an extra augmentation step. Copy-Paste augmentation examples can be found in Appendix B.5, while Figure 4.4.3 illustrates the Copy-Paste augmentation pipeline.

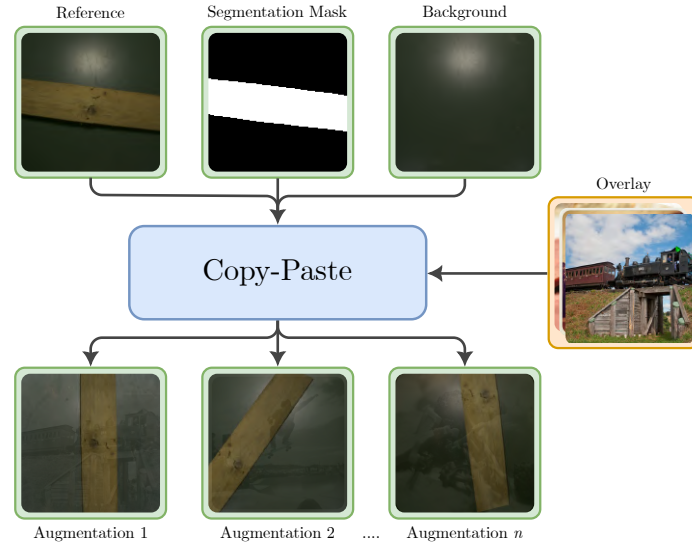


Figure 4.4.3: Copy-Paste data augmentation illustration.

Currently, ARIS Robotics uses the Copy-Paste method as their primary data augmentation method. They empirically observe that models trained with Copy-Paste are much more invariant to slight changes in each machine's background, as the background can differ in lighting conditions, wear, or include small debris. However, because the segmentation and background are sampled at two different instances, the lighting conditions can differ substantially between object and background. Additionally, the abrupt transitions between the segmented object and the background degrade the realism of the augmented images. It is therefore anticipated that the state-of-the-art diffusion models can mitigate this loss in realism.

Mixup, CutMix, and Copy-Paste conclude this section, which has given an overview of both modern and select classic data augmentation methods. The next section will delve into how these methods can be used in waste-sorting applications.

5 Methodology

With the principles of diffusion models covered, the following section delves deeper into the waste-classification use-case. Firstly, the ARIS waste classification dataset will be showcased, along with an explanation of the challenges in applying DiffDA to this dataset. Afterward, Section 5.2 covers the implementation details of a range of DiffDA methods selected for comparison. The methodology for training and generating datasets will be covered, as well as the evaluation methodology and the experimental setup.

5.1 Dataset

The main dataset used in this project is the ARIS waste classification dataset. The ARIS waste classification dataset consists of labelled images across more than 30 waste classes collected through their waste classification machine, internally known as the Waste Detective (WADE). Figure 5.1.1 shows a sketch of an image-capturing machine. These stations are placed around recycling centres in Denmark where the public can use them. By placing a piece of waste in the middle of the machine and clicking a button, the machine takes an image, performs inference with the built-in classification ResNet, and tells the user which type of waste the object is.

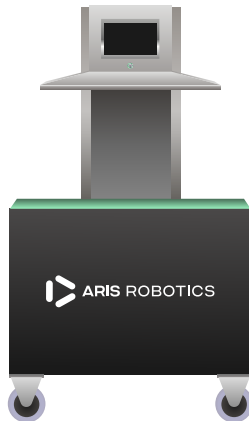


Figure 5.1.1: Sketch of the Waste Detective

A majority of waste classes in the dataset contain fewer than 100 images. If DiffDA proves significant, downstream performance gains can be achieved; it could be applied to these classes with low image counts. However, for development and experimental robustness, the classes with too few examples were considered insufficient in quantity for creating an exhaustive train/validation/test dataset. Therefore, only the four most populous classes will be used. Figure 5.1.2 shows a collection of samples from the four largest classes in the dataset.

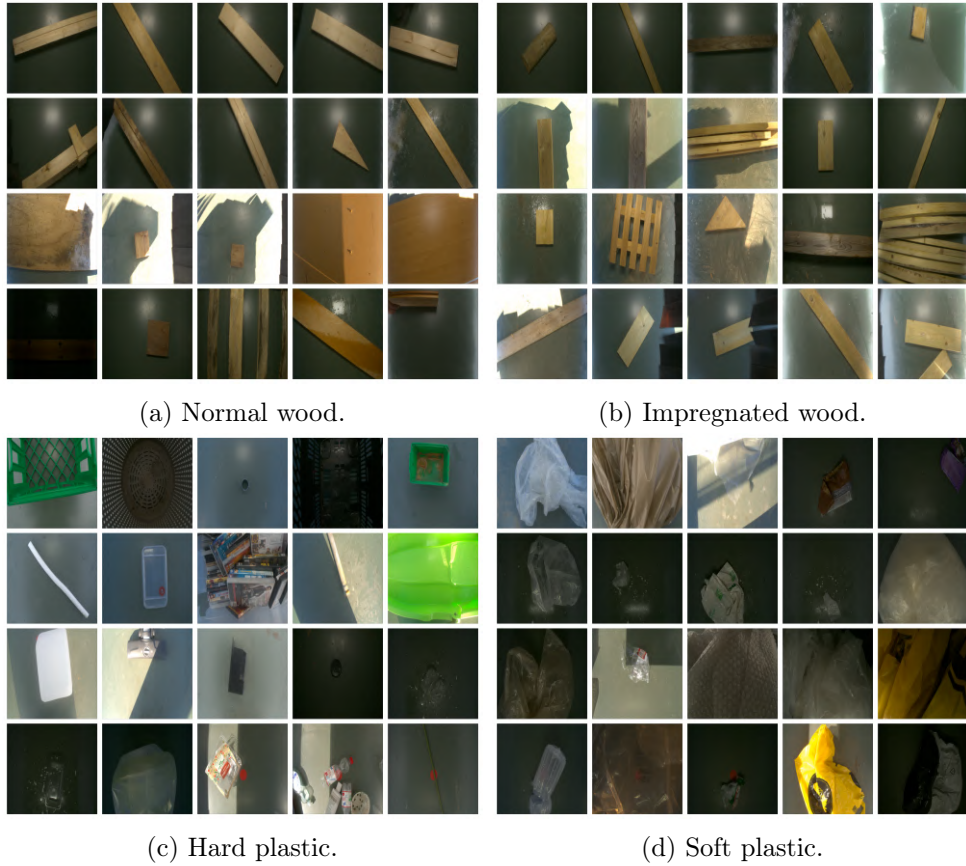


Figure 5.1.2: Primary 4 classes of the ARIS waste classification dataset.

As expected, the two wood classes are comparable, and the two plastic categories have similar features. The dataset has the following characteristics.

- **Image structure:** The ARIS dataset has a predefined structure relative to unconstrained datasets, such as ImageNet and CIFAR100 [42][43]. All images feature a green-hued surface with varying degrees of scratches, damage, and occasional debris. On the surface, a single piece of waste is visible from a top-down perspective. All objects are seen from the same perspective. Due to the constrained image structure, waste classification

should explicitly classify based on the minute details and structure of the waste object. This means the ARIS dataset is a fine-grained dataset, compared to a coarse-grained dataset.

- **Different lighting:** All images are captured following the same procedure; however, the lighting conditions of the images vary greatly. Since the image-capturing machines are expected to operate in outdoor waste disposal centers, the dataset includes images under harsh lighting conditions. Some samples of waste are shown with a low-hanging sun shining onto the capturing surface, partially blocked by the rubber slats. This causes portions of the images to be overexposed relative to the object of interest. The dataset also has interclass lighting tendencies. Both wood classes have comparable lighting conditions for the majority of images, whereas both plastic classes have a larger collection of less-intensely lit images. This has occurred since multiple groups of classes have been collected on different groups of machines.
- **Different machines:** In general, both samples from both wood classes have been collected on different machines than the plastic classes. This is due to machines being used exclusively to classify between wood types or plastic types. This results in hidden class information in other parts of the images relative to the class-defining objects, potentially leading to biases in diffusion and downstream classification tasks. Some of the machines used to capture images of plastic feature a bright red marker indicating where to place the object. This creates a semi-consistent feature across all plastic classes, influencing not just the classification process but also the generation process.
- **Interclass balance:** Although there are many classes of waste, diffusion-based data augmentation could be useful for the four largest and most general classes, which have been chosen as they provide sufficient images for testing and validation. The chosen classes contain the following number of images: {hard plastic: 5288, soft plastic: 1186, impregnated wood: 1507, normal wood: 3081}. For all subsequent training, classes are limited to the same number of images, with the remainder used for validation/testing; the decided dataset splits will be covered later in Table 5.4.2.
- **Intraclass object variance:** As the goal of diffusion-based data augmentation is to generate samples representing the total variance of a given class, it is important to understand the different levels of variation internally per class. Both wood classes have a low variation of object geometry and colour, making classification between these classes difficult. The majority of objects are singular wooden planks with a natural wooden texture. For both classes, there are a few larger objects which

are visually significantly different from the common plank. Both plastic classes have a higher variation of geometry and colours. If a diffusion model experiences *mode collapse*, it generates only the most common class. For a trained diffusion model to avoid mode collapse, it must be able to generate images that are representative of the whole distribution of the original dataset.

- **Reflection:** In most of the images of the dataset, a clear reflection can be seen just north of the centre of the image as a reflection of the blitz used to light up the object before capturing the image. This light source is the predominant source of light in all the pictures and therefore all generated shadows should be cohesive with this lighting.

Due to the method used to collect images for the ARIS dataset, there is a potential underlying bias in the images that may influence the training of the diffusion and classification models. The images collected by ARIS are captured by a fleet of machines across different recycling centers. As the number of customers grows, so does the number of machines. This naturally means that early machines have captured significantly more data than others. As shown in Table 5.1.1, which presents the distribution of images across machines, a few machines have captured significantly more images than others. Although precautions have been taken to keep lighting conditions consistent across machines, there is still a noticeable difference in lighting. The reason why different lighting conditions could be a bias factor is due to the imbalance of images across classes. Both normal wood and hard plastic have significantly more images than impregnated wood and soft plastic. If there are lighting differences across machines and some machines are correlated with more images of a given class, logically, classification models might be informed by the lighting conditions of the images rather than the object.

This overview of the data, the different classes, and the machine distribution from which they are sampled covers the context around what is required for ARIS with regard to image generation algorithms.

Table 5.1.1: Dataset distribution across machines and classes. Where each machine name is an adjective followed by an animal. * denotes the machines contributing disproportionately to the dataset size, which also contain 75% of all segmentation masks.

Machine	Normal Wood	Impregnated Wood	Hard Plastic	Soft Plastic
blushing-lion	0	0	288	50
bold-eagle*	1827	345	0	0
brave-panther	0	0	1502	30
cheerful-swan	320	46	0	0
curious-fox	167	10	0	0
daring-leopard	587	143	0	0
energetic-kangaroo	330	82	0	0
fierce-wolf*	1061	514	0	0
gallant-stag	92	126	0	0
gleaming-tiger	231	38	0	0
jolly-giraffe	328	108	0	0
majestic-horse	208	43	0	0
mysterious-owl	100	10	0	0
playful-monkey	52	97	0	0
radiant-bear	133	98	0	0
smiling-donkey	66	22	60	0
smoldering-whale*	0	0	3986	1230
Total	5502	1682	5836	1310

5.2 DiffDA model implementation

With the necessary background on DiffDA models provided, the following section details the implementation and hyperparameters of a range of DiffDA methods used to assess whether DiffDA yields significant downstream performance improvements on the ARIS dataset. The DiffDA models are categorised into two groups: base models and foundation models. Base diffusion models refer to models without a pre-trained backbone. Foundation models are models pre-trained on massive datasets, such as ImageNet [44]. Each presented diffusion model will showcase its pipeline and a preview of generated images; a qualitative analysis will be performed in the evaluation, Section 6.3. Table 5.2.1 summarizes the model architectures implemented.

Table 5.2.1: Summary of implemented DiffDA methods and their key technical designs. We use + to denote technical details related to model training and ♦ to denote sample generation at inference time.

Model type	Method	Key Technical Designs
Base	Base Unconditional [16]	+ No pre-training + One model trained per class. ♦ No control at inference time
	Class Conditioned [26]	+ No pre-training ♦ Class label
	Mask Conditioned [26]	+ No pre-training ♦ Mask, class label
	SDEdit [25]	+ No pre-training ♦ SDEdit, class label
Foundation	Real Guidance [8]	+ No fine-tuning ♦ SDEdit, class prompt
	DA-Fusion [5]	+ Textual Inversion ♦ SDEdit, class prompt
	Diff-Aug [9]	+ Textual Inversion + LoRA ♦ SDEdit, class prompt
	Diff-Mix [9]	+ Textual Inversion + LoRA ♦ SDEdit, class prompt ♦ Inter-class image mixup
	Diff-II [45]	+ Textual Inversion + LoRA ♦ Class prompt ♦ DDIM inversion ♦ Latent interpolation

Base Unconditional Diffusion

The base unconditional diffusion is implemented as the lowest-complexity diffusion model [16]. A base unconditional refers to a diffusion model without any conditioning or pre-training. In order for a base unconditional diffusion model to be used on a classification dataset, a separate diffusion model is trained for every class in the dataset. For dataset generation, each of the four unconditional diffusion models generates n images without any guidance or control. In principle, if an unconditional diffusion model is trained correctly, it should randomly generate an image sample from the training data distribution. Key parameters that influence the quality of generated images are the number of layers in the U-Net and the number of layers that use self-attention. An ablation of these parameters will be conducted in Section 6.2. Figure 5.2.1 shows the pipeline for generating images through base unconditional diffusion. Figure 5.2.2 shows a selection of images generated through unconditional diffusion.

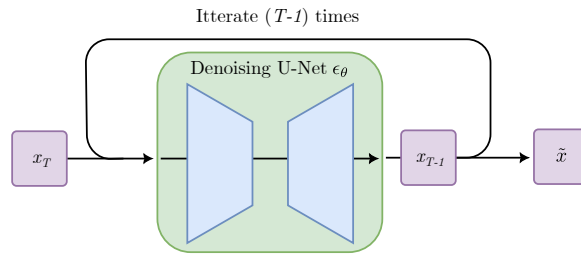


Figure 5.2.1: Base unconditional diffusion pipeline.

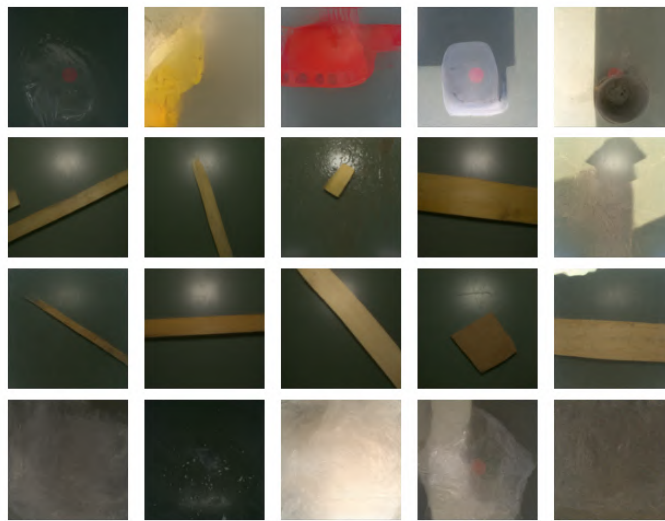


Figure 5.2.2: Base unconditional diffusion generated images. Each row contains images from one of the four classes.

Base Conditional Diffusion

The implemented base conditional diffusion model builds upon the unconditional version by conditioning it with class label embeddings [26]. This allows a single monolithic model to represent the distributions across all classes. This simplifies the training and inference processes, as only one model needs to be trained. Figure 5.2.3 highlights the addition of the class label y to the diffusion pipeline. Figure 5.2.4 shows an array of generated images through conditional diffusion.

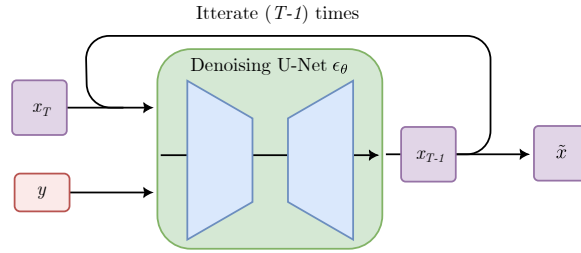


Figure 5.2.3: Base conditional diffusion pipeline.



Figure 5.2.4: Base conditional diffusion generated images.

Masked Conditioning

Masked conditioning is implemented based on the principles of conditional diffusion and the Copy-Paste classic augmentation technique using masks. The masked conditioning model is trained conditionally on both the class label and an object mask. The motivation behind masked conditioning is to counteract images generated as either completely background or exclusively foreground. The pipeline architecture is shown in Figure 5.2.5, and the generated images are shown in Figure 5.2.6.

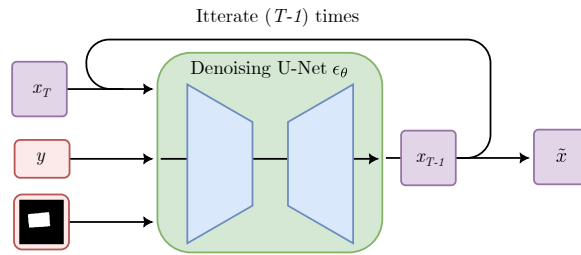


Figure 5.2.5: Base mask-conditioned diffusion pipeline.



Figure 5.2.6: Base mask-conditioned diffusion generated images.

SDEdit

SDEdit is the last of the base diffusion models. A base conditional diffusion model is used with the SDEdit guidance applied at inference time [25]. Figure 5.2.7 shows the added SDEdit method taking a guidance image and, via forward diffusion, creating an informed initial guess of x_T . Figure 5.2.8 shows a section of images generated through SDEdit with an augmentation strength of 0.8. The generated wood images remain close to the reference image with small permutations.

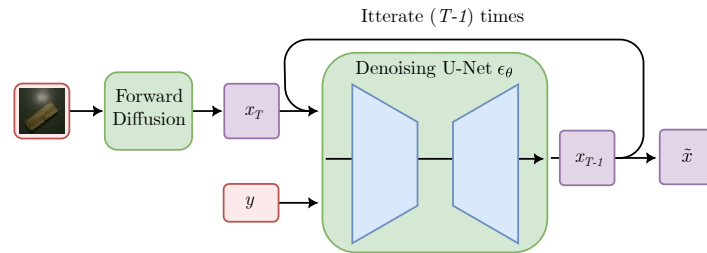


Figure 5.2.7: Base conditional diffusion pipeline with added SDEdit for image guidance.

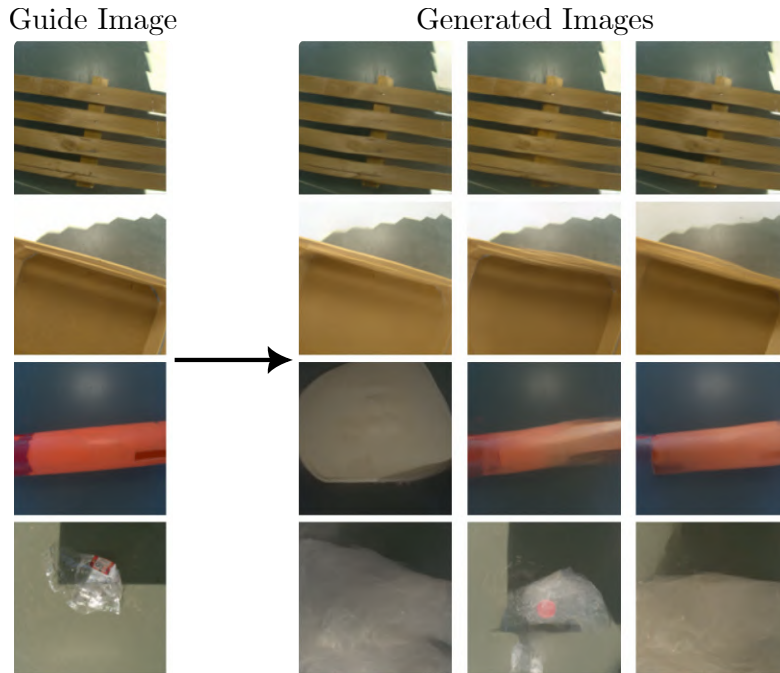


Figure 5.2.8: Images generated through SDEdit guidance. SDEdit augmentation strength: 0.8

Real Guidance

Real Guidance is the simplest of the foundation models, receiving no fine-tuning [8]. It uses the publicly downloadable "stable-diffusion-v1-5" model [46]. As SD1.5 is a text-to-image model, Real Guidance conditions image generation by passing a class label through a text encoder via the text string "a photo of y " where y represents one of the class labels: {"impregnated wood", "normal wood", "hard plastic", "soft plastic"}. Real Guidance generates images based on its default understanding of the words in the class labels. With SDEdit, Real Guidance is also guided via an image input. Real Guidance uses the SD1.5 pipeline architecture as presented in Figure 5.2.9 to generate images as seen in 5.2.10.

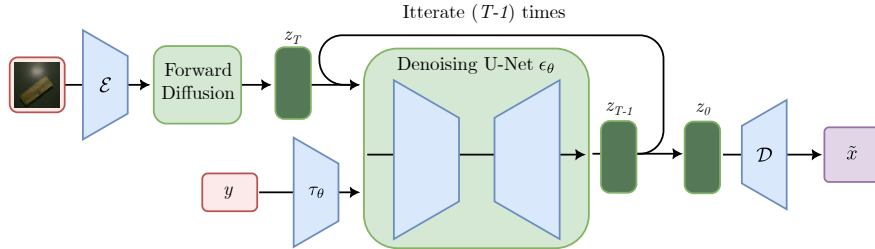


Figure 5.2.9: Real Guidance pipeline for latent image generation with added SDEdit for image guidance.

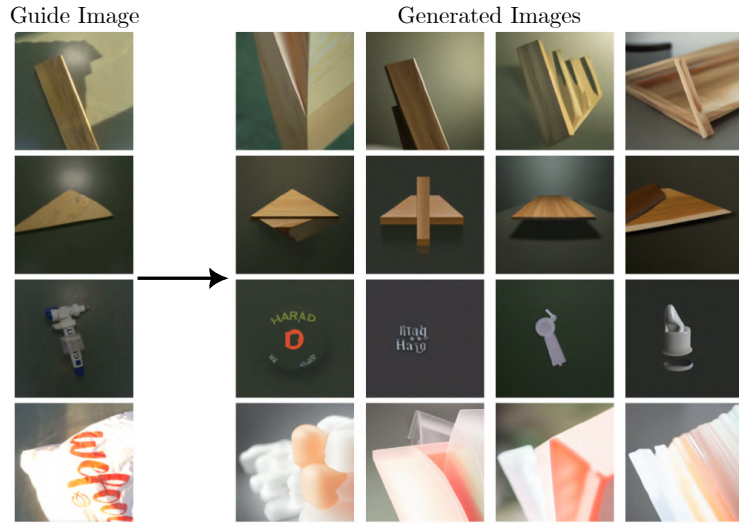


Figure 5.2.10: Real Guidance generated augmentations of real images in the first column. SDEdit augmentation strength: 0.9

DA-Fusion

DA-Fusion builds upon Real Guidance by using textual inversion to train the text embedding model to understand the distinct class labels [5]. Otherwise, it uses the same architecture and hyperparameters. Figure 5.2.11 shows images generated through DA-Fusion with fine-tuned textual inversion.

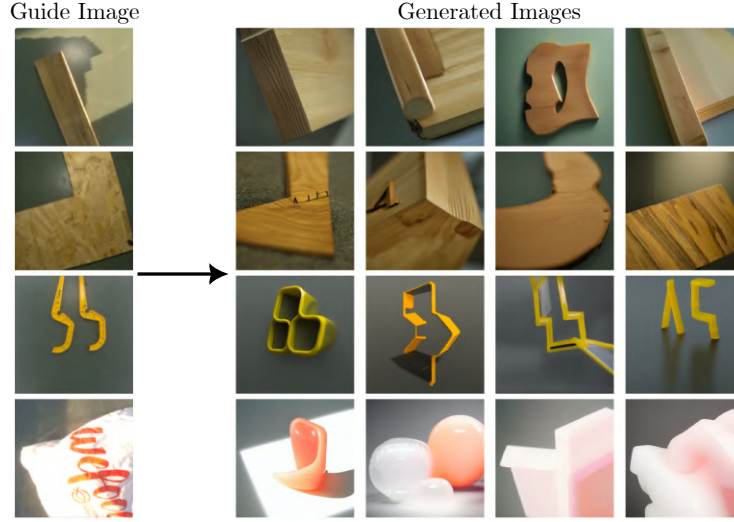


Figure 5.2.11: DA-Fusion generated augmentations of real images in the first column. SDEdit augmentation strength: 0.9

Diff-Aug

Diff-Aug utilises both textual inversion and LoRA to fine-tune the foundation model [9]. The fine-tuned pipeline is shown in Figure 5.2.12. Figure 5.2.13 shows images generated through Diff-Aug.

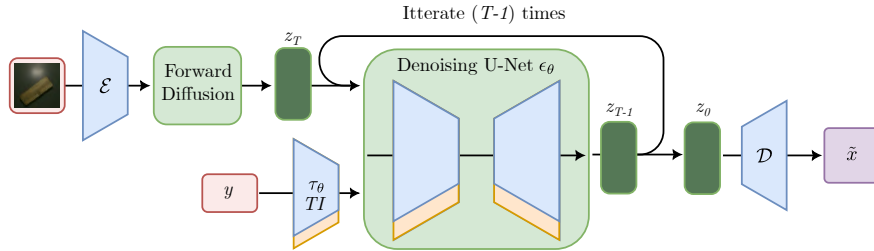


Figure 5.2.12: Diff-Aug pipeline for latent image generation. The text embedding, τ_θ , is fine-tuned via textual inversion. Both the text embedding and U-Net are finetuned via LoRA, marked in orange.

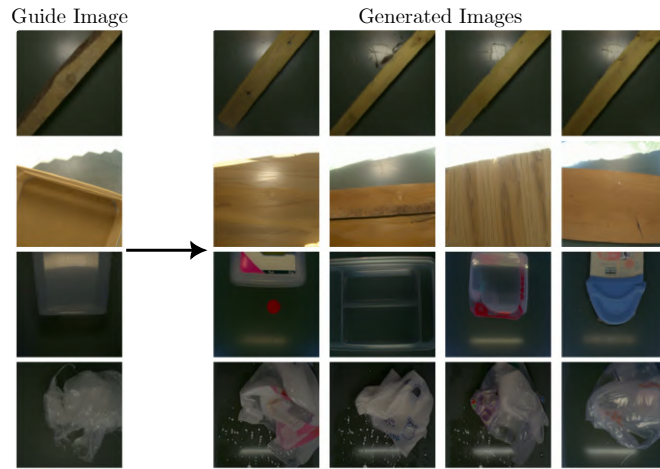


Figure 5.2.13: Diff-Aug generated augmentations of images in first column. SDEdit augmentation strength: 0.9

Diff-Mix

Diff-Mix uses the same architecture and training as Diff-Aug; however, it employs inter-class image mixup to increase image diversity [9]. Images generated with inter-class mixup use images from all classes as guidance, not just from their own class. Figure 5.2.14 shows a sequence of augmented images generated via SDEdit. The image composition and lighting conditions remain similar to the input image, but the object class shifts depending on the class label. This highlights the behaviour Diff-Mix tries to achieve. For any given image from a given class, generate a similar-looking image in terms of composition, while changing the subject to match a different class label.

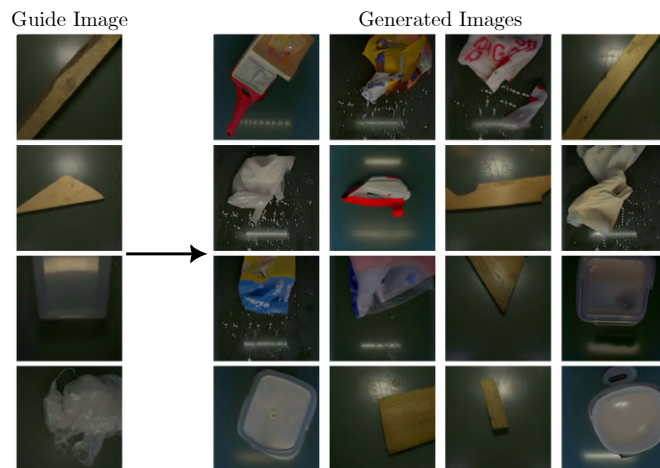


Figure 5.2.14: Diff-Mix generated augmentations of images in the first column. SDEdit augmentation strength: 0.9

Diff-II

Diff-II uses the same fine-tunings as Diff-Mix and Diff-Aug, and therefore the same pipeline shown in Figure 5.2.12; however, it uses a different image guidance method [45]. It hypothesizes that SDEdit is not a good method for fully exploring the distribution of images in a given dataset, as it guides the diffusion process toward an image close to the guiding image. Diff-II suggests that latent interpolation of the guidance images would ensure better diversity and better representation of the distribution of images in a dataset. Latent interpolation uses two guidance images, which use DDIM inversion to forward-diffuse the guidance images into the latent noise space, and linearly interpolates between the two images' latent noise representations. In principle, this should generate an image that is a direct mix of the two guidance images. Figure 5.2.15 shows images generated through Diff-II. Due to the constrained nature of the guidance images, the latent interpolation method does not generate images that are visually cohesive with the guidance images.

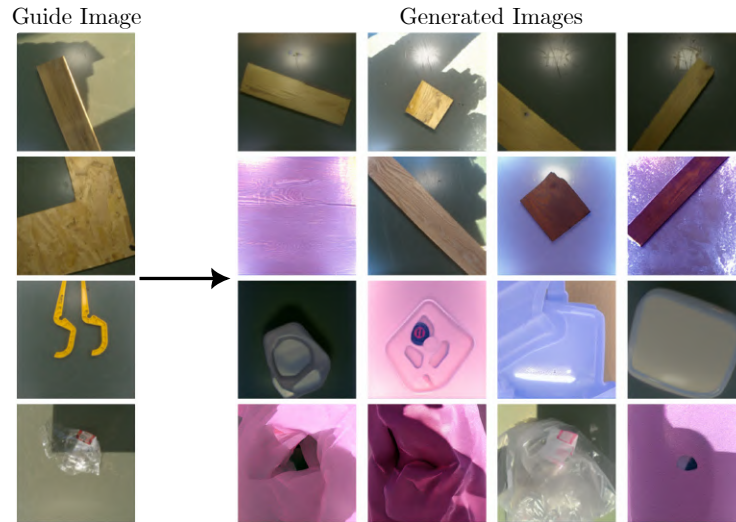


Figure 5.2.15: Diff-II generated augmentations of images in the first column

Larger collages with more images generated by all diffusion models can be found in Appendix B. By observing the shown generated images, some models generate images closer to the original dataset, but they all have visual cues that show they are generated. These artifacts are explored in Section 6.3. With the diffusion architectures from Table 5.2.1 implemented, the choices regarding the necessary hyperparameters for training and generating images using base and foundation models must be addressed.

5.3 Training & Generation parameters

Each diffusion model evaluated in this work has a range of hyperparameters unique to diffusion models. Because manually tuning every hyperparameter for every diffusion model would be too computationally expensive, hyperparameters are informed by prior research [22, 47].

As multiple diffusion models are to be trained on different datasets, an optimal trade-off between training time and downstream model performance must be found to make evaluation efficient. One of the biggest contributors to longer training time is the image resolution. However, because downstream performance is heavily dependent on the image resolution, a study will be conducted to find a balance between good downstream performance and low image resolution (Section 6.1). Another major contributor to training time is the model size; therefore, ablation studies will be conducted on the base U-Net architecture (Section 6.2). Firstly, the effect of model depth on visual quality will be evaluated both qualitatively and using the FID score. After determining the optimal trade-off between training time and model depth, an ablation study will be conducted across different self-attention configurations in the U-Net, yielding another training time-performance trade-off.

Besides the aforementioned studies, key primary hyperparameters have been explored in their influence on the visual quality of synthetic images. Using an iterative approach, the values for the following hyperparameters are chosen: training duration, batch size, U-Net width, learning rate, and learning rate scheduler. The various hyperparameters used for the base diffusion models are listed in Table 5.3.1.

Table 5.3.1: Training and sampling hyperparameters of the base diffusion models.

Parameter	Value
Batch size	16
Training steps	375000
Learning rate scheduler	Constant
Learning rate	0.0001
Sampling algorithm	DPM-Solver++
Denoising iterations	50
Adam β_1	0.95
Adam β_2	0.999
SDEdit strength	0.8
U-Net layer widths	[128, 128, 256, 256]
U-Net parameter count	2.84M

Both a cosine learning rate scheduler and a constant learning rate have been explored during training. When comparing the two qualitatively, no significant visual difference was observed. Therefore, a constant learning rate schedule was chosen, as it allowed the training run to continue without inter-

rupting the scheduler. For the sampling algorithm, the DPM-Solver++ was selected for its efficiency. A batch size of 32 was considered for smoother loss gradients; however, qualitatively, no significant difference was observed between 16 and 32. With a batch size of 32, training would frequently run out of VRAM; therefore, a batch size of 16 has been selected for future diffusion model training. Section 6.2 covers the training of a diffusion model across 10+ days of training to determine any convergence and/or training cut-off point.

The parameters for fine-tuning the foundation models and generating images have not been explored as extensively as the base diffusion model, but instead are informed by Wang et al. [9]. The primary parameters can be seen in Table 5.3.2.

Table 5.3.2: Training and sampling hyperparameters of the foundation diffusion models.

Parameter	Value
Batch size	16
Training steps	2000
Learning rate scheduler	Constant
TI Learning rate	5e-4
LoRA Learning rate	5e-6
Sampling algorithm	DPM-Solver++
Denoising iterations	50
Adam β_1	0.9
Adam β_2	0.999
SDEdit strength	0.9
VAE	True
U-Net layer widths	[320, 640, 1280, 1280]
U-Net parameter count	859M

Key parameters are the layer widths and the total number of parameters of the foundation diffusion model. The foundation SD1.5 model contains ~ 300 times as many parameters as the base diffusion model. Considering the base diffusion model cannot get larger due to VRAM constraints during training, it puts into perspective the immense size of foundation models. Based on the hyperparameters discussed for both base diffusion models and foundation models, the experimental procedure for determining whether DiffDA can improve downstream performance on the ARIS dataset is presented next.

5.4 Experimental Setup

Based on the DiffDA methods covered, the experimental section is described as follows. First, an overview will be given of how the diffusion models will be evaluated by downstream performance; then, the chosen dataset splits will be explained; next, the evaluation metrics will be explained; and finally, deep ensembles will be explored as a more rigorous way of evaluating diffusion models. Figure 5.4.1 gives an overview of the downstream model evaluation of the DiffDA methods across dataset size and augmentation ratio.

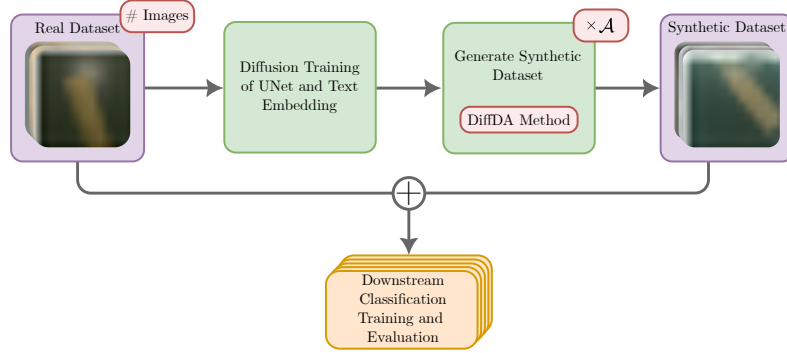


Figure 5.4.1: Downstream performance evaluation pipeline of Diffusion Models. For every combination of DiffDA Method, $\#$ Images, and $\times \mathcal{A}$, downstream classification accuracy is found via deep ensembles.

A set of image groups is selected with varying image counts, and a set of augmentation ratios, ' $\times \mathcal{A}$ ', is chosen. Every DiffDA Method is trained across all groups of $\#$ Images, and images are generated to match the maximum augmentation ratio $\times \mathcal{A}$. After image generation, a deep ensemble of ResNets is trained to estimate the mean downstream performance and variance of the given DiffDA method, applied to a given number of images.

Dataset split

To evaluate the effectiveness of diffusion-based augmentation across different dataset sizes, 7 sizes ranging from 10 to 1000 images per class are evaluated. As each additional dataset requires retraining the diffusion models and downstream evaluation, the computational expense becomes high; therefore, only 7 dataset sizes will be explored. The 7 dataset sizes are centered on smaller sizes, as shown in Table 5.4.2.

(a) 10 images per class				(b) 25 images per class			
Category	Train	Val	Test	Category	Train	Val	Test
Normal Wood	10	2746	2746	Normal Wood	25	2739	2738
Impregnated Wood	10	836	836	Impregnated Wood	25	829	828
Hard Plastic	10	2913	2913	Hard Plastic	25	2905	2906
Soft Plastic	10	650	650	Soft Plastic	25	642	643

(c) 50 images per class				(d) 100 images per class			
Category	Train	Val	Test	Category	Train	Val	Test
Normal Wood	50	2726	2726	Normal Wood	100	2701	2701
Impregnated Wood	50	816	816	Impregnated Wood	100	791	791
Hard Plastic	50	2893	2893	Hard Plastic	100	2868	2868
Soft Plastic	50	630	630	Soft Plastic	100	605	605

(e) 250 images per class				(f) 500 images per class			
Category	Train	Val	Test	Category	Train	Val	Test
Normal Wood	250	2626	2626	Normal Wood	500	2501	2501
Impregnated Wood	250	716	716	Impregnated Wood	500	591	591
Hard Plastic	250	2793	2793	Hard Plastic	500	2668	2668
Soft Plastic	250	530	530	Soft Plastic	500	405	405

(g) 1000 images per class			
Category	Train	Val	Test
Normal Wood	1000	2251	2251
Impregnated Wood	1000	341	341
Hard Plastic	1000	2418	2418
Soft Plastic	1000	155	155

Figure 5.4.2: Dataset split distribution for each class across different numbers of training images per class.

As previously mentioned, the dataset is unbalanced; this effect is greatest in the *1000 images per class* dataset (Table 5.4.2g), where the relative sizes of the minority classes are smallest. For this reason, the macro accuracy metric is chosen for evaluation. As the Copy-Paste method requires segmentation masks for data augmentation, the training data is structured to also contain segmentation masks. However, due to segmentation masks only being available from a subset of the dataset, with 75% coming from three of the early machines: fierce-wolf, bold-eagle, and smoldering-whale, the training dataset consists mostly of images from these machines. This bias could influence the strength of the conclusions drawn in this report. Therefore, a brief analysis of biases will be conducted in the evaluation, Section 6.1. As the synthetic-to-real image ratio that yields the greatest downstream performance improvement for this use case is unknown, four augmentation ratios, $\times\mathcal{A}$, will be explored: $1\mathcal{A}$, $2\mathcal{A}$, $5\mathcal{A}$, and $10\mathcal{A}$. An augmentation ratio of $5\mathcal{A}$ means every training

image is augmented five times. It is also hypothesized that the preferred augmentation ratio for each method can vary between dataset sizes; therefore, a single optimal augmentation ratio might not exist for each method. For this reason, every combination of dataset size, augmentation ratio, and DiffDA method will be explored. For each combination mentioned, a deep ensemble of five downstream ResNet models will be trained and evaluated on the test dataset, with a reported macro accuracy. Five models strike a balance between low computational effort and sufficient coverage of model variances[48].

Evaluation Metrics

To quantify the data augmentation performance of the diffusion models, the aforementioned FID score and downstream model performance will be used. As the FID is sensitive to the sample size [49], each reported FID score will be evaluated on 10000 images against the reference training dataset. Since it is a unitless, domain-dependent metric, it can only be compared to other works that use the same reference dataset. So to provide a point of reference for this work, the synthetic dataset generated by the Copy-Paste method achieves an FID score of 31.70 when compared with the training dataset. Because the Copy-Paste method uses realistic segments directly copied from the training dataset, it has an inherent advantage. Although the FID score remains a standard within diffusion model evaluation, it is more oriented towards quantifying the perceptual realism of a wide variety of images. Since images from the ARIS dataset have a very constrained image structure, it is suspected that a good FID metric does not necessarily translate to good downstream performance, and vice versa. As this work is more interested in the final downstream performance, metrics that reflect that performance will be much more representative. The FID score will therefore only be used sparingly as a proxy for downstream performance to accelerate the development of the U-Net architecture and its hyperparameters.

To evaluate downstream performance, a standard pretrained ResNet18 will be used, as it is a robust architecture commonly used in ARIS. Although other state-of-the-art models outperform ResNet18, absolute downstream accuracy is less important than relative performance compared to other DiffDA models. The original ResNet18 architecture will remain unchanged, except for the output layer, which will be reconfigured to match the dataset’s four classes. The hyperparameters used to configure the training process will be kept constant across all tests to minimize the network’s influence on performance across different augmentation methods and dataset sizes. The hyperparameters seen in Table 5.4.1 have been chosen empirically to balance high performance with stable convergence.

Table 5.4.1: Downstream ResNet18 hyperparameters.

Parameter	Value
Number of epochs	100
Learning rate	0.0001
Batch size	16
Loss	Cross entropy
Optimizer	Adam
Weight decay	0.0001
Learning rate scheduler	Cosine annealing
Warmup epochs	3
$T_{\max}$	97
$\eta_{\min}$	10^{-8}

During the training loop, the model is evaluated on the validation dataset at the end of every epoch. Whenever a gain in validation accuracy is observed, a model checkpoint is saved. The model is evaluated on the test set, where the accuracy, macro accuracy, F1-score, and macro F1-score are logged. Since the validation and test datasets are class-imbalanced (Table 5.4.2g), the macro accuracy (eq. 4.27) provides a more balanced performance metric, as it averages class-wise accuracies. Due to the imbalance, it will be used as the primary evaluation metric for downstream model performance; however, because it treats minority classes the same as majority classes, it can misrepresent the models’ overall performance.

Deep ensembles

In traditional deep learning research the efficacy of a model is generally analysed via a mean accuracy and variance across several trainings, also known as a deep ensemble [48]. By training and testing multiple models on the same architecture and dataset, the inherent variability of local minima in neural network training can be accounted for. The architecture’s mean accuracy and variance can therefore be more robustly estimated. Diffusion model training and validation for dataset augmentation is a three-step process of training a diffusion model, generating an augmented dataset, and training a classifier network on said dataset. A resource-intensive process, which takes multiple GPU-days. The holistic approach to estimating the downstream mean accuracy and variance of the diffusion model is to train a deep ensemble of diffusion models, generate a dataset for each of them, followed by one ResNet training for each generated dataset. To utilize the finite amount of resources efficiently, a preliminary test will be conducted to determine whether there is a significant difference in variance between deep ensembles at the diffusion training step or at the downstream classification step. A Diff-Mix model is trained 5 times, and a dataset is generated from each training. A ResNet18 model is

then trained 5 times on each augmented dataset. Figure 5.4.3 shows the mean ResNet accuracy across the 5 diffusion model trainings. Notably, the variance of each model varies.

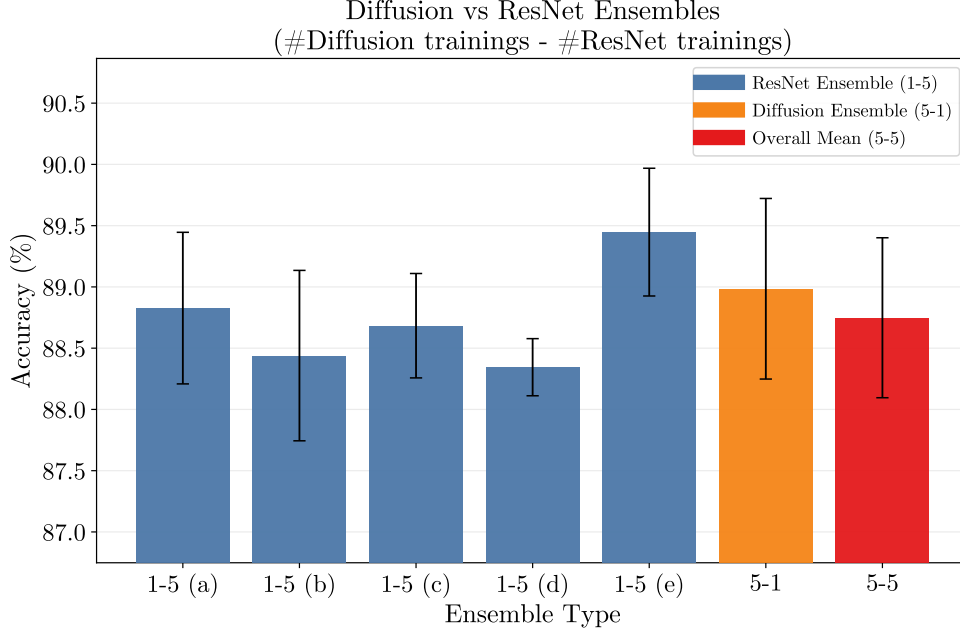


Figure 5.4.3: Deep Ensemble exploration. **1-5** (a)-(e): 1 diffusion model training and generation followed by 5 ResNet18 trainings. **5-1**: 5 diffusion model trainings and generations followed by 1 ResNet18 training per dataset. **5-5**: 5 diffusion model trainings and generations followed by 5 ResNet18 trainings per dataset.

The overall mean is the closest estimate available to the downstream accuracy of a diffusion-augmented dataset. Each (1-5) diffusion training varies in mean accuracy, showing uncertainty in the diffusion training process. As the uncertainty of the diffusion process is of more significance than the ResNets, and the (5-1) deep ensemble is close to the overall mean and variance, deep ensembles at the diffusion training level are considered the most rigorous estimate of the downstream performance of DiffDA methods. Although this method for analysing deep ensembles is not the most statistically rigorous, it allows this thesis to make an informed decision for future training. Due to the high compute cost of training deep ensembles of diffusion models and the relative time constraints, deep ensembles of ResNets will be the main method for determining the mean accuracy and variance of downstream dataset augmentation. For select top-performing diffusion models, where a true representation of performance is essential, deep ensembles will be performed at the diffusion model training step.

6 Evaluation

With the implementation and evaluation methodology covered, the following section will showcase the results obtained in this thesis. Firstly, an analysis of the nuances of the ARIS dataset will be explored, then ablation studies will be conducted to refine the base diffusion model architecture, and finally a cross-model comparison will be performed between the covered DiffDA methods and reference classical DA methods.

6.1 Dataset Analysis

Machine distribution analysis

The datasets chosen for this thesis are based on the assumption that the images captured by any one machine represent the same distribution as the combined distribution across all machines. It is important to confirm this assumption, as the majority of the training data is collected from a few machines. The FID score can provide insight into the dataset similarity between the train, validation, and test datasets. Table 6.1.1 compares the different dataset splits.

Table 6.1.1: FID scores between the 1000 images/class validation, training, and test datasets.

Dataset comparison	Train / Val	Train / Test	Val / Test
FID	30.04	32.02	10.19

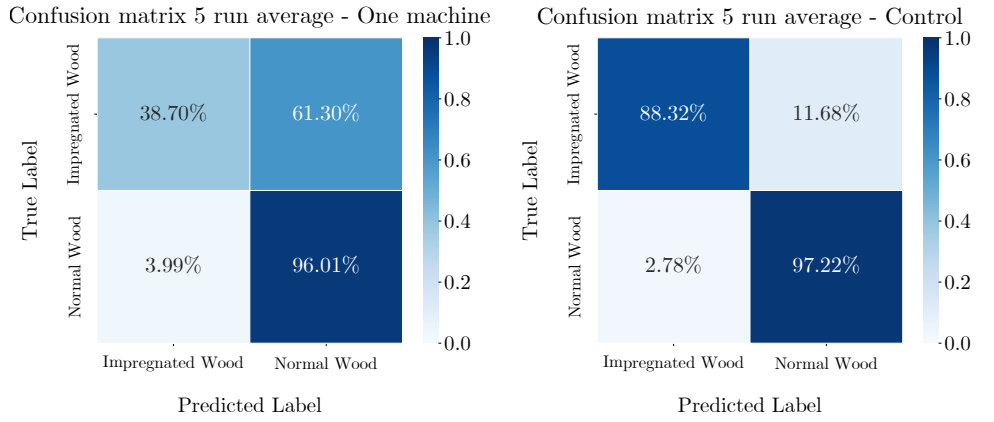
The FID score indicates a distribution shift between the training split and the validation/ test splits. This mismatch leads to the belief that images from different machines are drawn from different distributions. To test whether the images collected by one machine represent the overall image distribution, a ResNet18 will be trained solely on images from that machine, then validated and tested on images from the remaining distribution. To compare performance, a control model is also trained on the same number of images per class as the selected machines, and then validated and tested on the remaining images.

From Table 5.1.1, the only machine with images from both the wood and plastic classes is the smoldering-whale machine, but because each class has fewer than 100 images, it is deemed unfit. However, the two remaining majority machines, fierce-wolf and bold-eagle, do offer sufficient training data to train a binary wood model. A dataset preview of each of the machines plus a control dataset can be found in Appendix C. Two machines are selected because the fierce-wolf is suspected of misrepresenting the dataset. During the early development of the Waste Detective, the fierce-wolf machine was used to collect the initial data for ARIS’ wood classification model. The first data was collected indoors in a hall, in contrast to the machines at recycling centers today, which are subject to different lighting and weather conditions, potentially causing a distribution shift from the sterile environment of the fierce-wolf machine. Table 6.1.2 shows the results of training a ResNet18 purely on the data produced by the fierce-wolf and bold-eagle machines.

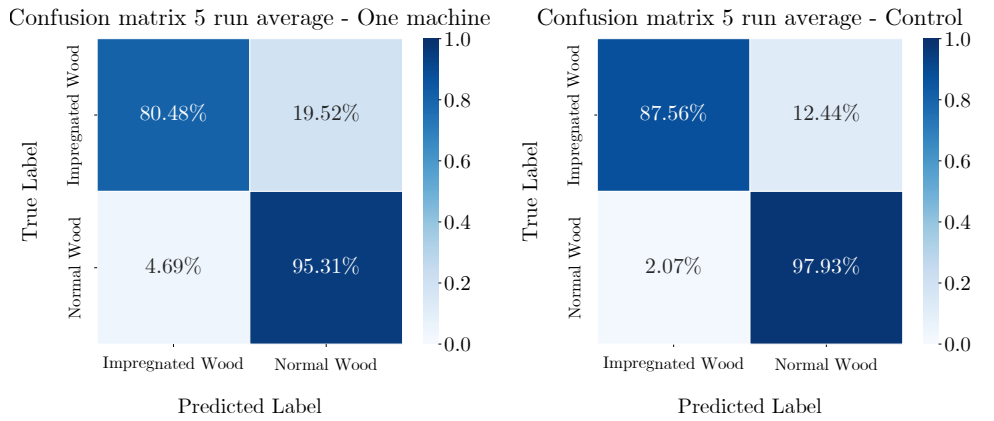
Table 6.1.2: Test performance comparison of the fierce-wolf and bold-eagle machines and their respective control datasets. Additionally, the FID is computed between the training and test datasets within the same evaluation.

Method	Macro Accuracy $\uparrow$	Accuracy $\uparrow$	FID $\downarrow$
Control $\div$ fierce-wolf	92.77% $\pm$ 1.02%	95.37% $\pm$ 0.13%	17.11
fierce-wolf	67.35% $\pm$ 2.41%	84.08% $\pm$ 1.42%	58.62
Control $\div$ bold-eagle	92.75% $\pm$ 0.41%	95.17% $\pm$ 0.30%	14.61
bold-eagle	87.89% $\pm$ 1.84%	91.36% $\pm$ 1.00%	24.43

The table shows a substantial downstream performance difference between the fierce-wolf machine and the control dataset on both macro accuracy and overall accuracy. The FID score supports this conclusion, indicating a large distributional shift between the training and test datasets. When looking at the performance difference for the bald eagle, it is much less pronounced, though a performance gap is still noticeable. The FID score similarly indicates a distribution shift between the training and test datasets, though severely smaller compared to the fierce-wolf dataset. To further investigate where the trained models struggle, confusion matrices are shown in Figure 6.1.1.



(a) fierce-wolf model trained purely on data from the fierce-wolf machine and evaluated on the remaining machines. (b) fierce-wolf control model trained on a random distribution with the same dataset size as available for fierce-wolf.



(c) bold-eagle model trained purely on data from the bold-eagle machine and evaluated on the remaining machines. (d) bold-eagle control model trained on a random distribution with the same dataset size as available for bold-eagle.

Figure 6.1.1: Confusion matrices for the fierce-wolf and bold-eagle evaluations. The left column shows models trained purely on data from one machine, while the right column shows the corresponding control models.

The confusion matrix in Figure 6.1.1a shows that the fierce-wolf model has collapsed to predicting the normal wood for the majority of testing samples, while the control model in Figure 6.1.1b performs much better with very little class confusion. This indicates that the indoor environment of the fierce-wolf causes a distribution shift. In the bold-eagle confusion matrix in Figure 6.1.1c, the model does not exhibit the same class collapse as the model trained on the fierce-wolf dataset. However, the control dataset (Figure 6.1.1d) shows downstream improvements over the bold-eagle dataset, suggesting that the bold-eagle machine does not fully represent the general machine distribution.

This analysis suggests that images captured by any one machine do not represent the same distribution as the combined distribution across all machines. A distribution shift was observed for both machines, though it was considerably worse for the fierce-wolf machine, likely due to the different environment in which the machine has been collecting data. With these observations in mind, they could explain the distribution shift shown by the FID score between the training and test/validation datasets. Therefore, training models solely on indoor or single-machine data is not recommended in the future, if possible.

This observation does give the Copy-Paste method an advantage over the DiffDA methods, as the background images used for the augmentations are sampled from all machines. This means that the bias introduced by the background is minimized by the Copy-Paste method, giving it an advantage over the alternative methods. Although a distribution shift was observed between the training and the validation & test datasets, for the purposes of this thesis, meaningful conclusions can still be drawn, as this work is more interested in relative performance improvements rather than maximum performance. And since the same datasets will be provided across the selected data augmentation methods, their downstream performance can still be compared.

With an analysis of the various machines used for capturing the dataset, the next step becomes to analyse the best trade-off between image resolution, computational requirements and downstream performance.

Performance across image resolutions

Diffusion models are very computationally demanding, both in VRAM requirements and training time. As image sizes grow with n^2 total pixels, so do the computational requirements. A 6-layer U-Net trained to denoise 512px images with a batch size of 16 requires 30 GB of VRAM. As this project is constrained to 24 GB of VRAM, a compromise must be made. Since image resolution significantly affects downstream performance, a test was conducted to assess its extent. 5 ResNets are trained for each of the 14 resizings of the 1000-image/class dataset.

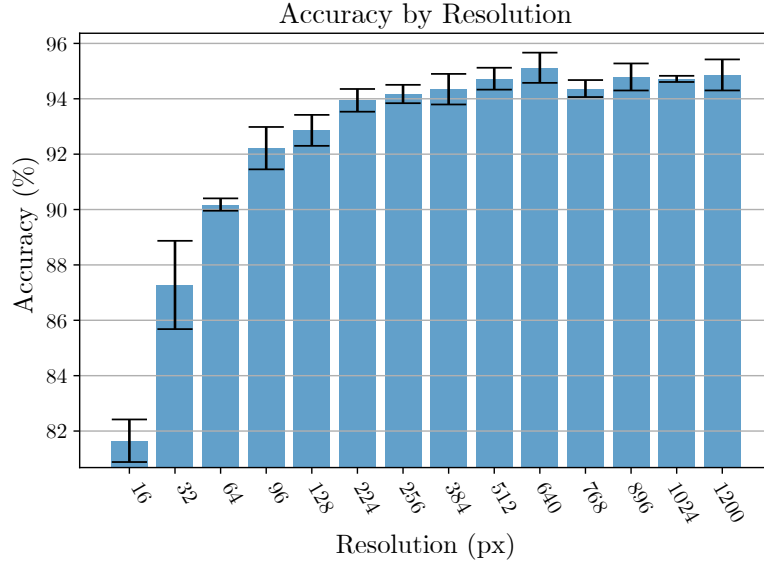


Figure 6.1.2: Overall test accuracy vs image resolution based on 5 ResNet trainings. The x-axis shows the quadratic sidelength for each image resolution. *: The last datapoint, 1200px, depicts the shortest side length of the full resolution dataset at a resolution of 1920×1200.

Figure 6.1.2 shows a clear correlation between accuracy and image resolution, with an elbow point around 224px and 256px. This is not surprising, as these resolutions are commonly used in image classification research. For reference, 224px images contain ~ 3 times more pixels than a 128px image. As this thesis is time- and resource-constrained and focuses more on relative performance rather than maximum task performance, a lower resolution of 128px is chosen to accelerate development.

6.2 Base diffusion model evaluation

This section evaluates the self-trained diffusion model architecture and determines the best trade-off among layer count, self-attention, and training time. Furthermore, it will determine the ideal timing for training termination and, finally, compare performance between a class-conditioned diffusion model and separate unconditional models for each dataset class.

Layer ablation study

Finding the right number of layers for any neural network is crucial to balancing the model’s attention to detail and the risk of overfitting. This section will evaluate a standard U-Net with 1-6 down- and up-convolution layers. It will

be trained as an unconditional model on the impregnated wood class for 1000 epochs, as it finds a good balance between image synthesis quality and training time for unconditional models. After each training session, 10000 images are generated and evaluated qualitatively and quantitatively by comparing them with the training data using the FID metric. Figure 6.2.1 shows a qualitative comparison of the six U-Net architectures.

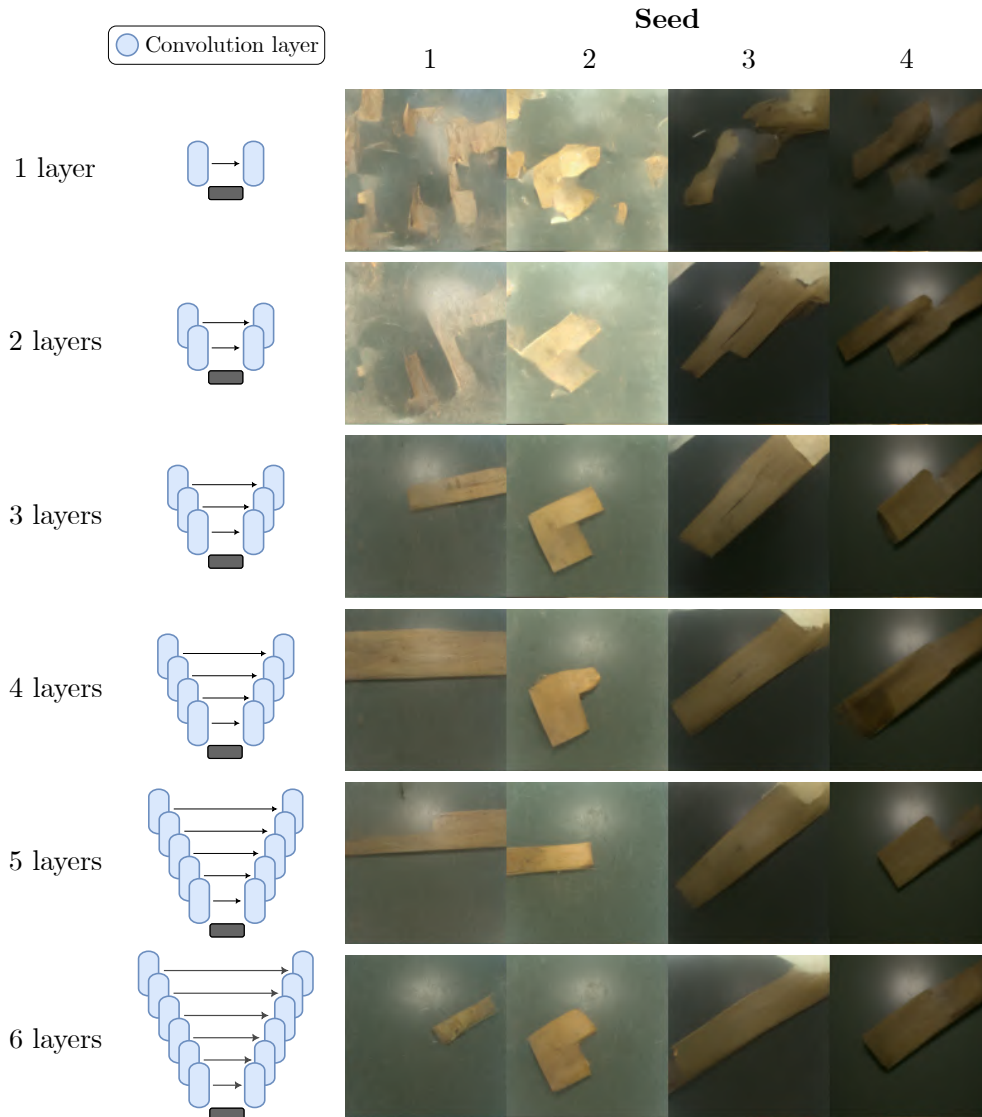


Figure 6.2.1: Four images sampled from the same seed from a U-Net with a varying number of convolution layers. Each architecture is labeled with its layer number and an illustration of the U-Net architecture for that evaluation.

When looking at the U-Net architectures with 1 and 2 layers, the model

has not captured the image distribution and appears to be underfitting. At 3+ layers, a large perceptual improvement is visible, with some samples looking indistinguishable from real training images. However, no large differences in realism are observed among U-Net architectures with 3-6 layers. A similar observation can be made when looking at the FID scores in Table 6.2.1.

Table 6.2.1: Ablation across number of U-Net layers.

Number of layers	FID	Training Time
1	224.84	4 hours 20 min
2	155.51	4 hours 42 min
3	76.80	5 hours 35 min
4	64.68	5 hours 39 min
5	63.72	6 hours 09 min
6	61.90	6 hours 24 min

The FID scores show substantial improvement with each additional layer up to the fourth, with diminishing returns thereafter. As the FID only serves as a proxy for downstream performance, the additional layers offer limited value relative to the additional training time. For that reason, a 4-layer U-Net will be the preferred layer count for base diffusion models.

With a fixed layer count, the next step is to determine which layers should use self-attention.

Self attention layer ablation study

Self-attention is vital for the network to relate different parts of an image together. The trade-offs between self-attention layers and training time should therefore be studied. Figure 6.2.2 shows a qualitative comparison between the U-Net without self-attention and 5 different self-attention configurations in a 4-layer U-Net. The 5 U-Nets are all trained as unconditional models on the impregnated wood dataset.

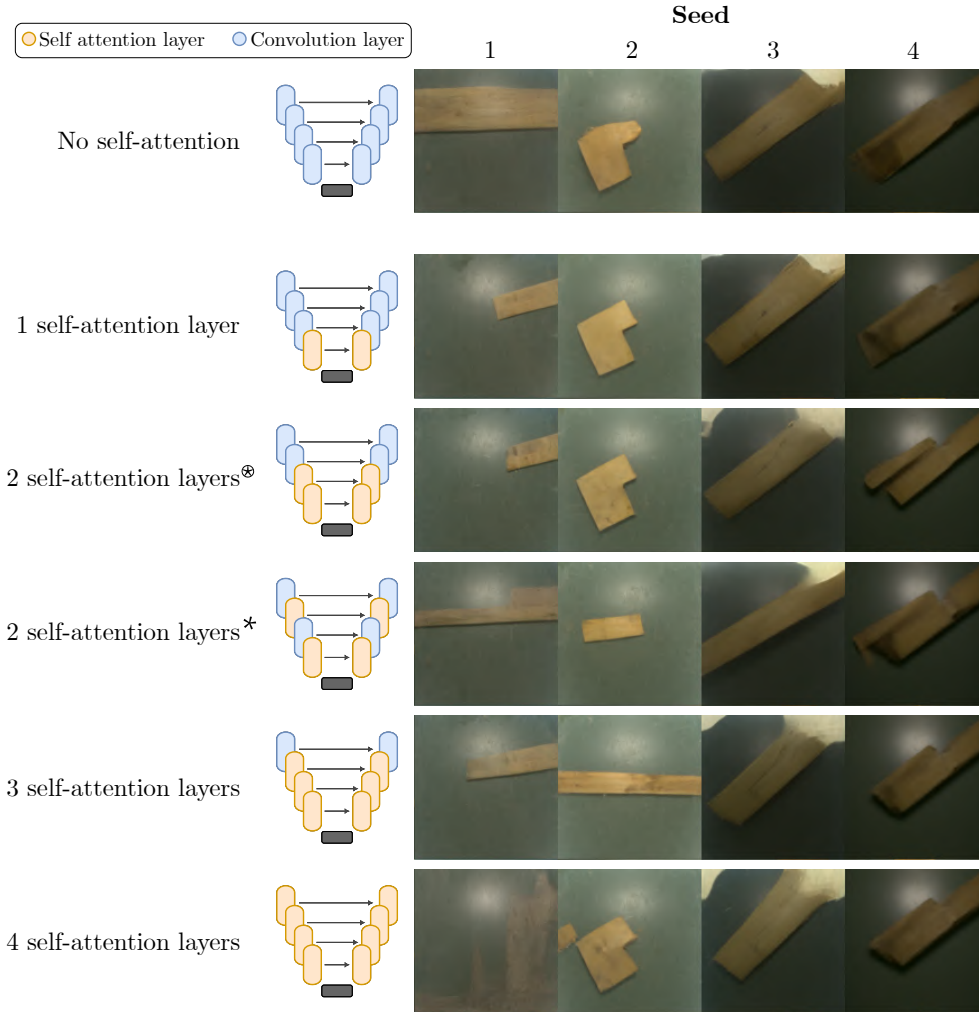


Figure 6.2.2: Sampled images from U-Nets with varying self-attention layers.

When looking at Figure 6.2.2, no overwhelming difference can be seen. Although some self-attention examples show finer details, this is not consistent across all examples. Given the very small sample size, the self-attention layer decision will be based on the FID scores shown in Table 6.2.2. The FID scores have been evaluated on 10000 synthetic images from each of the architectures and compared to the training dataset as a reference to calculate the FID score.

Table 6.2.2: Self-attention ablation FID results, tested against the validation set with a total of 4 layers. $\otimes$: Self-attention is added to the two deepest layers. $*$: Self attention is added to the 2nd and 4th layer. See Figure 6.2.2 for a visualization of the architecture difference.

# Self Attn.	FID	Training Time
0	64.68	5 hours 42 min
1	54.33	5 hours 47 min
2 $\otimes$	56.37	6 hours 42 min
2 $*$	55.02	10 hours 20 min
3	49.86	11 hours 17 min
4	51.08	3 days 4 hours

The FID scores show a much clearer improvement than what was visible qualitatively, with the biggest improvement from the first added self-attention layer. While three self-attention layers yield the lowest FID score, the doubled training time makes the architecture too expensive for this thesis’s time and computational budget. Therefore, the chosen U-Net architecture is a 4-layer U-Net with 1 self-attention layer at the deepest layer. This ablation does not fully explore all possible self-attention configurations, and additional combinations could have been evaluated. These selected combinations provided a broad understanding of self-attention and a brief exploration of different permutations with the same number of layers. The 2 $*$ self-attention permutation gave minimal performance increases over the 2 $\otimes$ self-attention permutation whilst significantly increasing training time.

Training time evaluation

As the convergence of diffusion models is notoriously hard to assess, this evaluation aims to assess downstream performance across an extended training period. This evaluation has two purposes: first, to find the optimal trade-off between training time and downstream task performance, and secondly, to investigate any signs of model convergence or emergent behaviour after extended training. A base conditional diffusion model is trained with 1000 images/class across 2375k training steps. Figure 6.2.3 shows sample images generated with the same seed across training steps. Images generated before the model has converged experience intense random colour filters. Furthermore, the generated waste pieces are blurry and lack visual cohesion. Around 200k training steps, both the colour, texture, and geometry of the generated images become comparable to images found in the training dataset. No visually significant changes occur between 200k-2000k training steps.

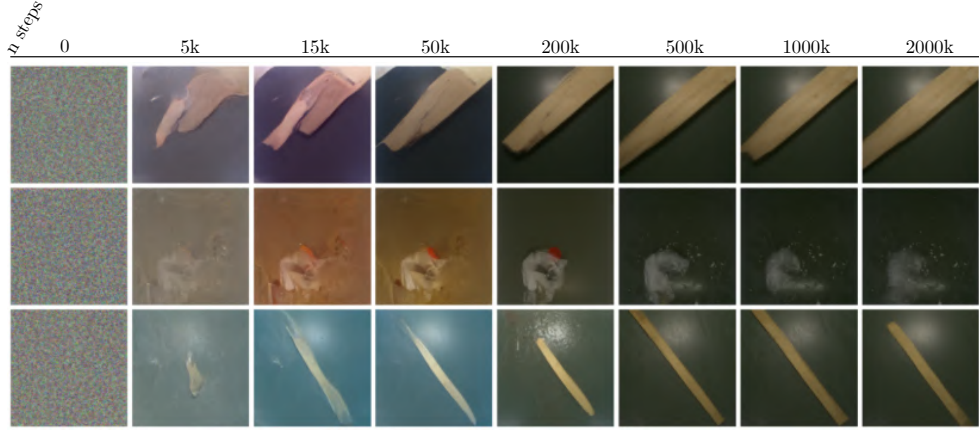


Figure 6.2.3: Base diffusion-generated images across training steps. The x-axis is logarithmic.

At every 125k-step interval, the checkpoint with the lowest validation loss is selected for evaluation. With an augmentation ratio of 2.5, each evaluation checkpoint generates 10000 synthetic images, which are used to train 5 ResNets with both the real training images plus 2500 synthetic images/class. Secondly, the FID score will be evaluated with the training data as a reference.

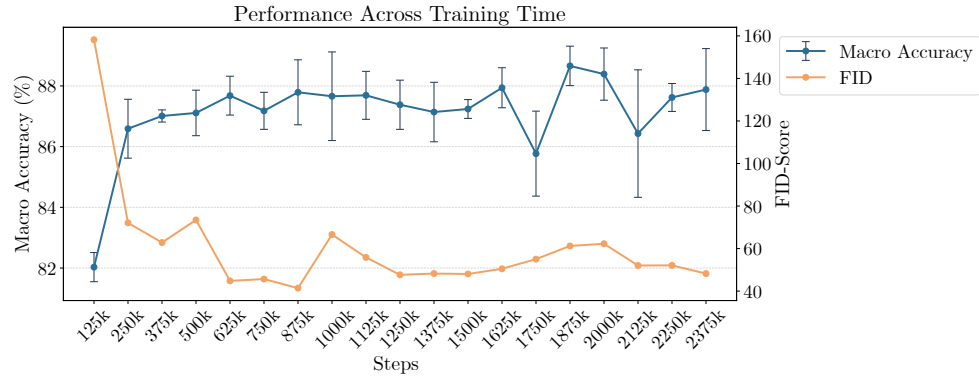


Figure 6.2.4: Class conditioned diffusion model performance across training steps. The left axis shows the downstream test F1 Macro, and the right axis shows the FID score computed in reference to the training dataset.

For the first 625k steps, Figure 6.2.4 shows a positive trend between training time and macro accuracy. However, after 1M steps, the performance fluctuates around a macro accuracy score of 87.5%. Based on these findings, there is a clear elbow point beyond which downstream performance no longer increases significantly. With a margin, roughly 400k steps, or 3000 epochs, is chosen as a sweet spot between good macro accuracy and a limited training time of 42 hours per base model; it will therefore be used as the default training duration

for the remaining base class-conditional diffusion models. The full 2375k steps took 11 days and 2 hours to train on an RTX 3090 Nvidia GPU.

For the FID graph, it converges to ≈ 82 after 1250k. By observing the relationship between FID and macro accuracy, it can be seen that there is no direct correlation between accuracy and FID scores. As the FID converges towards its lowest score, around 1500k steps, there is a drop in accuracy. Furthermore, the best downstream accuracy is achieved at a point where the FID has degraded. This suggests that FID and downstream accuracy are not correlated; therefore, FID cannot serve as a proxy for downstream accuracy. All further tests will therefore forgo FID and use downstream macro accuracy as the primary validation metric. The next section will compare the downstream performance between a class-conditioned diffusion model and multiple unconditioned diffusion models.

Class conditional diffusion vs unconditional

A class-conditioned diffusion model has the same number of parameters as an unconditional model, but one class-conditioned model should be able to generate all classes from a dataset. Unconditional diffusion models would require n trained models for an n -class dataset. Unless there is a significant performance difference between training n unconditional models and one conditional model, the single conditional diffusion model is preferred. Both models are trained on the 1000-image/class dataset (Table 5.4.2g) for 1000 epochs, where the class-conditioned models are trained on all four classes directly, and the unconditional models are trained separately on each class. Results can be seen in Figure 6.2.5.

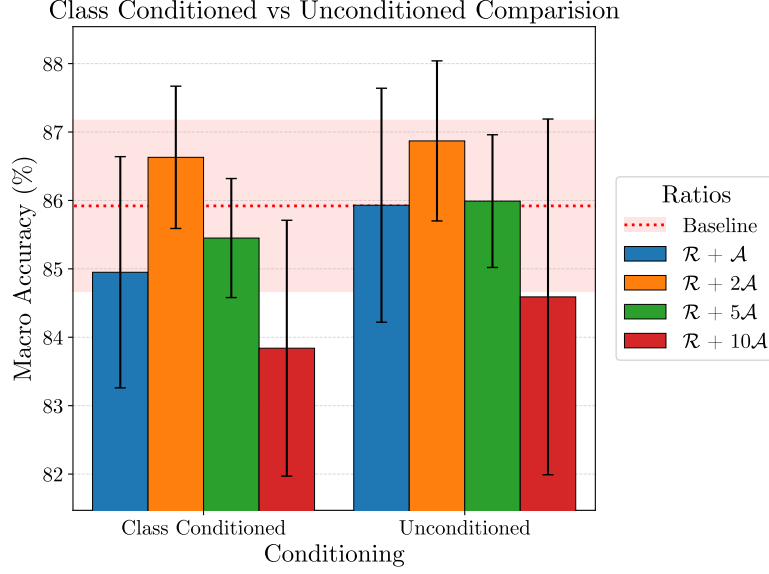


Figure 6.2.5: Class conditioned diffusion model vs unconditioned diffusion model across 4 augmentation ratios

Both models have the same development across augmentation ratios. The unconditional models perform $\sim 0.5\%$ better across all ratios, but due to the uncertainties, this is not considered a significant performance increase. Therefore, the class-conditional model is chosen as it provides a more unified model structure.

Base diffusion conclusion

Based on the ablations covered and the comparison between conditional and unconditional, the base diffusion model selected for comparison with other models is a 4-layer, 1-self-attention model conditional diffusion model trained for 400k steps, or 3000 epochs. This ablation is not fully exhaustive and does not cover other parameters, such as layer width, learning rate, number of inference steps, and more.

6.3 Diffusion image qualitative analysis.

As images generated through diffusion models can be used for much more than just DiffDA, this section looks into individual images to qualitatively evaluate them. For simple diffusion models trained purely on waste images without pretraining, the model will expectably overfit after enough training. When trained on a limited selection of images per class, such as only 10 images per class for the lowest example, the model is expected to converge toward generating similar images to the ones found in the original dataset. Ideally,

the model should learn the features such as object shape and placement in the image but generate new permutations of object texture. Figure 6.3.1 shows a selection of generated images that closely match one of the original images used for training.

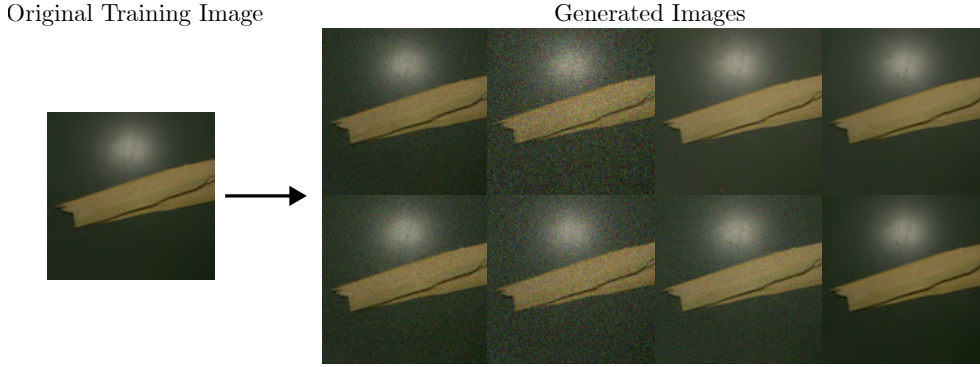


Figure 6.3.1: Synthetic images generated through an overfitted unconditional diffusion model. Trained on 10 images per class.

Overfitting of the diffusion model results in near-one-to-one generation of images from the original dataset. The only significant differences are slight variations in lighting intensity and the amount of noise in the generated images. A subset of generated images do not generate a fully noiseless image. This one-to-one generation of an image from the original dataset is a recurring trend across other base diffusion models trained only on 10 images. This points towards base diffusion models trained from scratch on a small amount of waste classification images overfitting, and therefore not achieving higher downstream performance because the generated images are copies of the original dataset. Quantitative results supporting this will be shown in Section 6.4.

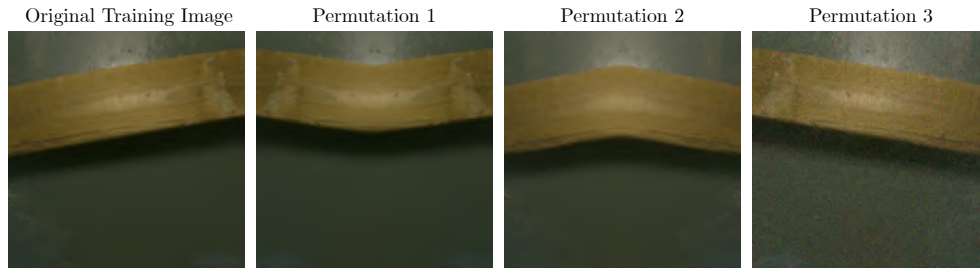


Figure 6.3.2: Unconditional generated images show a mirroring artifact when trained on scarce datasets. Trained on 10 images per class.

Figure 6.3.2 shows an emergent behaviour seen in diffusion models trained on a low amount of training images. As part of the training process for ResNet classifier networks, random horizontal flips have been included to increase the

diversity of the training data. The unconditional diffusion models trained on datasets with 10 images per class have deliberately been overfitted to reduce image noise and to mimic the original dataset images closely. A mirroring artifact occurs due to the horizontal flipping utilised in the training process, causing a portion of the generated images to be mirrored permutations of the original training image, as shown in Figure 6.3.2 permutations 1-2.

The consequence of this artifact is semantically unstructured images that do not accurately reflect the object structure of the original image class, such as crooked planks of wood. Horizontal flipping under the training process was intended to add diversity in object placement and texture. It has, however, produced image artifacts not present in the original dataset. Whether these mirror artifacts lead to downstream performance increases or decreases will not be explored. Besides training artifacts, such as overfitting and mirroring artifacts, other general features can be found across groups of generated images. Figure 6.3.3 shows a range of the most common artifacts.

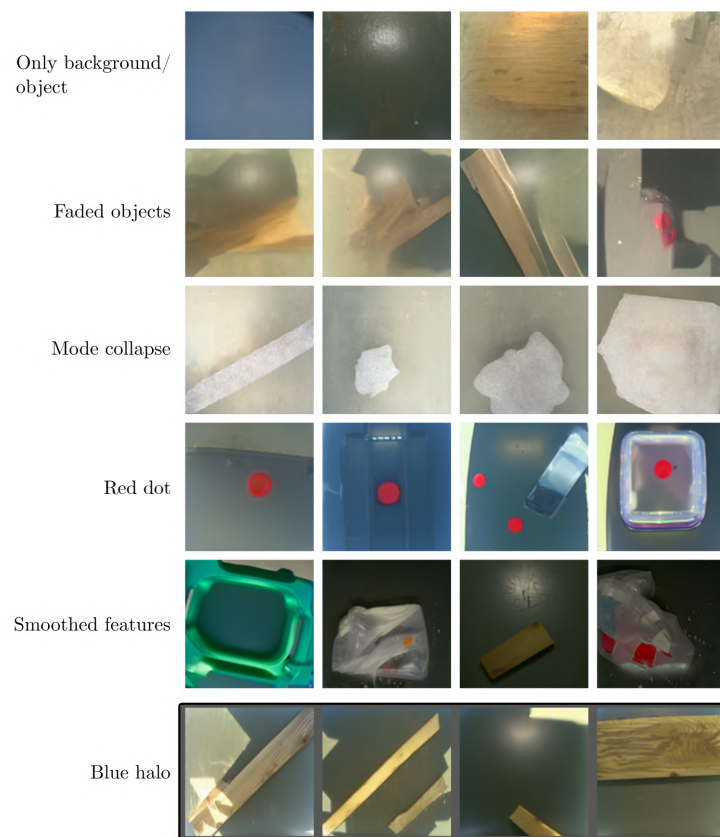


Figure 6.3.3: Commonly occurring types of generated images. The grey box around the blue halo images is intended to assist the reader in observing the blue halo at the perimeter of the images.

Base diffusion models without guidance produce a large number of images that contain the background or the foreground exclusively. Only foreground images can still inform the classification process, however, background images are erroneous as they do not contain any class information. Furthermore, for base diffusion models, faded objects appear. These objects do not have a clear boundary between foreground and background. This phenomenon is less present in guided diffusion models. Most prevalent in the masked diffusion model, mode collapse occurs when the generated object is a featureless, flat, monotonous colour.

The phenomenon labeled red dots, which is most prominent in Diff-Aug and Diff-Mix, is due to the red dots on the machines used to collect images of plastic. This visual cue becomes a dominating visual feature of many plastic images, but as shown in Figure 6.3.3, the placement of the red dot is not consistent and the dot is sometimes interpreted as a plastic bottle lid. Images generated by latent diffusion models generally have a smooth aesthetic. Generated soft plastic images do not capture the high-frequency wrinkles present in the real dataset. There is also a blue halo phenomenon that occurs in wood images generated through Diff-Mix. Because one of the machines used for capturing images of plastic has a blue-tinted background, when Diff-Mix uses inter-class mixup to generate images, artifacts from the blue-tinted background spill into images generated of wood.

The presence of several of these artifacts indicates concrete situations in which DiffDA methods can be further improved. Having an understanding of these artifacts occurring in generated images helps put into context the downstream performance of DiffDA models.

6.4 Cross-model comparison

With the various DiffDA models covered in section 5.2, a cross-model comparison is performed to determine which DiffDA methods perform best and whether any DiffDA methods outperform the baseline and/or classic DA methods. Figure 6.4.1 shows images generated by all implemented DiffDA methods with the same guidance image, when applicable. Full-page collages of images from all DiffDA models are found in Appendix B. Based on the Real Guidance column, images generated without fine-tuning the foundational model are visually different from the real dataset. Images generated by DA-Fusion also perceptually lack the constrained structure of the training images. The Images generated by Diff-Aug and Diff-Mix are visually coherent and comparable to those in the original dataset. Once the foundation model is fine-tuned, the images generated are directly comparable to the guidance image. Diff-II, which attempts to create diverse samples via latent interpolation, suffers from generating blurry pink images. The reason is attributed to the interpo-

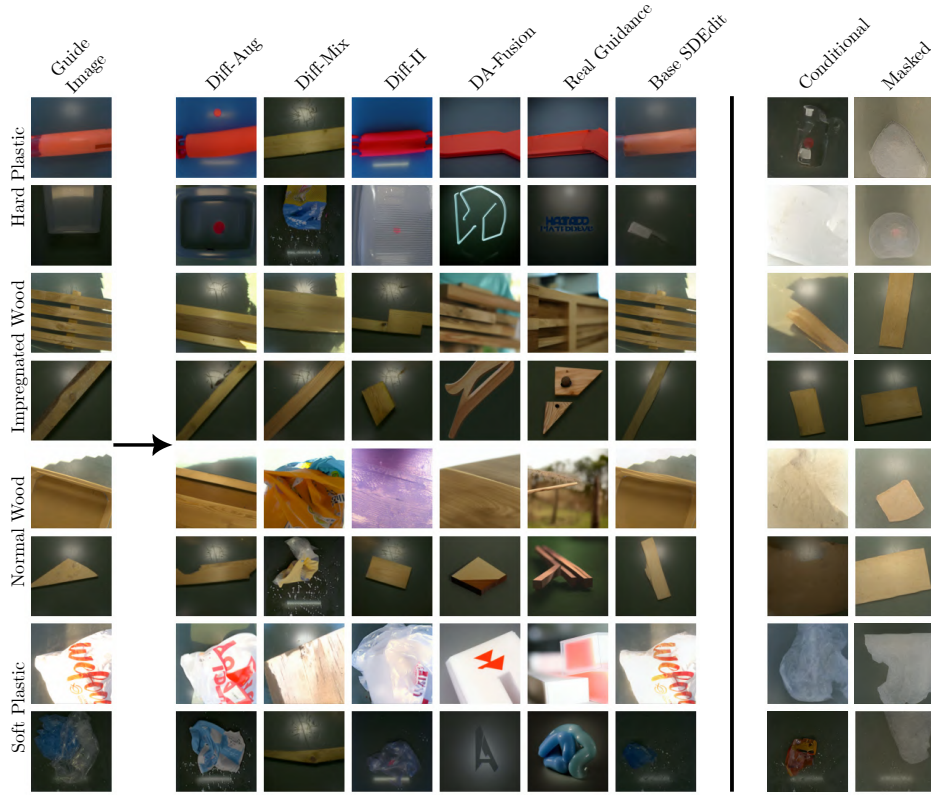
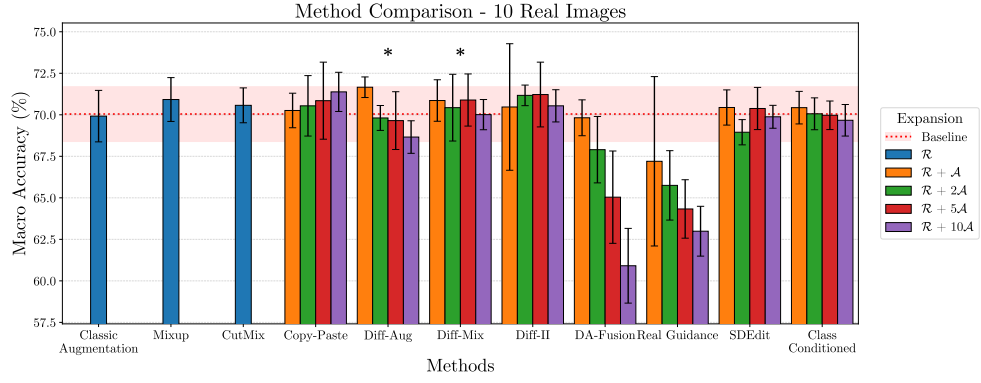


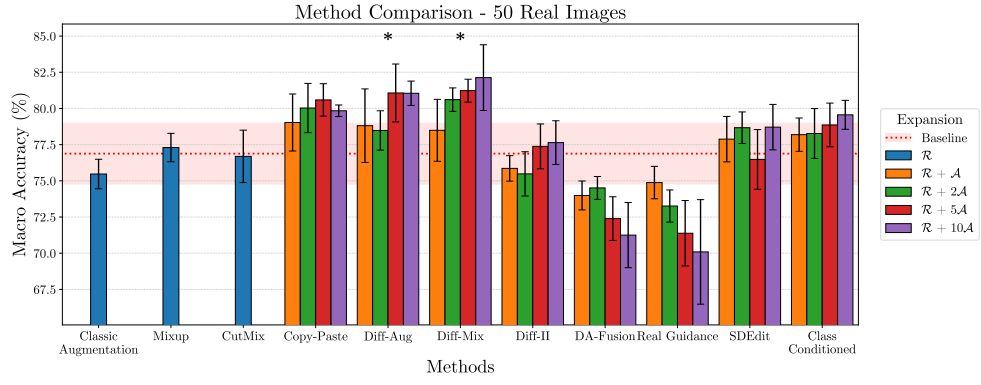
Figure 6.4.1: Cross model comparison of all trained models. Images to the left of the black line are guided by the guide images. Images on the right are without guidance.

lation between the two latent noisy representations of guidance images. The cause of the pink hue is not explored further. All base diffusion models generate images that visually look like images from the dataset distribution. The conditional model generation frequently generates images with objects covering the whole screen. This problem is fixed by the mask-conditioned model; however, the mask-conditioned model experiences mode collapse. The mode collapse is recognised by all the images containing flat, blurry representations of an object. Due to this mode collapse, masked diffusion will not be explored in the quantitative analysis.

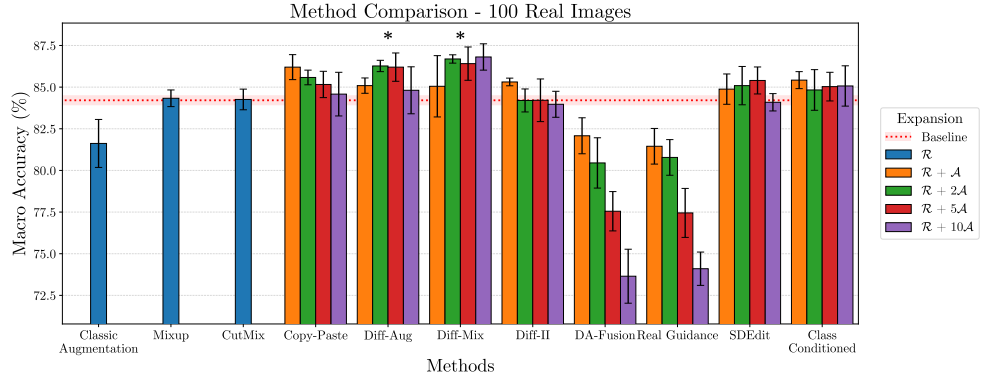
The motivation for performing quantitative model comparisons across different numbers of training images is to make sure any conclusions drawn are not falsely based on a single instance of training data, but across multiple permutations with varying image counts. Figure 6.4.2 shows the cross-model comparison for training datasets with 10, 50, 100, and 1000 real images per class. All cross-model comparison figures, including Figures containing 25, 250, and 500 images per class, can be found in enlarged scale in Appendix D.



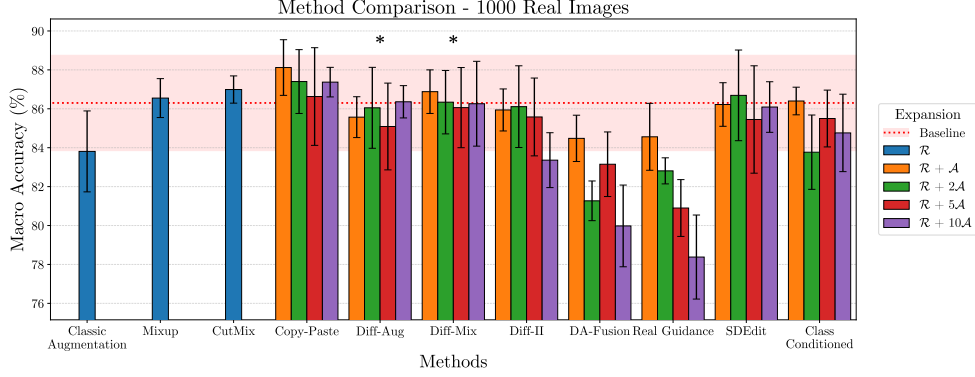
(a) 10 real images per class.



(b) 50 real images per class.



(c) 100 real images per class.



(d) 1000 real images per class.

Figure 6.4.2: Data augmentation comparison for different numbers of real images available per class. $\mathcal{R}$ denotes the real dataset used in the training process and $\mathcal{A}$ denotes an augmented dataset of the same size as $\mathcal{R}$ so that $\mathcal{R} + 5\mathcal{A}$ has 5 augmented images for each real image. Models denoted with * have been trained via deep ensembles of the diffusion model.

General observations across all numbers of training images in Figure 6.4.2 are as follows. DA-Fusion and Real Guidance significantly degrade downstream accuracy, and a higher augmentation ratio results in worse performance. This tendency can be seen across all numbers of training data. Therefore, we conclude that foundation diffusion models without fine-tuning do not lead to performance improvements on the ARIS waste classification dataset.

Classic augmentation, which refers to applying random hue shifting, brightness, colour jitter, and more, performs worse across all numbers of images. This correlates with the machine distribution biases found in Section 6.1

The classical augmentation methods do not lead to performance increases over the baseline dataset. This is unexpected, as the whole purpose of classical augmentation methods is to improve performance.

It is that the classic augmentation method merely shifts the training distribution further away from the validation and test datasets.

When evaluating the performance of CutMix and mixup, no significant improvements are observed in this industrial application. When analysing CutMix samples from the Appendix in Figure B.4, CutMix takes a random snippet from one image and attaches it to another. Due to the constrained image structure and the limited information relevant to the class label, there is a high chance that CutMix uses a pure background snippet to weight an image towards another class, thereby providing minimal training signal for the waste class. Although mixup has been proven to be a valid augmentation method in other research [39], it has not shown significant downstream improvements. The top-performing diffusion models are the fine-tuned foundation models Diff-Aug and Diff-Mix. When looking at Figure 6.4.2a with 10

real images per class, all methods generally struggle to improve downstream models significantly over the baseline.

By extracting the top-performing augmentation ratios from Figure 6.4.2, any general trends in augmentation ratios can be observed. Table 6.4.1 shows which augmentation ratios achieved the highest average downstream accuracy on the validation dataset. This figure does not consider the uncertain estimates of downstream performance, and the augmentation ratio is only analysed at values of $\{1, 2, 5, 10\}$.

Table 6.4.1: Augmentation ratio with the highest average downstream performance on the validation dataset. Augmentation ratio options are $\{1, 2, 5, 10\}$.

Training Images	Copy-Paste	Diff-Aug	Diff-Mix	Diff-II	DA-Fusion	Real Guidance	SDEdit	Class Cond.
10	10	1	1	2	1	1	1	1
25	2	2	2	10	2	1	5	2
50	5	5	10	10	2	1	10	10
100	1	2	2	1	1	1	2	1
250	1	5	1	1	1	1	2	1
500	5	1	2	2	1	1	1	1
1000	1	2	1	2	1	1	2	1

Most models achieve the highest downstream performance increases at low augmentation ratios, except at 50 training images. This general trend across all models suggests there is a maximum augmentation ratio before downstream performance becomes negatively affected. All further augmentation above that decreases downstream performance. As Diff-Aug, Diff-Mix, SDEdit, and class-conditioning show an upward trend in accuracy with higher augmentation ratios in Figure 6.4.2b, the maximum tested augmentation ratio of 10 does not seem to be the limit. Given this dependency, we conclude that there is a maximum augmentation ratio for DiffDA that depends on the number of training images in the dataset. Because this analysis is based on four discrete augmentation ratios, nothing more specific can be concluded.

6.5 Varying real image count ablation study

Based on the augmentation ratios achieving the highest downstream performance, as shown in Table 6.4.1, the Diff-Aug and Diff-Mix models are analysed across the number of training images available. The reason why we choose the DiffDA trainings, which has achieved the highest downstream performance on

the validation dataset, is to analyse the maximum potential of DiffDA. As the training process of diffusion models is unstable, by only looking at the best trainings, determined through the validation dataset, we assume that we have achieved a training close to the global optimum for each model. Figure 6.5.1 shows Diff-Aug and Diff-Mix across a sweep of training image numbers compared to ARIS’ Copy-Paste method and a baseline unaugmented training dataset.

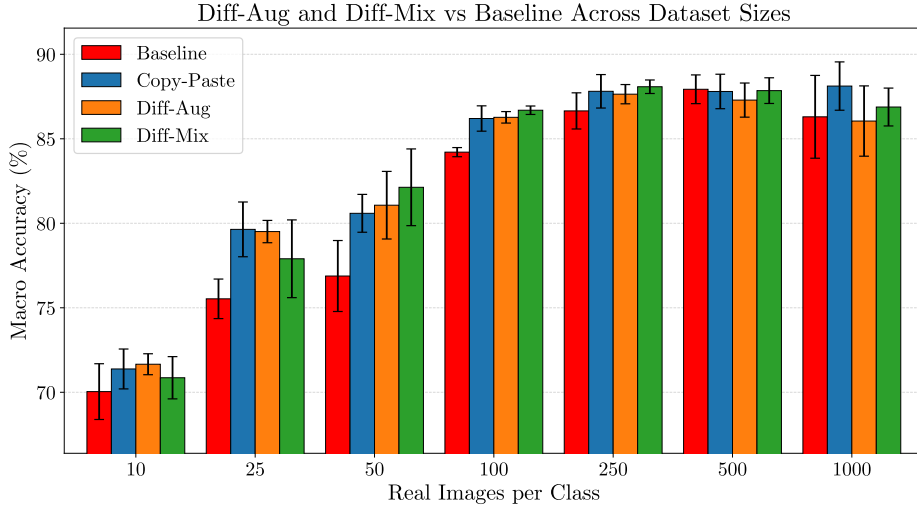


Figure 6.5.1: Diff-Aug and Diff-Mix comparison with baseline and Copy-Paste across dataset sizes, with test macro accuracy reported. For each dataset size, the augmentation ratio with the highest downstream validation accuracy is selected. A graph of the test overall accuracy can be found in Appendix E.2

By comparing Diff-Mix and Diff-Aug to the baseline accuracy, the DiffDA methods achieve a higher downstream performance than the baseline. Due to the high uncertainty estimates and the fact that the mean and variance are estimated from only 5 deep ensemble training runs, this thesis cannot conclude a *statistically significant* improvement over the baseline. With that in mind, both DiffDA methods increase downstream performance as the number of training images per class increases, from 10 to 250. From 500 images and up, DiffDA methods do not lead to downstream improvements. When comparing the DiffDA methods to ARIS’ Copy-Paste method, no significant improvements are observed.

Although fine-tuned foundation models outperform base diffusion models trained from scratch, a similar analysis of base diffusion models can show whether training exclusively on data present in the dataset yields downstream performance gains. Figure 6.5.2 shows base SDEdit and class-conditioned models in comparison to the baseline.

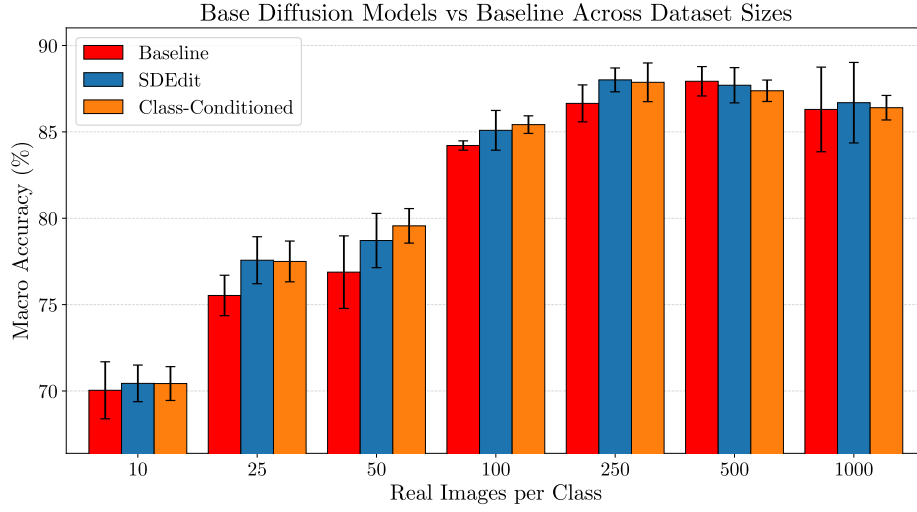


Figure 6.5.2: Base diffusion models comparison with baseline across dataset sizes, with test macro accuracy reported. For each dataset size, the augmentation ratio with the highest downstream validation accuracy is selected. A graph of the test accuracy can be found in Appendix E.4

There is no difference in performance between SDEdit and class-conditioned diffusion; however, both models achieve higher downstream accuracy than the baseline on the dataset between 25 and 250 images. In a similar fashion to Diff-Aug and Diff-Mix, no performance increase occurs after 250 images. Furthermore, no significant improvements are observed at 10 images. This suggests there is a lower limit of samples from the dataset distribution required for any meaningful dataset inflation via base diffusion. Based on the data shown, the following conclusion can be drawn. Base diffusion models trained exclusively on the given dataset achieve downstream performance increases in the range where there is enough data to partially represent the data distribution but not enough to fully represent it; the reported results suggest that the range seems to be between 10 and 250 images per class.

7 Discussion

DiffDA for ARIS waste classification

Diffusion-based data augmentation has shown itself to lead to downstream performance increases for select quantities of images for general visual datasets [7]. The best diffusion model training across different augmentation ratios was used to conclude whether DiffDA methods can outperform the baseline. Our findings suggest that DiffDA leads to downstream improvements for the ARIS dataset up to a maximum of 250 training images per class, beyond which it no longer leads to improved performance. With too few training samples, base diffusion models experience mode collapse, in which the generated images are direct copies of those in the training dataset. For more than 250 images per class, DiffDA methods reach a saturation point beyond which generated images no longer improve performance. If a mathematical expression for the expected performance increase is of interest. Finding a mathematical expression for the direct correlation between the number of dataset images and performance increase would require a different methodology, as we have based the methodology on 7 discrete image counts.

Our findings are based on deep ensembles of ResNets for all diffusion models and deep ensembles of diffusion fine-tunings for Diff-Aug and Diff-Mix, which provide uncertainty estimates for the models' downstream performance. As the uncertainties in the estimated accuracies are large relative to the cross-model differences in accuracy, we cannot rely on statistical significance or p-values to draw conclusions. This is further reinforced by the fact that every accuracy estimate is calculated from only 5 training runs. To determine normal distributions, we would, in principle, need more training runs per model; however, we have determined that development resources would be used more efficiently by refining the training process to reduce uncertainty in the accuracy estimate. The FID metric is found to be a poor proxy for estimating downstream performance. As we have based our base diffusion architecture decisions on the FID score, it would have been more representative to use the downstream task performance instead. Although the idea behind a metric to quantify the performance of a generated dataset is sound, it does not meaningfully capture downstream performance gains in industrial use cases.

Both the base diffusion models and the foundation models only saw minor

improvements across the ARIS dataset. This could be explained by the fact that ARIS’ dataset is a fine-grained dataset where small details and nuances differentiate one class from another. Adding a perspective back to the previous attempts at dataset inflation in Section 4.4, where Li et al. did not experience improvements of DiffDA models on fine-grained datasets, their findings support why significant broad improvements are not found.

Base Diffusion Compared to Pre-Trained Diffusion

We explored the application of diffusion models trained exclusively through the given training dataset and compared them to state-of-the-art models utilising pre-trained SD1.5 backbones. Although there is no significant downstream performance difference between base SDEdit and base conditional diffusion, SDEdit helps reduce the number of generated images that contain only background or only waste. Furthermore, image guidance helped mitigate mode collapse. Diff-Aug and Diff-Mix both outperformed the base diffusion models on downstream tasks. Besides downstream performance, other factors to consider include the technical overhead and ethical concerns associated with pre-trained diffusion models. Converting a class label into a text prompt for a text embedding adds technical overhead, whereas class-label conditioning performs, in principle, the same function. However, removing the text embedding and cross-attention layers does not guarantee that the foundation U-Net retains SD1.5’s current structural awareness. Furthermore, as the foundation model is trained on data scraped from the internet, it may include intellectual property or potentially personal information. The ethical implications of applying this to an industrial use case are up for debate.

From a training standpoint, base diffusion models without a pre-trained backbone take a significantly longer time to train. Since a 4000-image dataset took roughly 2 days to train, any larger dataset, or one containing more unconstrained images, would drastically increase the training time before convergence in visual quality is achieved. The fine-tuning process of a pre-trained backbone takes only ~4 hours to achieve comparable or better results.

Cost of diffusion models

Another consideration when discussing diffusion models and AI is the computational requirements. During the research, implementation, and evaluation of this thesis, over 1.27 M synthetic images have been generated. Just training base diffusion models on a dataset with few classes takes days. Training base diffusion models on datasets with many classes would take weeks. This is one of the biggest drawbacks of base diffusion models, which is why, for datasets with many classes, we recommend DiffDA methods that rely on pre-trained foundation models.

With months of development time spent on training and generating, two high-powered computers have had a high uptime. Assuming 40% uptime for the two workstations, with a power consumption of 500W [50] each over four development months, results in a conservative total power consumption estimate of 1.2 *MWh*. Assuming a *kWh* cost of 2.47kr [51], this comes to an estimated electricity cost of 2845kr. Finally, the total CO² cost from developing and using the diffusion pipeline comes out to 122 *kg* of CO², assuming 0.106 *g/kWh* CO² from 2024 by Energistyrelsen [52]. Given ARIS' mission for sustainability, the wasteful energy consumption of training and developing diffusion models should be considered.

Diffusion vs Classic Augmentation

ARIS' Copy-Paste method was the best-performing classical DA method. Although DiffMix and DiffAug do not yield significant improvements over ARIS' current advantaged Copy-Paste method, they do offer one key advantage. The top-performing DiffDA methods do not rely on segmentation masks on the training data. They only require guidance images and class labels. Therefore, we would recommend that ARIS explore DiffAug or DiffMix further.

Current Pitfalls

The reported evaluations in this work have all been based on one selected dataset split. Since this thesis aims to compare the data augmentation method against the Copy-Paste method currently used at ARIS, the training data had to have segmentation masks. Because segmentation masks are only available for select machines, the training dataset will be forced to have a majority of its data come from a few machines. Our machine distribution evaluation (Figure 6.1.1 and Table 6.1.2) shows that the machine distribution of the training data influenced the classification performance. This effect is increased for larger datasets, such as 250, 500, and 1000 images/class, because a majority of the validation and test data will come from different machines than those on which the model was trained. This influence can be seen in classic augmentation methods, which consistently perform worse than the baseline, indicating that the augmentations only shift the training data further away from the validation and test distributions. Though this distribution shift is problematic, the data still show the strength of the strongest augmentation methods, as they manage to augment the biased training data to better represent the val and test data than the baseline.

Future work

Based on our findings, future work includes evaluating DiffDA methods on balanced and unbiased industrial datasets to better understand their generalisability. This could be done either by expanding and modifying the ARIS dataset or by finding other public few-class classification datasets that represent classes with nuanced differences similar to those in the presented ARIS dataset. Furthermore, with unconstrained time and resources, we would want to strengthen the conclusions drawn in this project by training and evaluating DiffDA on more classes and a wider range of waste categories. For ARIS, it is necessary to estimate whether DiffAug and DiffMix truly outperform Copy-Paste without giving Copy-Paste a machine-specific advantage as mentioned in Section 6.1. Future avenues of research could be parameterising the image generation process to be robust across any machine, making the DiffDA method machine-agnostic, and training the diffusion models at higher image resolution, as this thesis accepted a performance compromise in return for faster development. As ARIS is interested in maximum classification performance, the reported results should be verified at higher image resolutions.

Additionally, a future adaptation of the Copy-Paste method could be to apply the interclass mixing method of CutMix. When looking at the dataset preview of CutMix in Appendix B.4, several augmented images have cutouts, purely of the background pasted onto the second image, which contain no meaningful information about the waste object class. With Copy-Paste as the base augmentation method, CutMix soft targets for the loss function could be computed from the segmentation masks used in Copy-Paste, potentially enhancing its robustness.

8 Conclusion

This project successfully implements and evaluates diffusion models for artificial inflation of the ARIS waste classification dataset. Both base diffusion models and text-to-image models using the foundation SD1.5 model are explored. Base diffusion models trained purely on images from the dataset improve downstream performance beyond that of the baseline training dataset. Using class labels as a text prompt, text-to-image foundation diffusion models, such as Real Guidance with an SD1.5 backbone, do not generate images consistent with those in a visually constrained industrial dataset. By fine-tuning through textual inversion and LoRA, DiffDA methods such as Diff-Aug and Diff-Mix generate images visually similar in structure and content to images in the ARIS dataset. Diff-Aug and Diff-Mix outperform the base conditional diffusion model and the baseline dataset without augmentation. The observed performance increase over the baseline is not a generalised statement. There is an interval of image numbers where diffusion models lead to performance increases. For the ARIS dataset, DiffDA methods showed no further downstream improvements beyond ~ 250 images in our tests. Furthermore, the ratio of augmented images added to the dataset was explored. No clear trends were observed to suggest a universal augmentation ratio. Data suggests the ideal augmentation ratio is dependent on the number of images in the training set. Classical DA methods were used for comparison; however, due to the image structure in the ARIS dataset, they did not improve training beyond the baseline, contrary to expectations. An analysis of the ARIS dataset reveals a hidden bias in the images based on the machines used to capture images. This bias causes the image distribution in the training dataset to differ sufficiently from that of the test and validation datasets, thereby influencing downstream performance. When comparing DiffDA to ARIS' Copy-Paste method, no significant difference was observed. Copy-Paste was, however, noted to have an advantage due to the machine distribution analysis. Because DiffDA methods perform comparably to the advantaged Copy-Paste method and do not require image masks for image augmentation, we recommend that ARIS add Diff-Aug or Diff-Mix to their workflow.

9 Bibliography

- [1] “Goal 12 | ensure sustainable consumption and production patterns,” <https://sdgs.un.org/goals/goal12>, [Accessed 30-05-2026].
- [2] “ARIS Robotics - Rethinking recycling — aris-robotics.com,” <https://www.aris-robotics.com/product>, [Accessed 30-05-2026].
- [3] A. Mumuni and F. Mumuni, “Data augmentation: A comprehensive survey of modern approaches,” *Array*, vol. 16, p. 100258, 2022. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2590005622000911>
- [4] Z. Li, Y. Shi, Y. Gao, and D. Xu, “Diffusion-based data augmentation for image recognition: A systematic analysis and evaluation,” *International Journal of Computer Vision*, vol. 134, no. 3, Feb. 2026. [Online]. Available: <http://dx.doi.org/10.1007/s11263-026-02754-x>
- [5] B. Trabucco, K. Doherty, M. Gurinas, and R. Salakhutdinov, “Effective data augmentation with diffusion models,” 2025. [Online]. Available: <https://arxiv.org/abs/2302.07944>
- [6] A. Bacchin, L. Barcellona, M. Terreran, S. Ghidoni, E. Menegatti, and T. Kiyokawa, “Wastegan: Data augmentation for robotic waste sorting through generative adversarial networks,” in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, Oct. 2024, p. 5080–5087. [Online]. Available: <http://dx.doi.org/10.1109/IROS58592.2024.10802403>
- [7] P. Alimisis, I. Mademlis, P. Radoglou-Grammatikis, P. Sarigiannidis, and G. T. Papadopoulos, “Advances in diffusion models for image data augmentation: A review of methods, models, evaluation metrics and future research directions,” 2025. [Online]. Available: <https://arxiv.org/abs/2407.04103>
- [8] R. He, S. Sun, X. Yu, C. Xue, W. Zhang, P. Torr, S. Bai, and X. Qi, “Is synthetic data from generative models ready for image recognition?” 2023. [Online]. Available: <https://arxiv.org/abs/2210.07574>

- [9] Z. Wang, L. Wei, T. Wang, H. Chen, Y. Hao, X. Wang, X. He, and Q. Tian, “Enhance image classification via inter-class image mixup with diffusion model,” 2024. [Online]. Available: <https://arxiv.org/abs/2403.19600>
- [10] H. Fang, B. Han, S. Zhang, S. Zhou, C. Hu, and W.-M. Ye, “Data augmentation for object detection via controllable diffusion models,” in *2024 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2024, pp. 1246–1255.
- [11] S. Nie, Z. Wang, X. Wang, and K. He, “Diverse generation while maintaining semantic coordination: A diffusion-based data augmentation method for object detection,” 2024. [Online]. Available: <https://arxiv.org/abs/2408.02891>
- [12] M. Zhang, J. Wu, Y. Ren, M. Li, J. Qin, X. Xiao, W. Liu, R. Wang, M. Zheng, and A. J. Ma, “Diffusionengine: Diffusion model is scalable data engine for object detection,” 2023. [Online]. Available: <https://arxiv.org/abs/2309.03893>
- [13] K. Li, S. Luo, and K.-K. Tseng, “Regiondiffusion: Generative data augmentation for object detection with diffusion models,” *Neurocomputing*, vol. 654, p. 131115, 2025. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0925231225017874>
- [14] A. Toker, M. Eisenberger, D. Cremers, and L. Leal-Taixé, “Satsynth: Augmenting image-mask pairs through diffusion models for aerial semantic segmentation,” 2024. [Online]. Available: <https://arxiv.org/abs/2403.16605>
- [15] N. Konz, Y. Chen, H. Dong, and M. A. Mazurowski, “Anatomically-controllable medical image generation with segmentation-guided diffusion models,” 2024. [Online]. Available: <https://arxiv.org/abs/2402.05210>
- [16] J. Ho, A. Jain, and P. Abbeel, “Denoising diffusion probabilistic models.” [Online]. Available: <http://arxiv.org/abs/2006.11239>
- [17] J. Song, C. Meng, and S. Ermon, “Denoising diffusion implicit models,” 2022. [Online]. Available: <https://arxiv.org/abs/2010.02502>
- [18] C. Lu, Y. Zhou, F. Bao, J. Chen, C. Li, and J. Zhu, “Dpm-solver++: Fast solver for guided sampling of diffusion probabilistic models,” *Machine Intelligence Research*, vol. 22, no. 4, p. 730–751, 2025. [Online]. Available: <http://dx.doi.org/10.1007/s11633-025-1562-4>
- [19] J. Ho and T. Salimans, “Classifier-free diffusion guidance,” 2022. [Online]. Available: <https://arxiv.org/abs/2207.12598>

- [20] A. Odena, C. Olah, and J. Shlens, “Conditional image synthesis with auxiliary classifier GANs,” in *Proceedings of the 34th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, D. Precup and Y. W. Teh, Eds., vol. 70. PMLR, 06–11 Aug 2017, pp. 2642–2651. [Online]. Available: <https://proceedings.mlr.press/v70/odena17a.html>
- [21] L. Zhang, A. Rao, and M. Agrawala, “Adding conditional control to text-to-image diffusion models,” 2023. [Online]. Available: <https://arxiv.org/abs/2302.05543>
- [22] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, “High-resolution image synthesis with latent diffusion models,” 2022. [Online]. Available: <https://arxiv.org/abs/2112.10752>
- [23] O. Ronneberger, P. Fischer, and T. Brox, “U-net: Convolutional networks for biomedical image segmentation,” 2015. [Online]. Available: <https://arxiv.org/abs/1505.04597>
- [24] B. Liu, C. Wang, T. Cao, K. Jia, and J. Huang, “Towards understanding cross and self-attention in stable diffusion for text-guided image editing,” 2024. [Online]. Available: <https://arxiv.org/abs/2403.03431>
- [25] C. Meng, Y. He, Y. Song, J. Song, J. Wu, J.-Y. Zhu, and S. Ermon, “Sdedit: Guided image synthesis and editing with stochastic differential equations,” 2022. [Online]. Available: <https://arxiv.org/abs/2108.01073>
- [26] P. Dhariwal and A. Nichol, “Diffusion models beat gans on image synthesis,” 2021. [Online]. Available: <https://arxiv.org/abs/2105.05233>
- [27] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, “Learning transferable visual models from natural language supervision,” 2021. [Online]. Available: <https://arxiv.org/abs/2103.00020>
- [28] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, “An image is worth 16x16 words: Transformers for image recognition at scale,” 2021. [Online]. Available: <https://arxiv.org/abs/2010.11929>
- [29] R. Gal, Y. Alaluf, Y. Atzmon, O. Patashnik, A. H. Bermano, G. Chechik, and D. Cohen-Or, “An image is worth one word: Personalizing text-to-image generation using textual inversion,” 2022. [Online]. Available: <https://arxiv.org/abs/2208.01618>

- [30] N. Ruiz, Y. Li, V. Jampani, Y. Pritch, M. Rubinstein, and K. Aberman, “Dreambooth: Fine tuning text-to-image diffusion models for subject-driven generation,” 2023. [Online]. Available: <https://arxiv.org/abs/2208.12242>
- [31] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, “Lora: Low-rank adaptation of large language models,” 2021. [Online]. Available: <https://arxiv.org/abs/2106.09685>
- [32] L. Perez and J. Wang, “The effectiveness of data augmentation in image classification using deep learning,” 2017. [Online]. Available: <https://arxiv.org/abs/1712.04621>
- [33] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” 2015. [Online]. Available: <https://arxiv.org/abs/1512.03385>
- [34] M. Sokolova and G. Lapalme, “A systematic analysis of performance measures for classification tasks,” *Information Processing Management*, vol. 45, no. 4, pp. 427–437, 2009. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0306457309000259>
- [35] A. Luque, A. Carrasco, A. Martín, and A. de las Heras, “The impact of class imbalance in classification performance metrics based on the binary confusion matrix,” *Pattern Recognition*, vol. 91, pp. 216–231, 2019. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0031320319300950>
- [36] M. Heusel, H. Ramsauer, T. Unterthiner, B. Nessler, and S. Hochreiter, “Gans trained by a two time-scale update rule converge to a local nash equilibrium,” 2018. [Online]. Available: <https://arxiv.org/abs/1706.08500>
- [37] J.-C. Su and S. Maji, “The semi-supervised inaturalist challenge at the fgvc8 workshop,” 2021. [Online]. Available: <https://arxiv.org/abs/2106.01364>
- [38] T. Kumar, A. Mileo, R. Brennan, and M. Bendeckache, “Image data augmentation approaches: A comprehensive survey and future directions,” 2023. [Online]. Available: <https://arxiv.org/abs/2301.02830>
- [39] H. Zhang, M. Cisse, Y. N. Dauphin, and D. Lopez-Paz, “mixup: Beyond empirical risk minimization,” 2018. [Online]. Available: <https://arxiv.org/abs/1710.09412>
- [40] S. Yun, D. Han, S. J. Oh, S. Chun, J. Choe, and Y. Yoo, “Cutmix: Regularization strategy to train strong classifiers with localizable features,” 2019. [Online]. Available: <https://arxiv.org/abs/1905.04899>

- [41] G. Ghiasi, Y. Cui, A. Srinivas, R. Qian, T.-Y. Lin, E. D. Cubuk, Q. V. Le, and B. Zoph, “Simple copy-paste is a strong data augmentation method for instance segmentation,” 2021. [Online]. Available: <https://arxiv.org/abs/2012.07177>
- [42] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, “Imagenet: A large-scale hierarchical image database,” in *2009 IEEE Conference on Computer Vision and Pattern Recognition*, 2009, pp. 248–255.
- [43] A. Krizhevsky, “Learning multiple layers of features from tiny images,” Tech. Rep., 2009.
- [44] R. B. et al, “On the opportunities and risks of foundation models,” 2022. [Online]. Available: <https://arxiv.org/abs/2108.07258>
- [45] Y. Wang and L. Chen, “Inversion circle interpolation: Diffusion-based image augmentation for data-scarce classification,” 2024. [Online]. Available: <https://arxiv.org/abs/2408.16266>
- [46] StabilityAI, “Stable diffusion hugging face repository.” [Online]. Available: <https://huggingface.co/stable-diffusion-v1-5/stable-diffusion-v1-5>
- [47] P. von Platen, S. Patil, A. Lozhkov, P. Cuenca, N. Lambert, K. Rasul, M. Davaadorj, D. Nair, S. Paul, W. Berman, Y. Xu, S. Liu, and T. Wolf, “Diffusers: State-of-the-art diffusion models,” <https://github.com/huggingface/diffusers>, 2022.
- [48] B. Lakshminarayanan, A. Pritzel, and C. Blundell, “Simple and scalable predictive uncertainty estimation using deep ensembles,” 2017. [Online]. Available: <https://arxiv.org/abs/1612.01474>
- [49] “Evaluating Diffusion Models · Hugging Face — huggingface.co,” <https://huggingface.co/docs/diffusers/conceptual/evaluation>, [Accessed 08-05-2026].
- [50] “NVIDIA GeForce RTX 3090 Specs — techpowerup.com,” <https://www.techpowerup.com/gpu-specs/geforce-rtx-3090.c3622>, [Accessed 31-05-2026].
- [51] “Energipriser — danmarks statistik — dst.dk,” <https://www.dst.dk/da/{S}tatistik/emner/miljoe-og-energi/energiforbrug-og-energipriser/energipriser>, [Accessed 31-05-2026].
- [52] “Nøgletal om energiforbrug og -forsyning — ens.dk,” <https://ens.dk/analyser-og-statistik/noegletal-om-energiforbrug-og-forsyning>, [Accessed 31-05-2026].

10 Appendix

A Code repository

All code used to implement the Diffusion and evaluation pipeline can be found in the following GitHub repository:

https://github.com/Crask21/aris_master

B Generated Image Collages

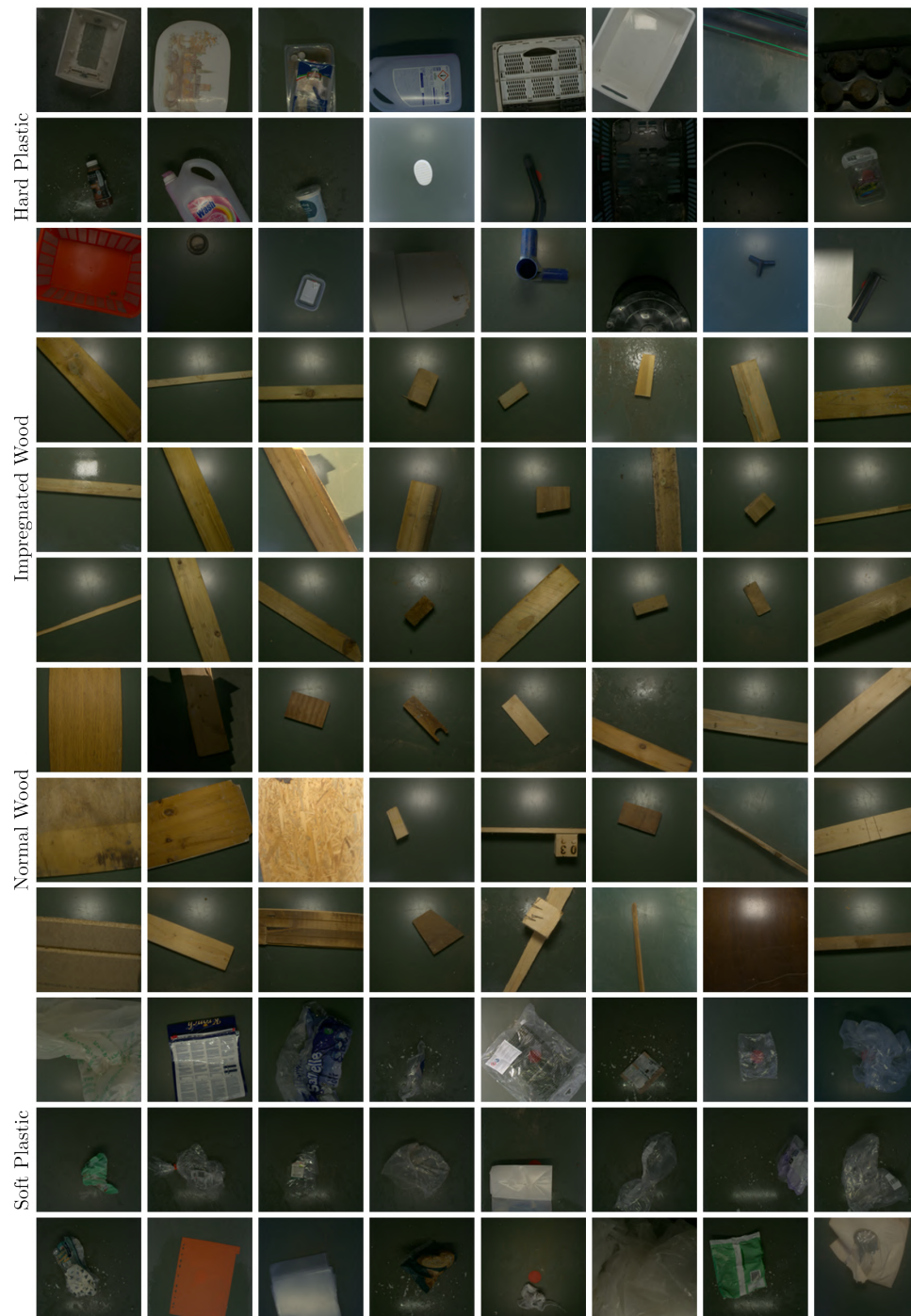


Figure B.1: Samples of real images of the Aris waste classification dataset

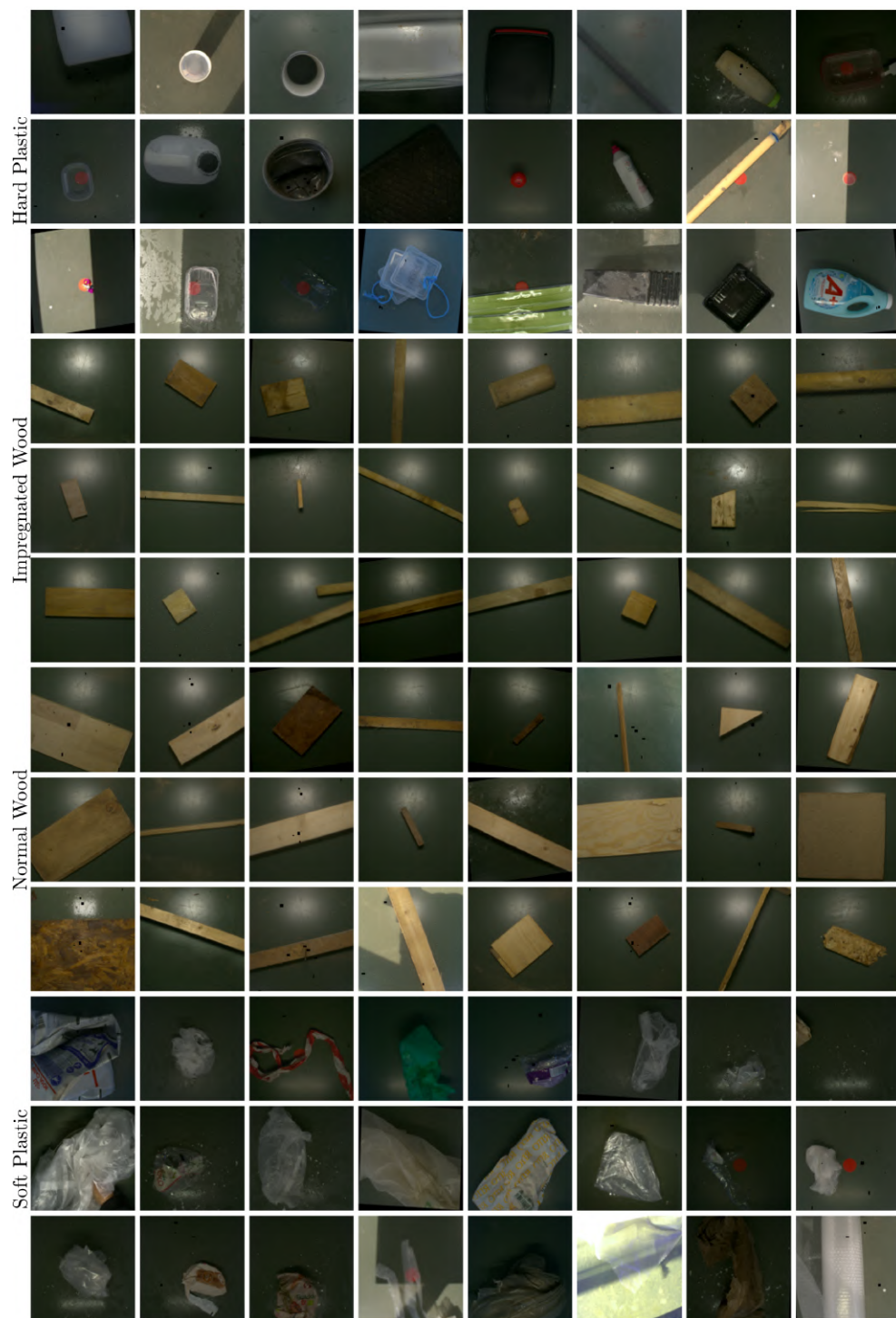


Figure B.2: Classic augmentation samples from the 1000 image/class dataset.

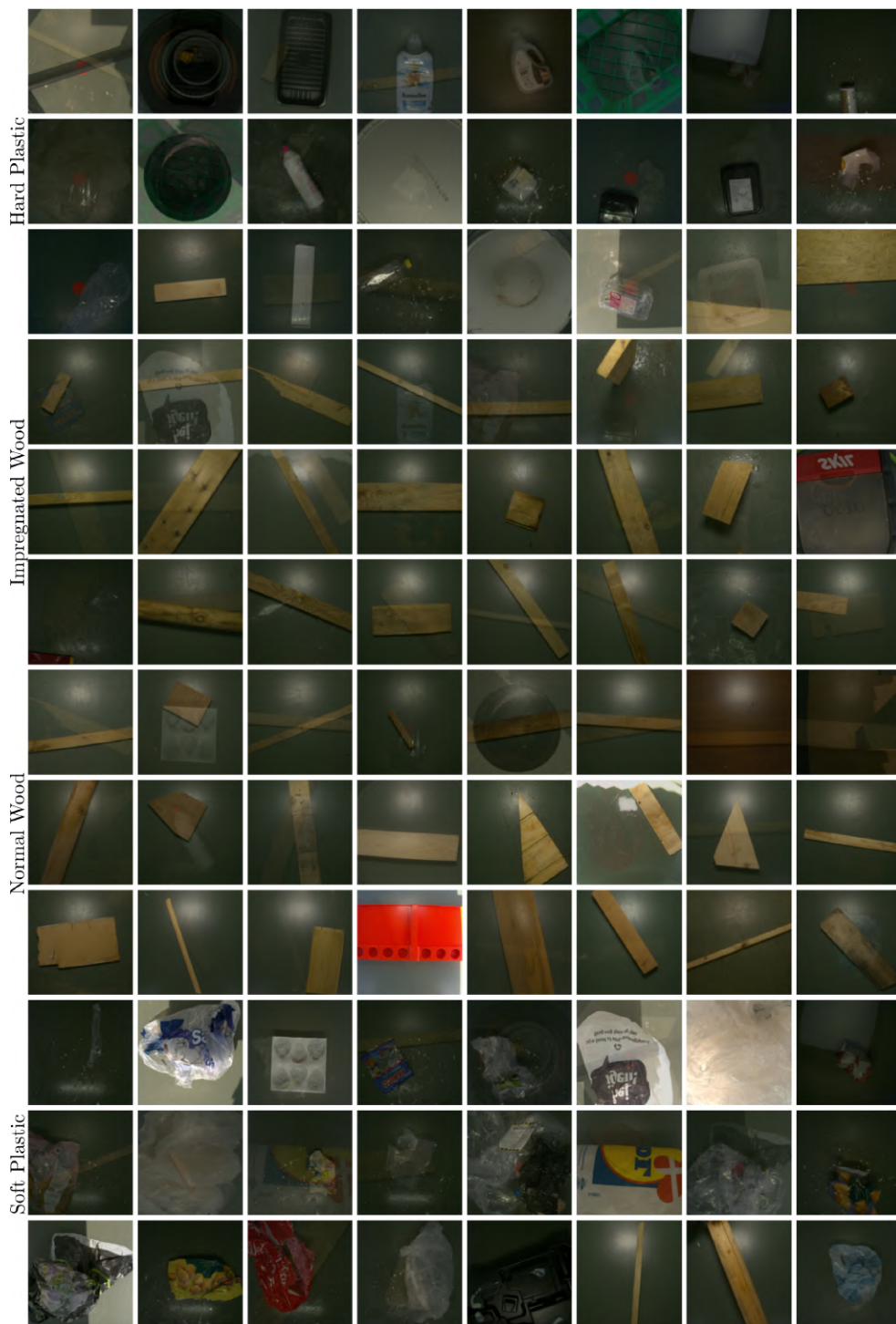


Figure B.3: Mixup samples from the 1000 image/class dataset

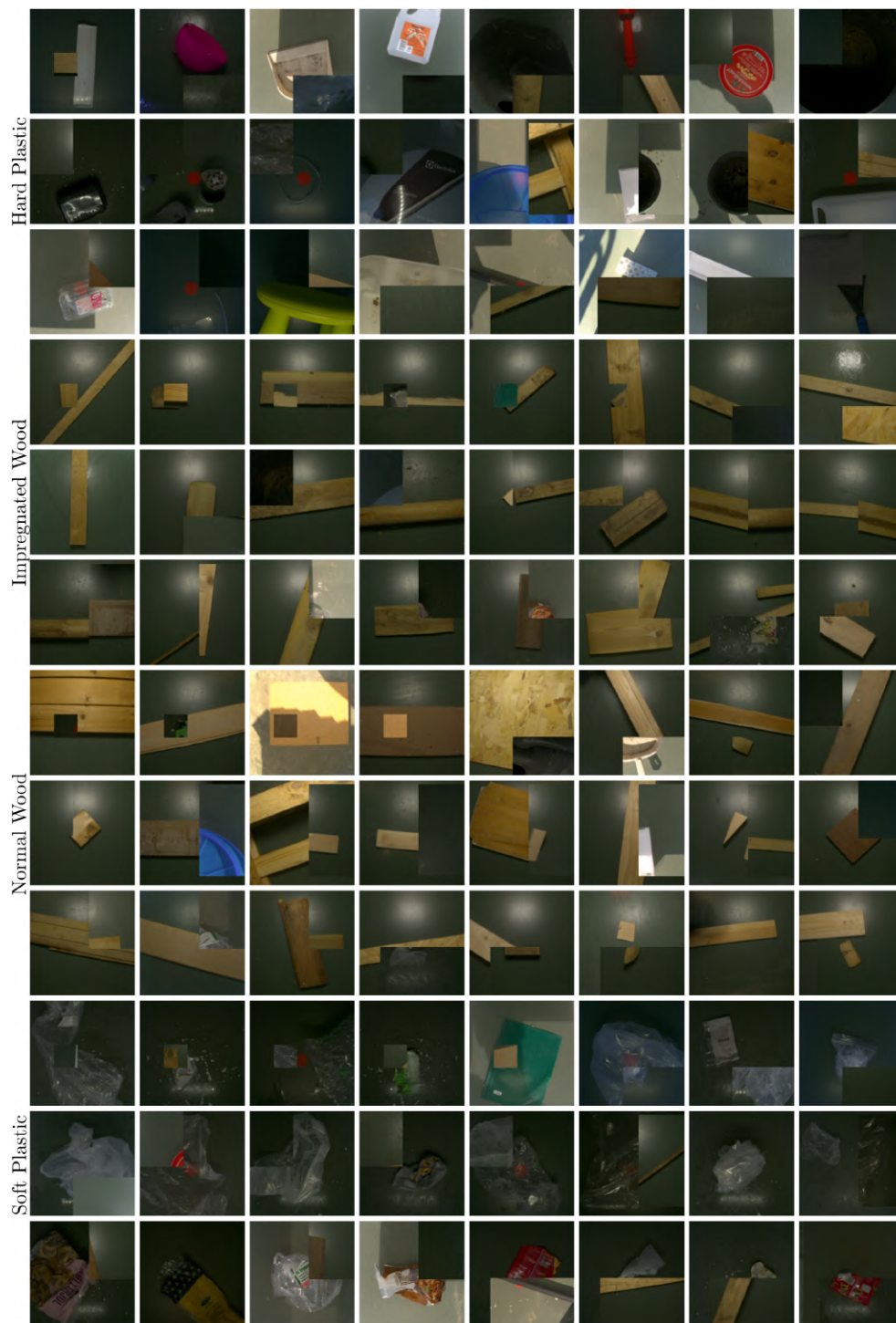


Figure B.4: CutMix samples from the 1000 image/class dataset



Figure B.5: Copy-paste method for synthetic image generation. Aris' current method.



Figure B.6: Diff-Mix generated images from 1000 image/class dataset



Figure B.7: Diff-Aug generated images from 1000 image/class dataset

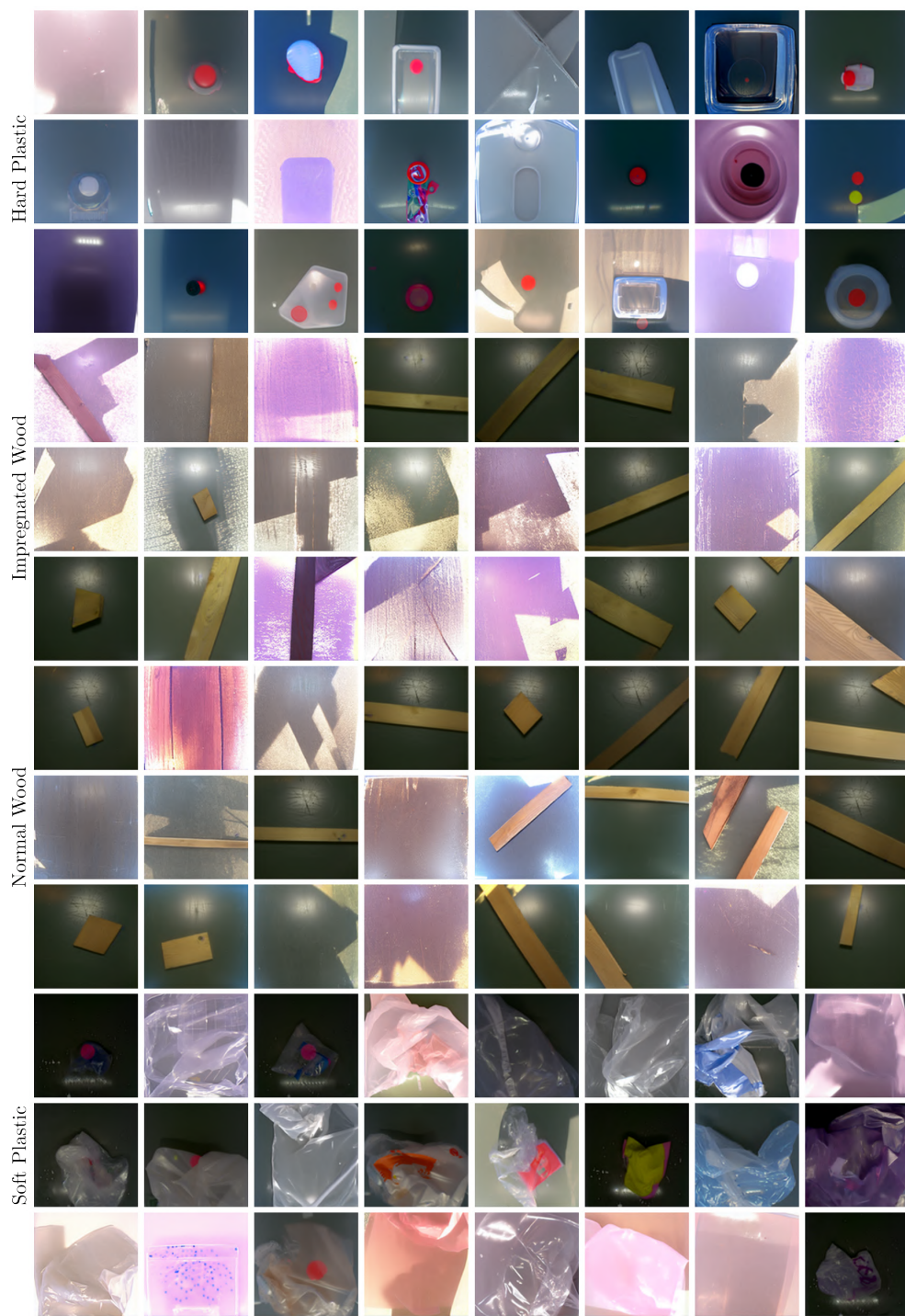


Figure B.8: Diff-II generated images from 1000 image/class dataset

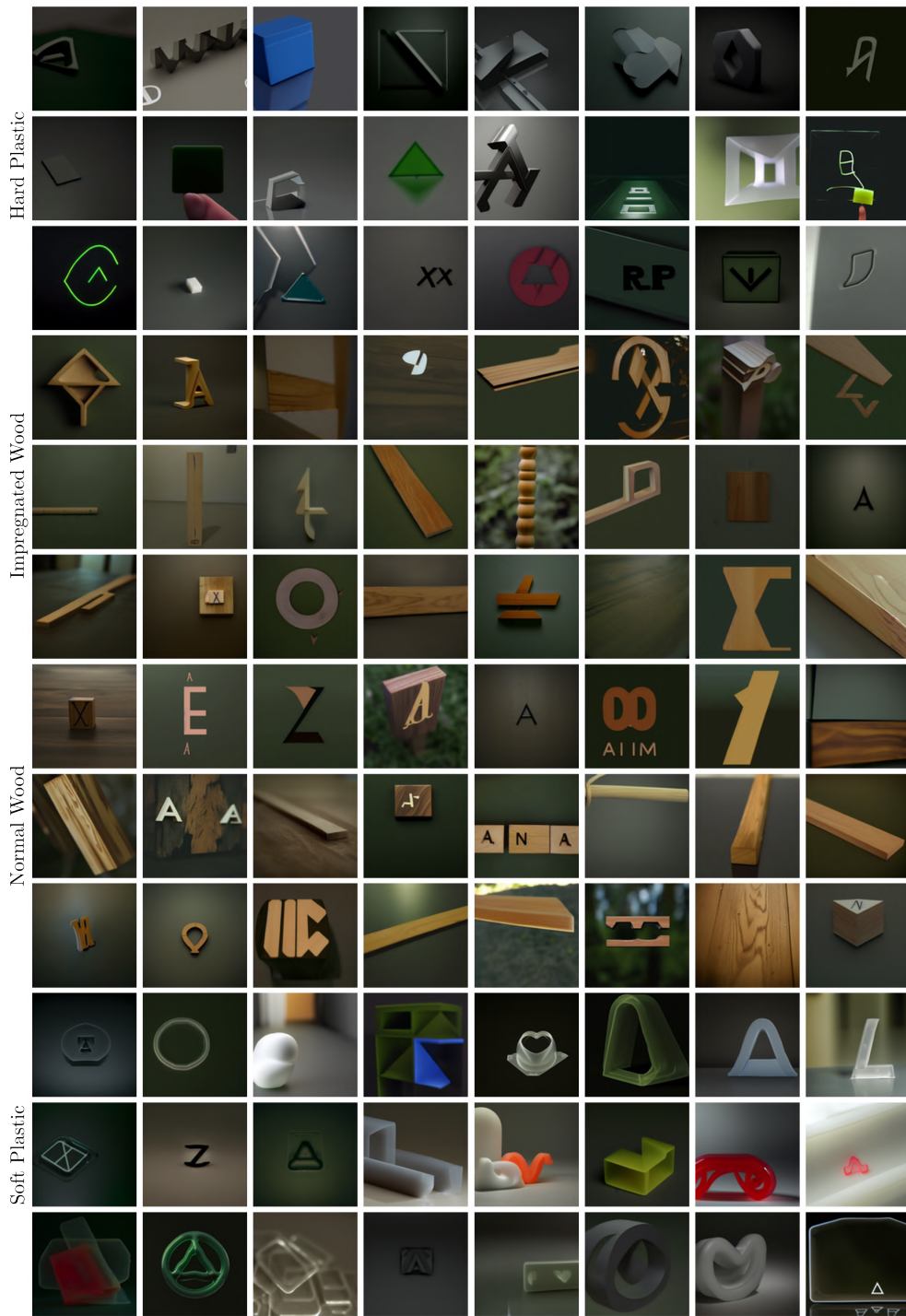


Figure B.9: DA-Fusion generated images from 1000 image/class dataset

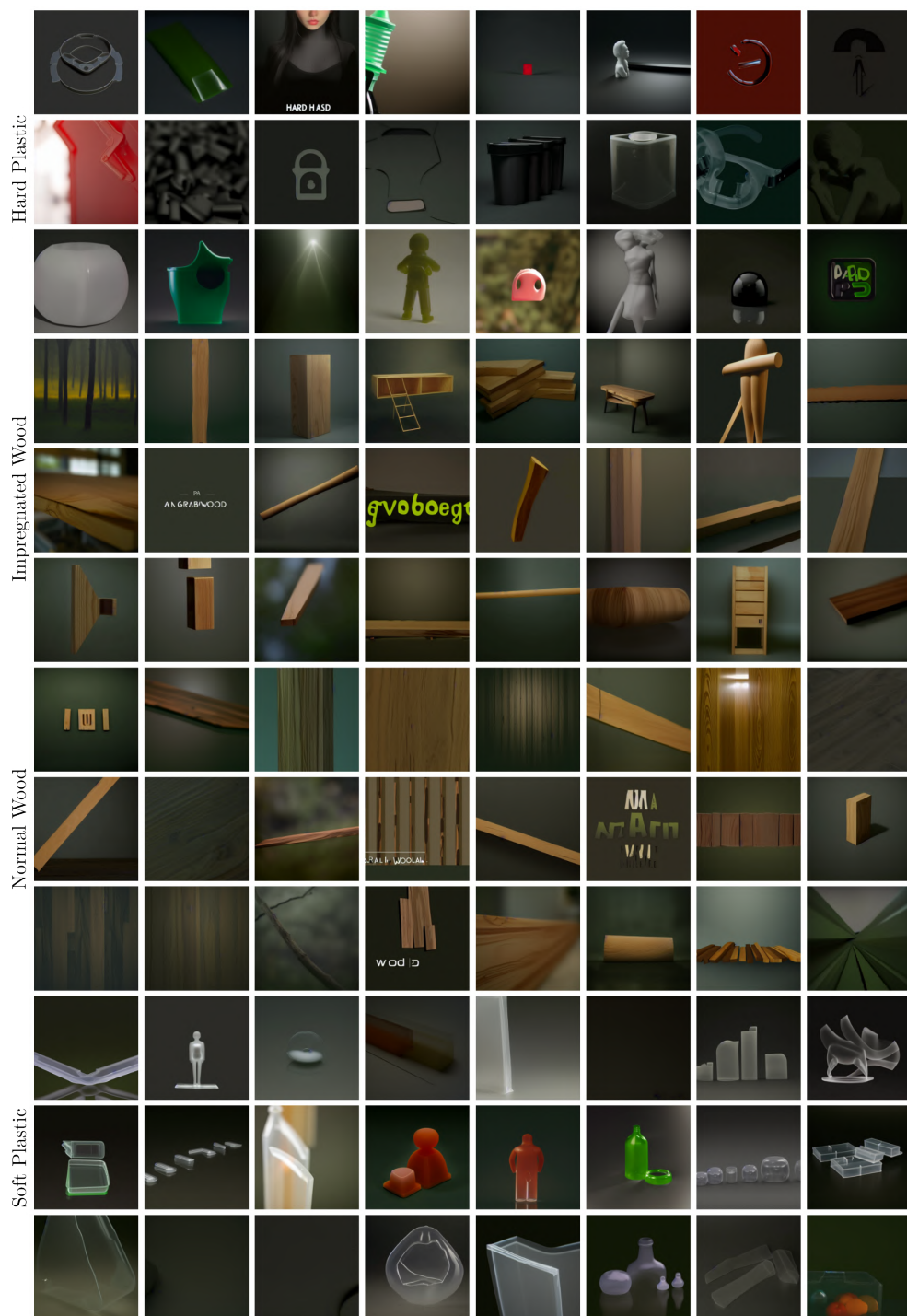


Figure B.10: Real Guidance generated images from 1000 image/class dataset



Figure B.11: SDEdit generated images using class-conditional model trained on 1000 image/class dataset



Figure B.12: Class-conditional generated images from 1000 image/class dataset

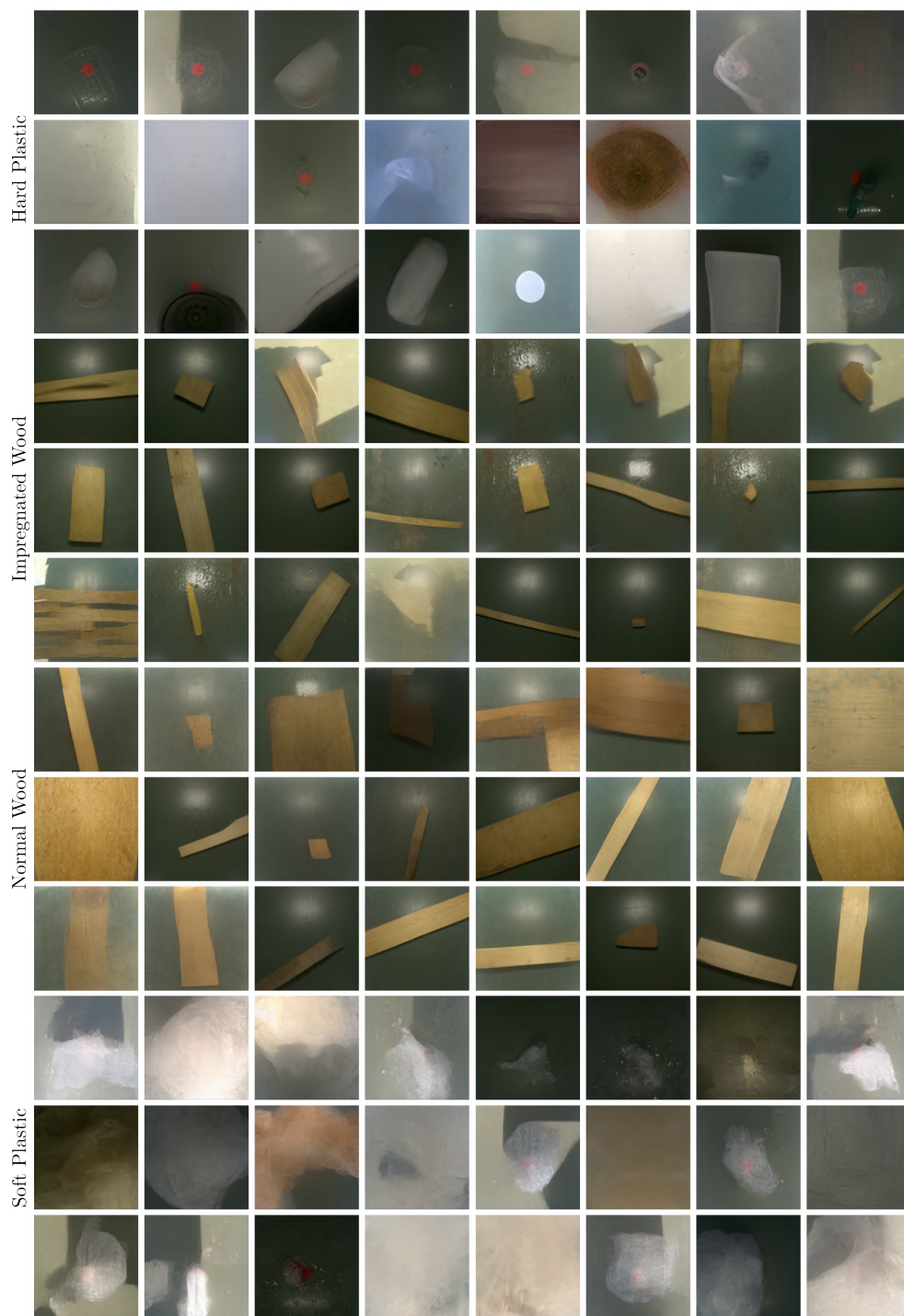


Figure B.13: Unconditional generated images from 1000 image/class dataset



Figure B.14: Mask-conditioned generated images from 1000 image/class dataset

C Machine dataset preview



Figure C.1: Dataset preview of the fierce-wolf dataset.



Figure C.2: Dataset preview of the bold-eagle dataset.

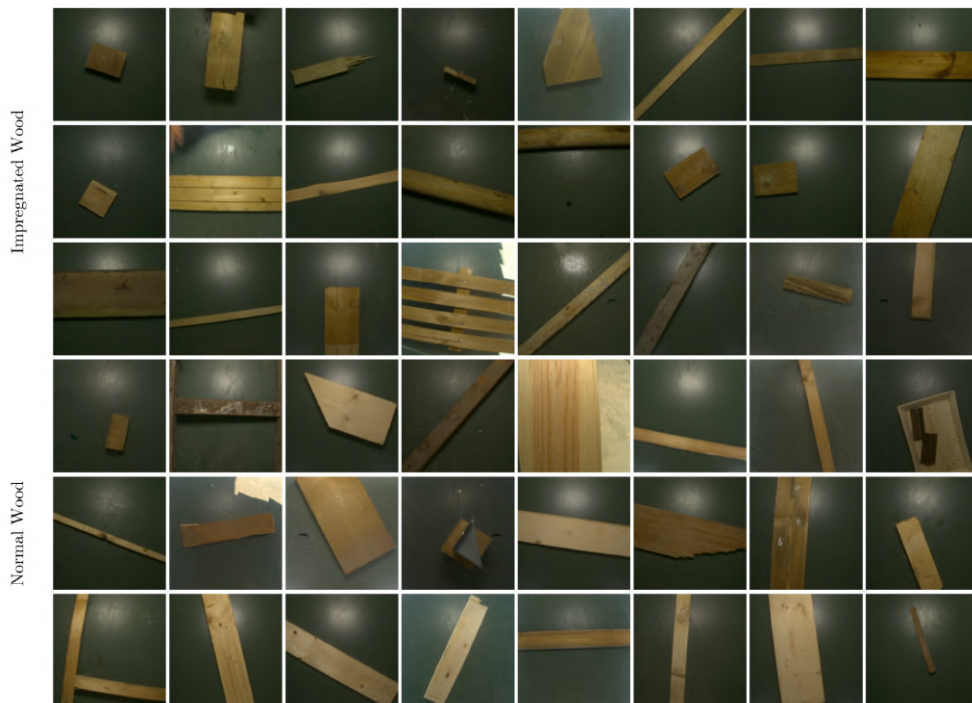


Figure C.3: Dataset preview of the control dataset.

D Cross-model comparison

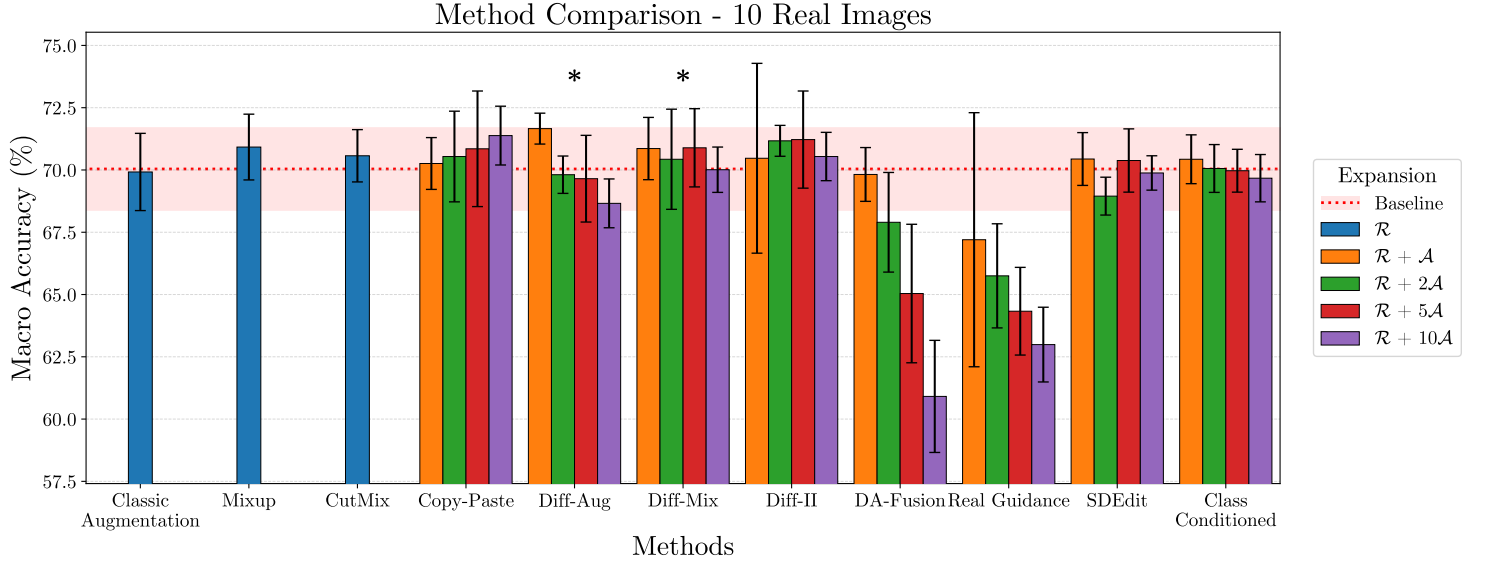


Figure D.1: Method comparison for 10 real images/class, evaluated on test macro accuracy.

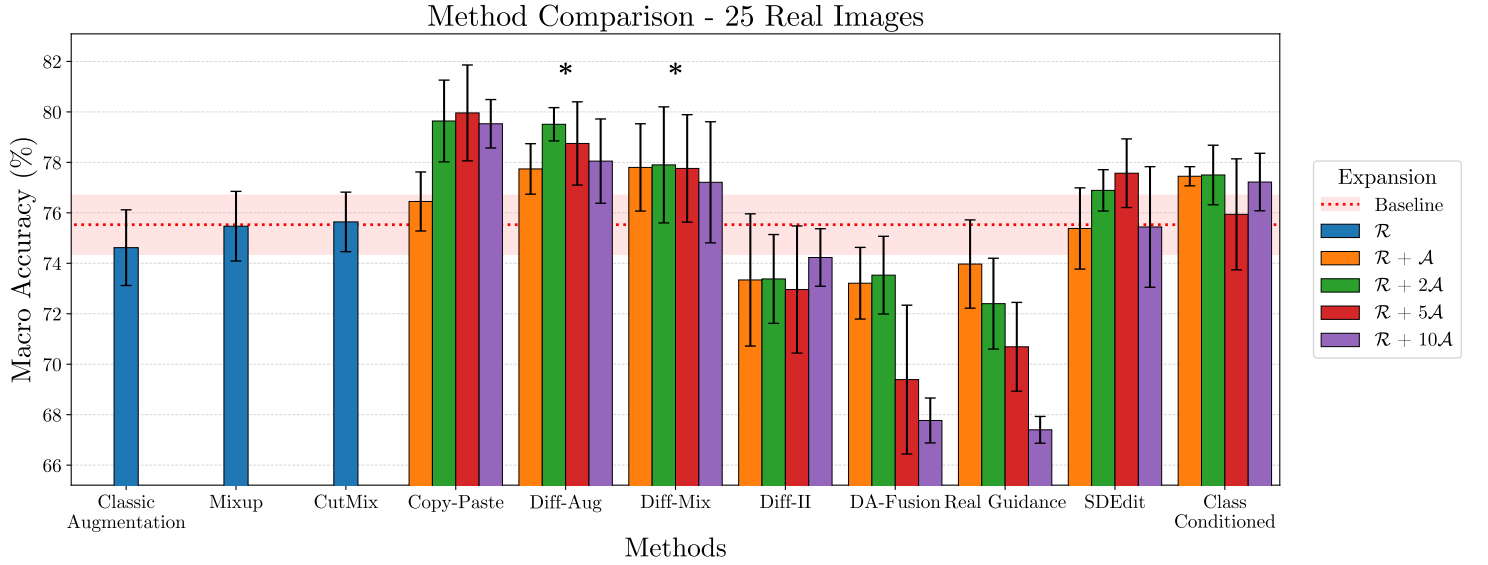


Figure D.2: Method comparison for 25 real images/class, evaluated on test macro accuracy.

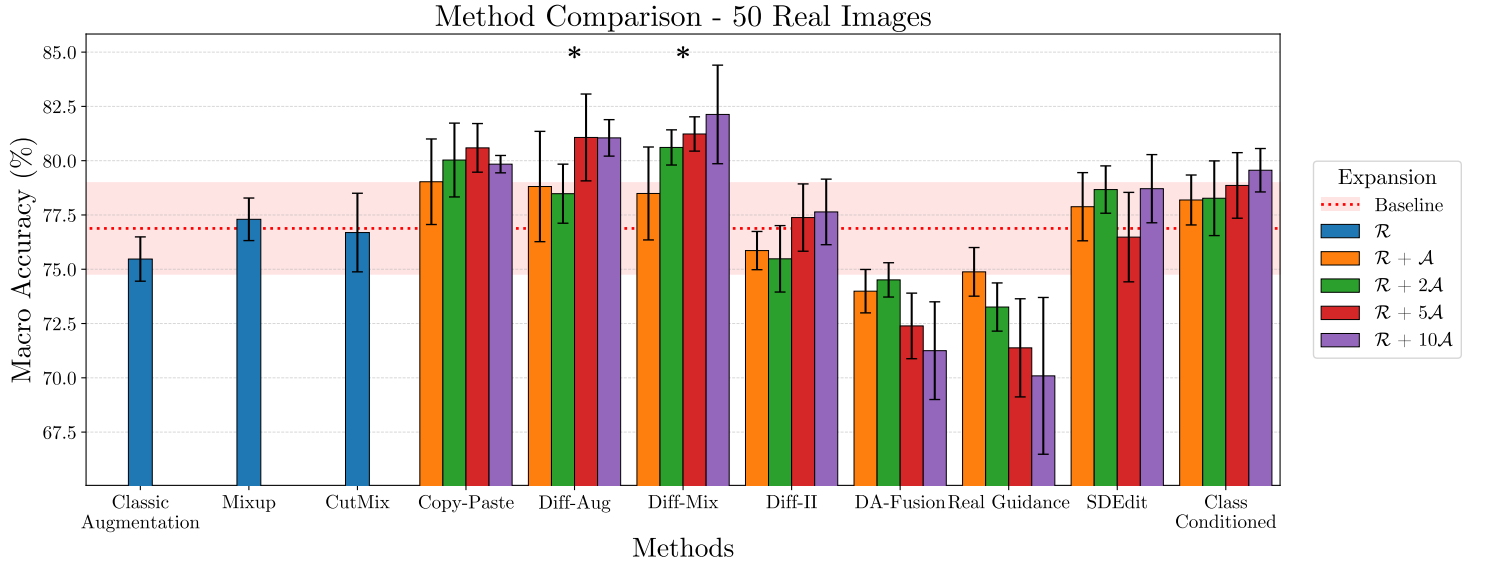


Figure D.3: Method comparison for 50 real images/class, evaluated on test macro accuracy.

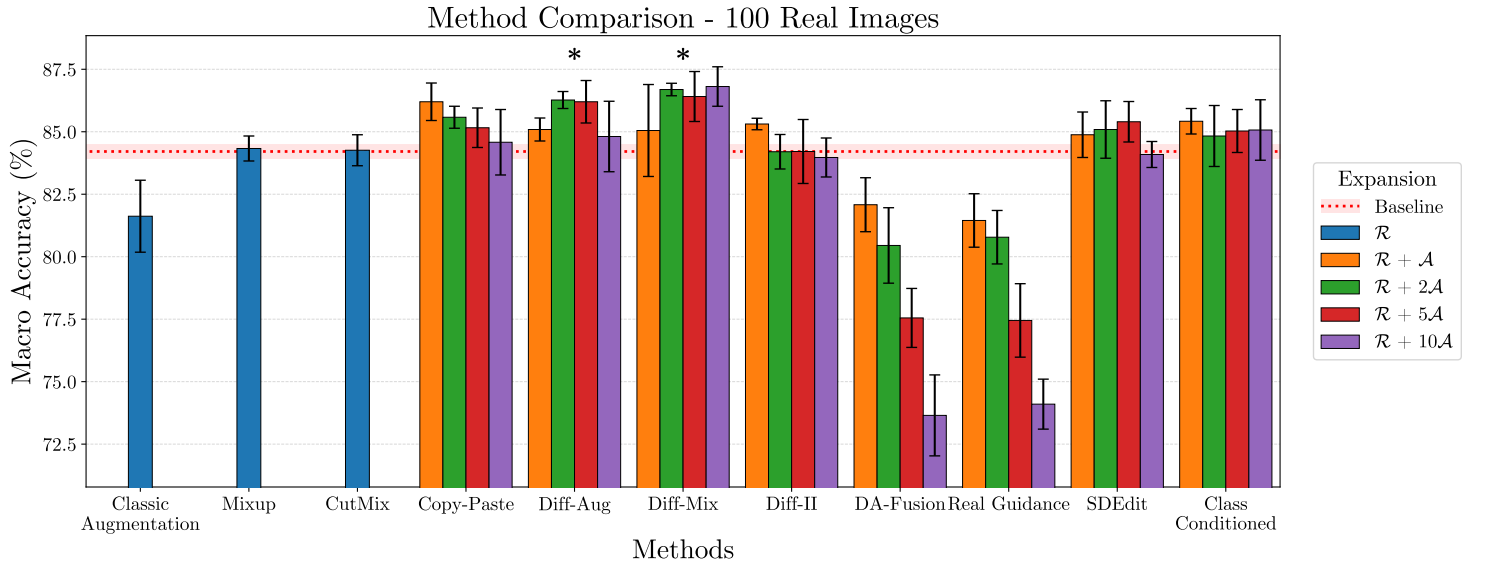


Figure D.4: Method comparison for 100 real images/class, evaluated on test macro accuracy.

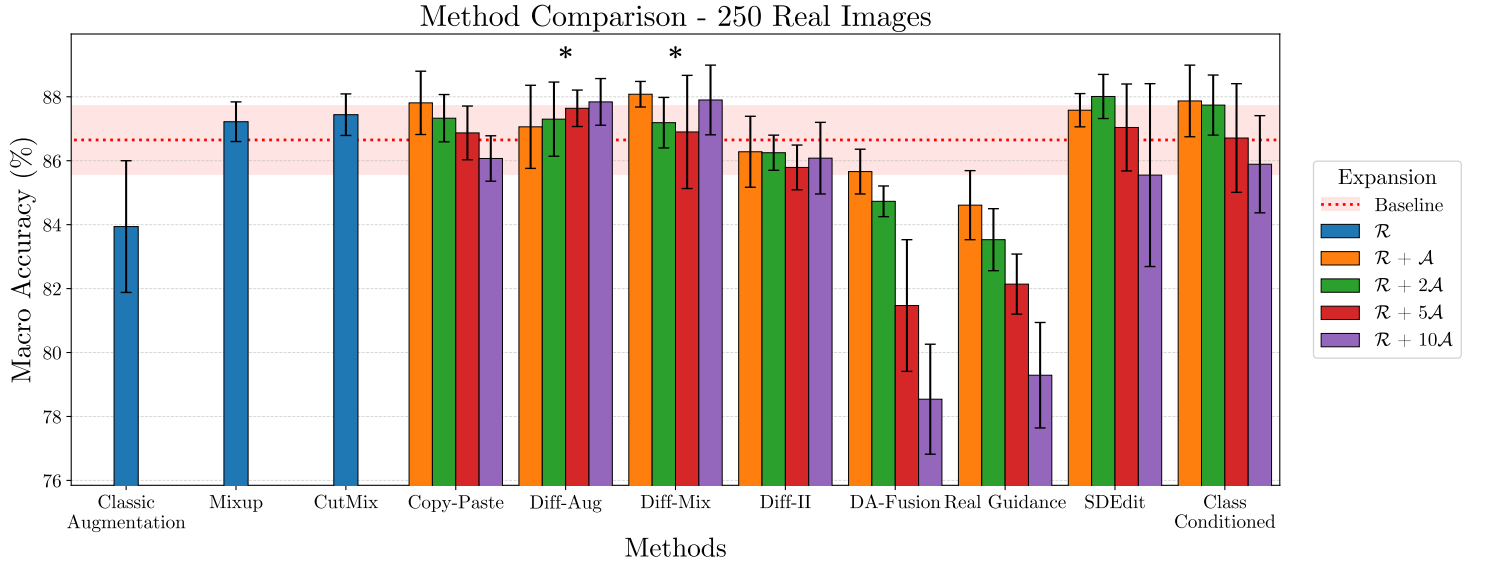


Figure D.5: Method comparison for 250 real images/class, evaluated on test macro accuracy.

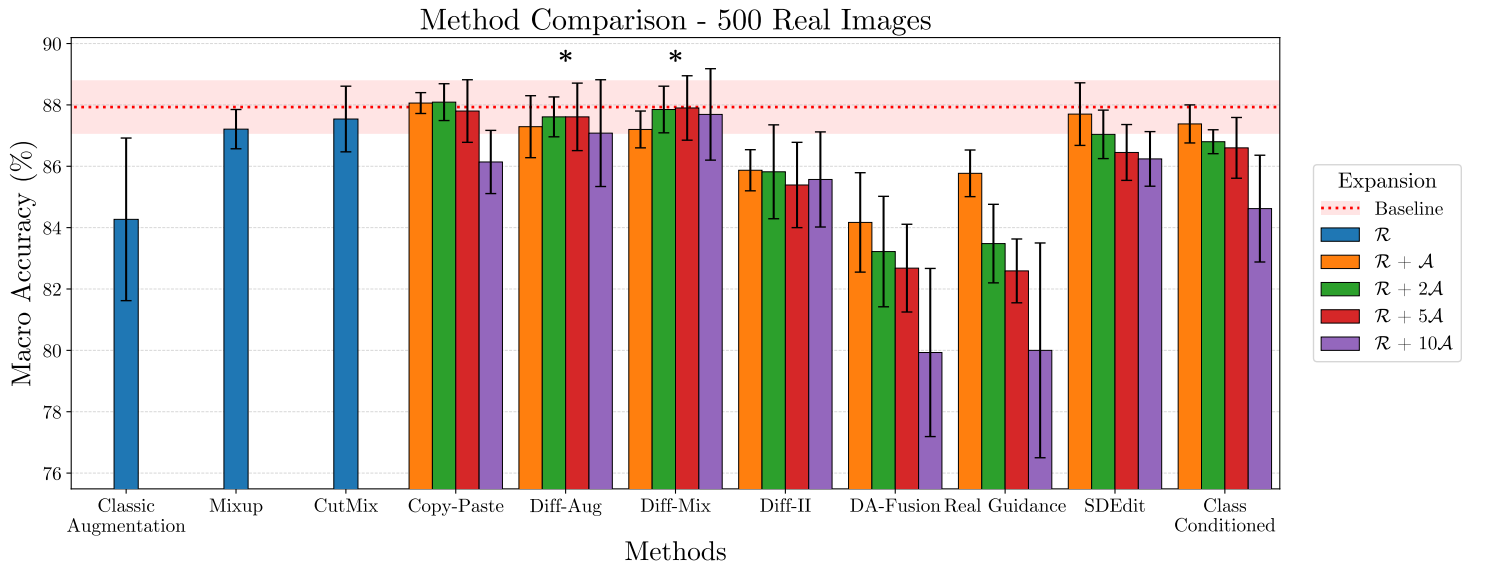


Figure D.6: Method comparison for 500 real images/class, evaluated on test macro accuracy.

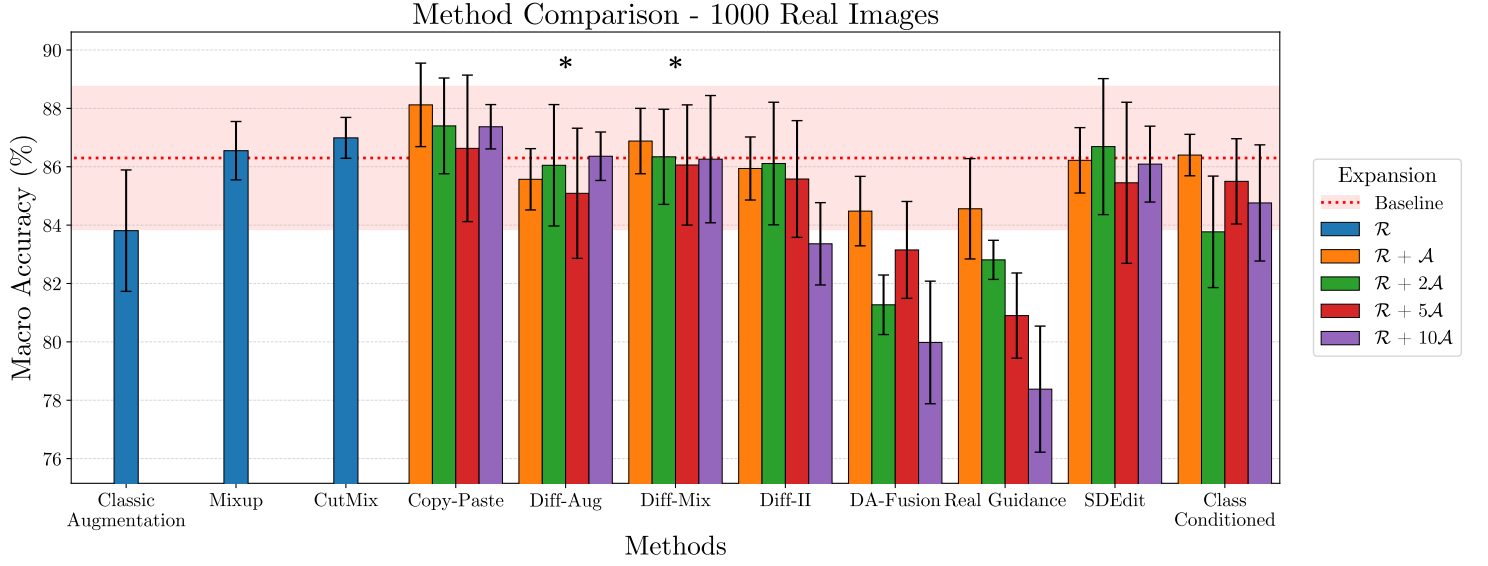


Figure D.7: Method comparison for 1000 real images/class, evaluated on test macro accuracy.

E Dataset size comparison

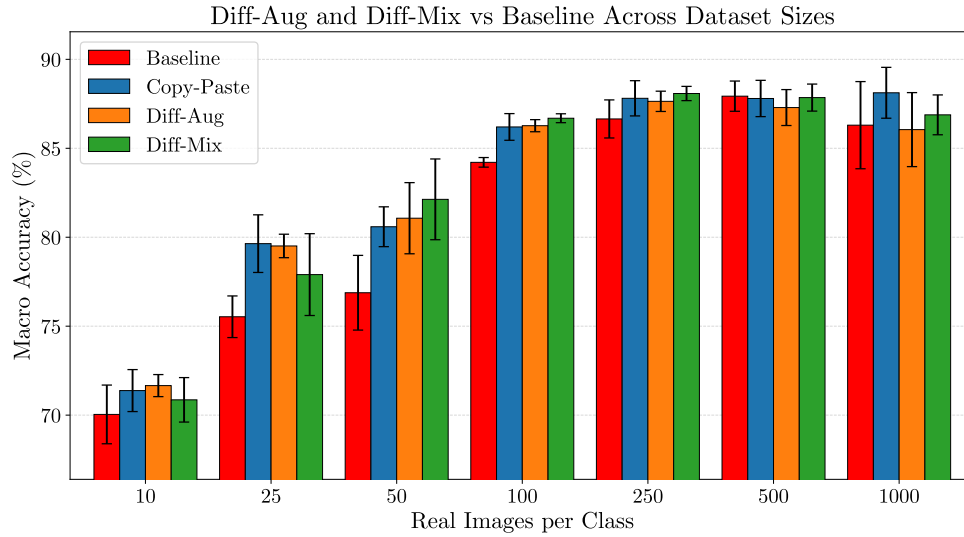


Figure E.1: Diff-Aug and Diff-Mix comparison with baseline and Copy-Paste across dataset sizes. For each dataset size, the best-performing synthetic split is selected. The test macro accuracy is reported.

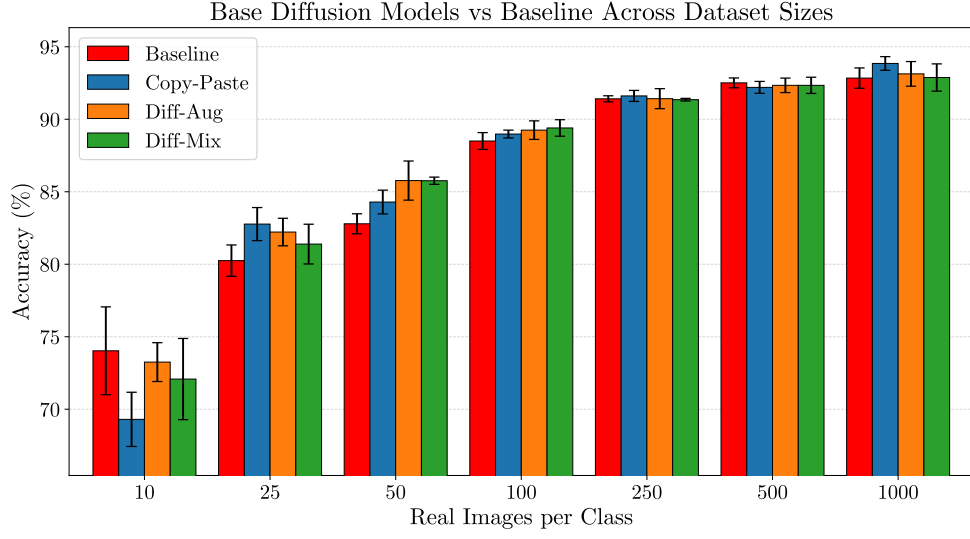


Figure E.2: Diff-Aug and Diff-Mix comparison with baseline and Copy-Paste across dataset sizes. For each dataset size, the best-performing synthetic split is selected. The test accuracy is reported.

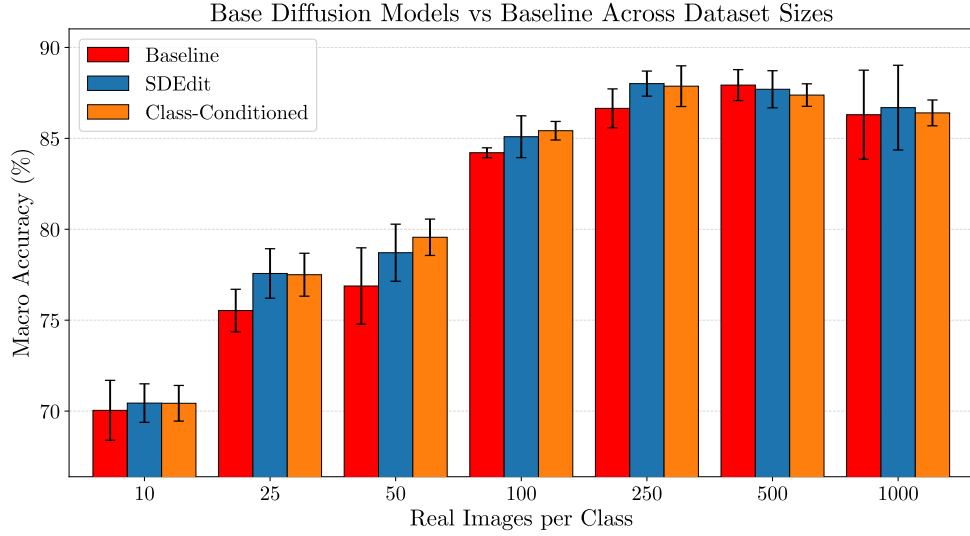


Figure E.3: Base diffusion model comparison with baseline across dataset sizes, herein class-conditioned diffusion and SDEdit. For each dataset size, the best-performing synthetic split is selected. The test macro accuracy is reported.

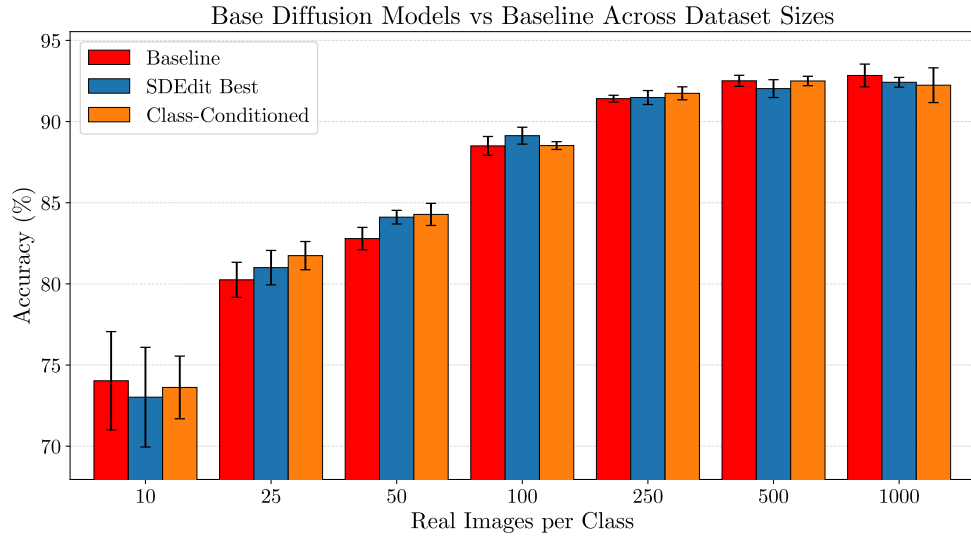


Figure E.4: Base diffusion model comparison with baseline across dataset sizes, herein class-conditioned diffusion and SDEdit. For each dataset size, the best-performing synthetic split is selected. The test accuracy is reported.